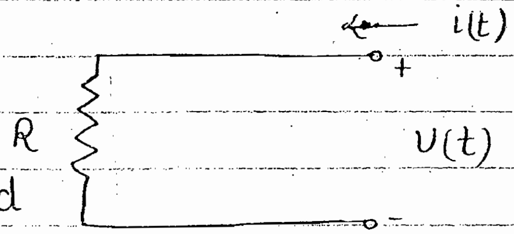
ANUPAM SHUKLA

Network Theory

Components of electrical networks :-

1. **Resistor**



linear and bileteral.

$$\boxed{\begin{aligned} V(t) &= R\, i(t) \\ i(t) &= \frac{1}{R} V(t) \end{aligned}}$$ ------ in time domain

$$\boxed{V(S) = R\, I(S)}$$

---- in s-domain

$$\boxed{I(S) = \frac{1}{R} V(S)}$$ S = complex freq

$S = \sigma + j\omega$

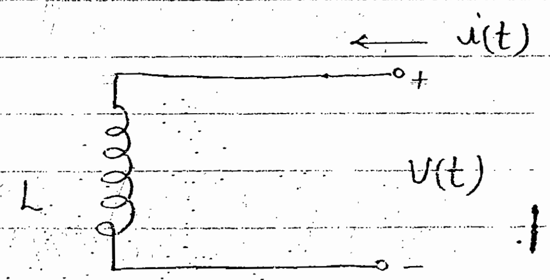
$$\boxed{\begin{aligned} V &= I R \\ I &= \frac{1}{R} V \end{aligned}}$$ using phasor

for sinusoidal excitation

$S = j\omega$

$V = \underbrace{|V|}_{\text{RMS value}} \angle\theta \qquad I = |I| \angle\theta$

2. **Inductor** -



$\boxed{\phi_m \propto i}$ (a)

$\phi_m = L i$ Inductor $\longrightarrow$ Linear

$V(t) = L \dfrac{d\, i(t)}{dt}$ (i)

--- in

$i(t) = \dfrac{1}{L} \displaystyle\int_0^t V(t)\, dt$ (ii) time domain from above 2 equa^n it is clear that they are not linear i.e they are not directly proportional but they are derive from equa^n (a) so Inductor is also considered as Linear element.

$$\boxed{\begin{aligned} V(S) &= SL\,I(S) \\ I(S) &= \frac{1}{SL} V(S) \end{aligned}}$$ --- In S domain assuming zero

above equa^n also shows (initial condi^n in linear equa^n. S domain) ICs

$V = j\omega L . I$ (a) --- using phasor ($S = j\omega$)

$I = \frac{1}{j\omega L} . V$ (b) -- for sinusoidal excitation

as $j = 90$ it means V is leading I by 90 and in equa^n (b) $j = 90$ i.e $I = -90 . V$ it means I is lagging V by 90

$Z_L = SL$, $Y_L = 1/SL$

$$\boxed{Z_L = j\omega L \quad \& \quad Y_L = 1/\omega L}$$

In a linear element the variation betw^n the voltage and current variables is linear either in time domain or S domain or in both domain.

In a bilateral element the current flows

I/P voltage source
or I/P I "

in either direction irrespective of the the type of polarity of the voltage source applied between the two terminals of a given network.

The excitation to a given n/w may be non sinosoidal or sinosoidal in nature.

When the exceti^n is nonsinosoidal a given n/w can be analysed either in the time domain or in the laplase domain.

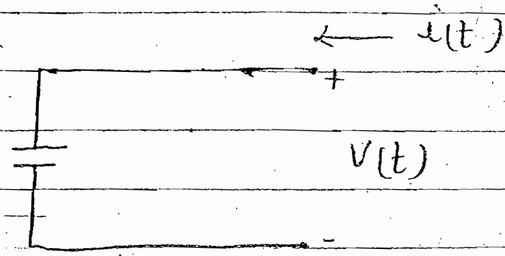
When the excitation is sinosoidal the n/w can be analysed only by using phasors by substituting $s = jw$

When any network is analysed in the laplase domain the advantages are :-

a) any differential or integral equa^n is transformed to a linear equa^n and the manipula^n becomes simpler.

b) Initial condi^n if any present in the element are automatically taken care of in the analysis.

Capacitor :



$$\boxed{Q \propto V}$$
$$\boxed{Q = CV}$$

$$\boxed{i(t) = C\frac{dV(t)}{dt}}$$

$$\boxed{v(t) = \frac{1}{C}\int_0^t i(t)\,dt}$$ in time domain

$I(s) = sCV(s)$

---- in s domain assuming zero ICs

$V(s) = \frac{1}{sC} \cdot I(s)$

$Y_C = sC$

$Z_C = 1/sC$

$$\boxed{\begin{array}{l} I = j\omega C V \\ V = \dfrac{1}{j\omega C} I \end{array}}$$

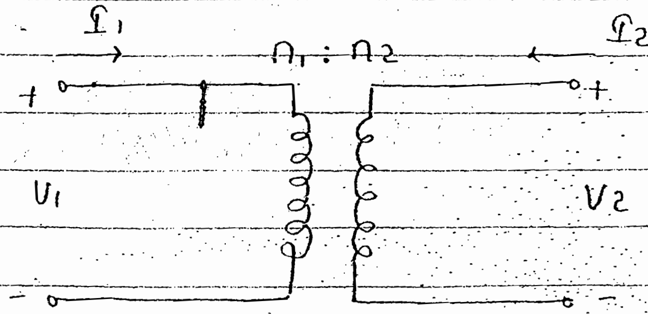
--- using phasors ($s = j\omega$)

--- for sinusoidal excitation

$\boxed{Y_C = j\omega C}$ & $\boxed{Z_C = 1/j\omega C}$

	L	C
$\omega = \omega$	$Z_L = j\omega L$	$Z_C = 1/j\omega C$
$\omega = 0$	S. ckt.	Open ckt.
$\omega = \infty$	O. ckt.	Short ckt.

Transformer :



$$\boxed{\frac{V_1}{V_2} = \frac{n_1}{n_2}}$$

$n_2 > n_1$ --- Step up

$n_2 < n_1$ --- Step down

$$I_1/I_2 = n_2/n_1$$

If $n_1 = 1$ & $n_2 = 10$

$$\frac{V_2}{V_1} = \frac{n_2}{n_1} = \frac{10}{1} \quad ; \quad (>1)$$

$$\frac{I_2}{I_1} = \frac{n_1}{n_2} = \frac{1}{10}$$

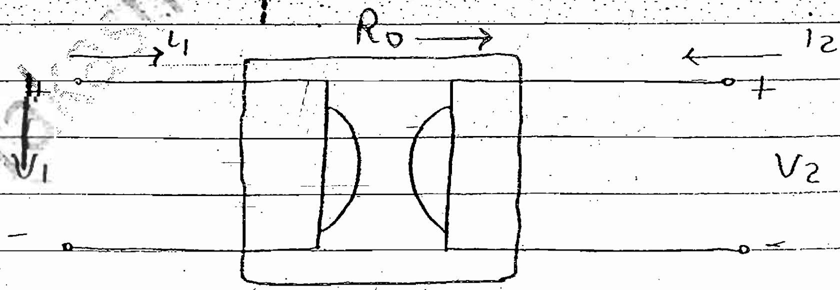
$$\frac{V_2}{V_1} \times \frac{I_2}{I_1} = 1$$

i.e $\boxed{V_2 I_2 = V_1 I_1}$

Any step up xmer cannot work as as an amplifier since power at the I/P and at O/P remains same.

For any device to work as an amplifier the power and the O/P terminals must always be more than the power at the input terminals.

Gyrator:



$R_0 \rightarrow$ Cofficient of gyrator

$$\boxed{\begin{aligned} V_1 &= +R_0 i_2 \\ V_2 &= -R_0 i_1 \end{aligned}}$$

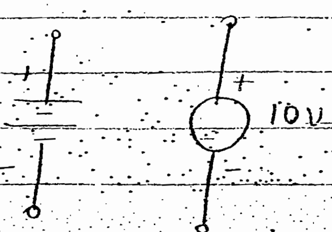
This device represent Impedance Inversion

A Gyrator is a four terminal or a two port device.

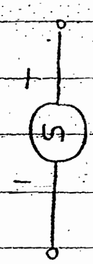
- It can be made by using an operational amplifier and some externally connected R & C components.
- The coffecient of the gyrator depends upon the parameters of the operational amplifier & the numerical values of R & C components.
  - for a given Gyrator Ro has a fixed value.
- It can be used to simulate inductance.
- It represents an impedence inversion. ∴ for a capacitive load the I/P impedance is inductive and vice versa.

6. <u>Voltage Sources</u> :

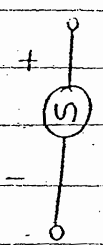
Independent V. Sources -



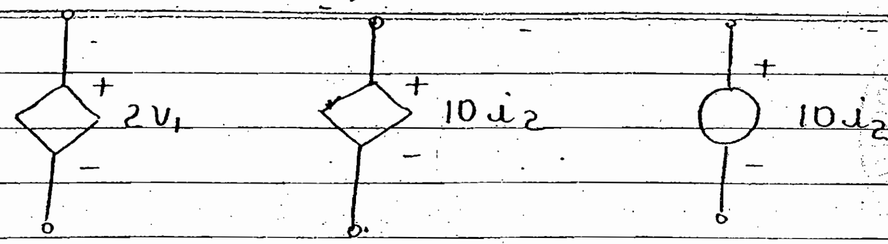
DC voltage source



$10t\,u(t)$
$5e^{-2t}\,u(t)$ ------ Time varying non-sinosoidal voltage sources.
$20te^{-5t}\cos 200t$



$2\angle 45^\circ$ -- V
$\underbrace{20\sin 200t}_{V_m \sin \omega t}$ -- V ------ Sinosoidal voltage sources.



VCVS CCVS CCVS

dependent or controlled voltage source

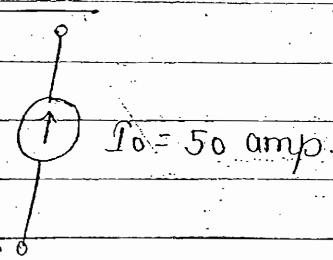
NOTE :-

While analysing any network containing independent and dependent sources the handling of these sources remains same while writing the KVL and KCL equations except the following two cases:-

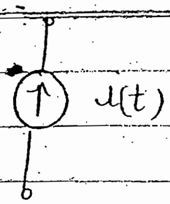
a) Analysis using the superposiⁿ theorem.
b) Analysis using thevenin's & norton's theo.

In such cases :- All independent voltage sources are short ckt or replaced by their internal impedances.
All independent current sources are open circuited or replaced by their internal impedances.
All dependent voltage and current sources remain as they are & ∴ these sources are neither short circuited nor open ckt

Current Sources :



---- Dc current source



--- $5te^{-2t}u(t)$

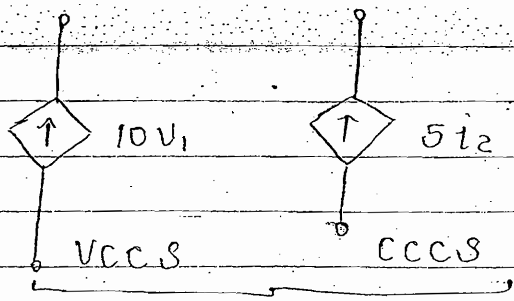
--- $10e^{-10t}u(t)$

$5te^{-2t}\sin 100t$ — Source

non-sinusoidal time varying current.

$20\angle 90°$ --- A

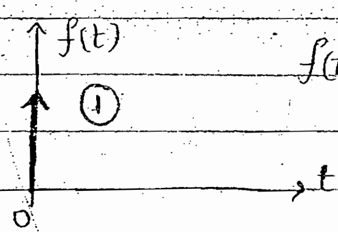
$5\sin 100t$ --- A — Sinusoidal current sources.



Controlled or dependent current sources

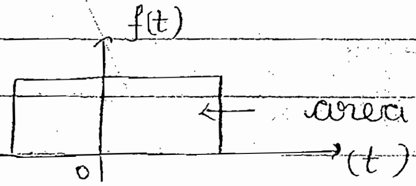
Basic functions :

Unit Impulse func^n



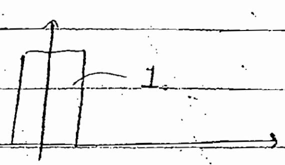
$$f(t) = \delta(t) = \infty \; ; \; t = 0$$
$$= 0 \; ; \; t \neq 0$$

$$\boxed{\int_{-\infty}^{\infty} \delta(t)\,dt = 1}$$



$$\int_{-\infty}^{\infty} g(t)\,\delta(t)\,dt = g(t)\Big|_{t=0} = g(0)$$

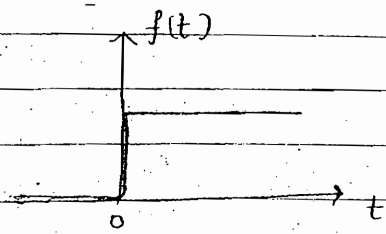
exists at $t=0$



$$\int_{-\infty}^{\infty} g(t)\,\underbrace{\delta(t-T)}_{\text{exists at } t=T}\,dt = g(t)\Big|_{t=T} = g(T)$$

$\delta(t) \to 1$

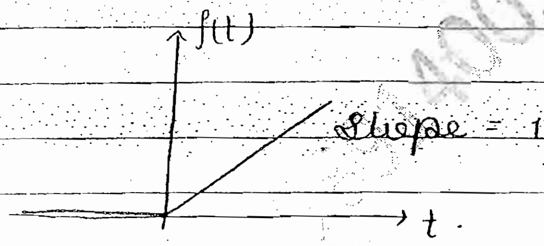
Unit Step function :



$$f(t) = u(t) = 0 \; ; \; t < 0$$
$$= 1 \; ; \; t > 0$$

$$\boxed{u(t) = 1/s}$$

Unit Ramp function :

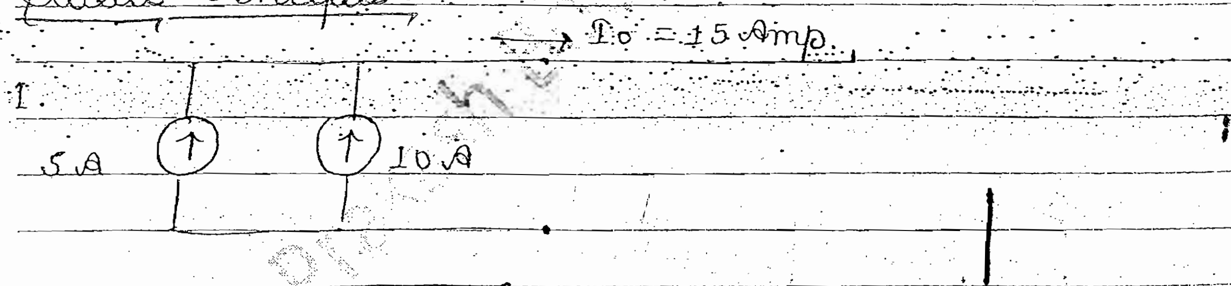


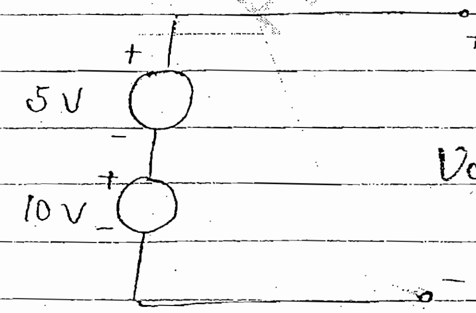
$$f(t) = 0 \; ; \; t < 0$$
$$= t \; ; \; t > 0$$

$$\boxed{f(t) = t\,u(t)} \; , \; \boxed{f(t) = r(t)}$$

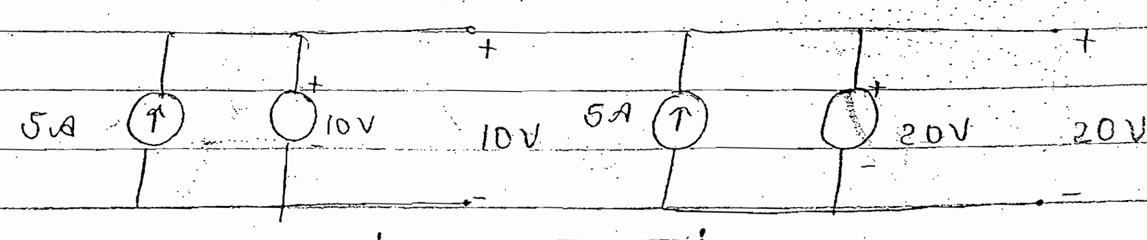
$$\boxed{r(t) = 1/s^2}$$

Basic Concepts :

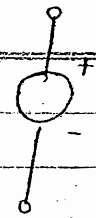




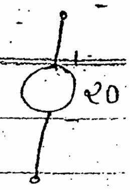
$V_o = 15V$



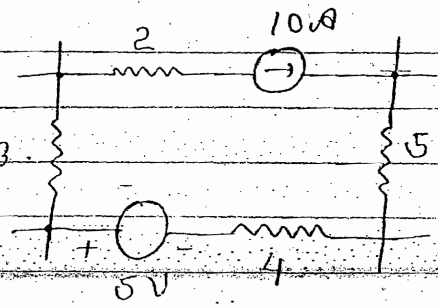
$\Rightarrow$



10V constt voltage source



In any loop if current source is present then

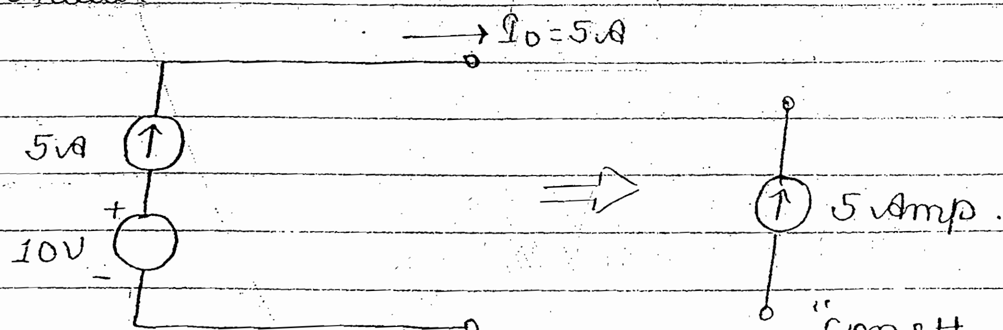


A parallel combina^n of a voltage source or a current is equivalent to a const voltage source.

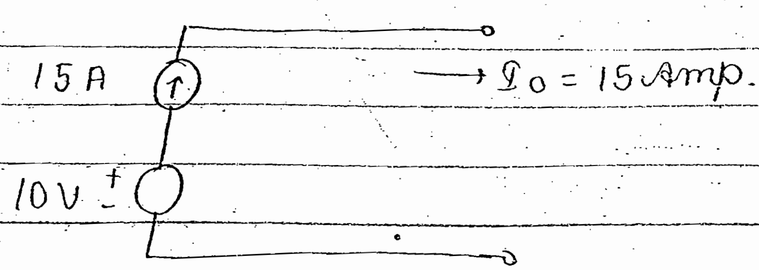
The voltage across a current source is purely arbitary and depends only upon externally connected voltage source or externally connected element.

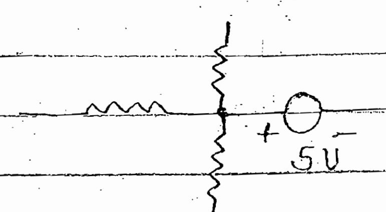
We cannot write any KVL equa^n in a closed loop which contains a current source since voltage across it is arbitrary and is unknown.

3.



"Constt current source."

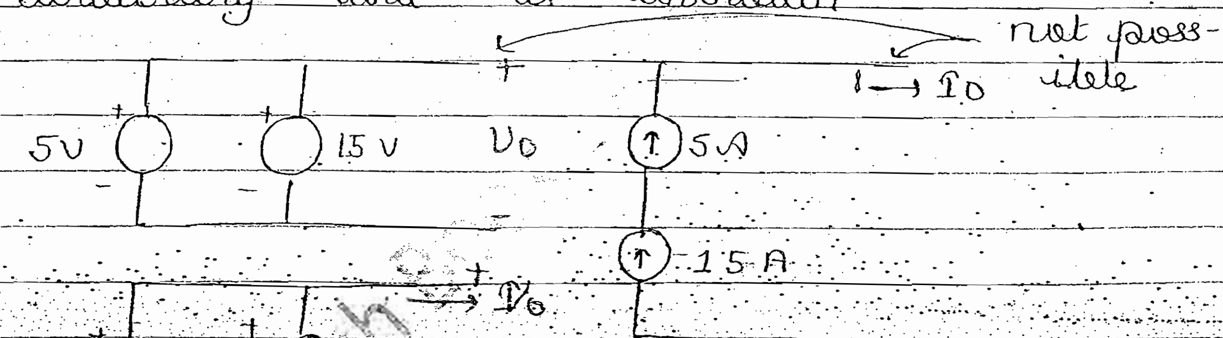


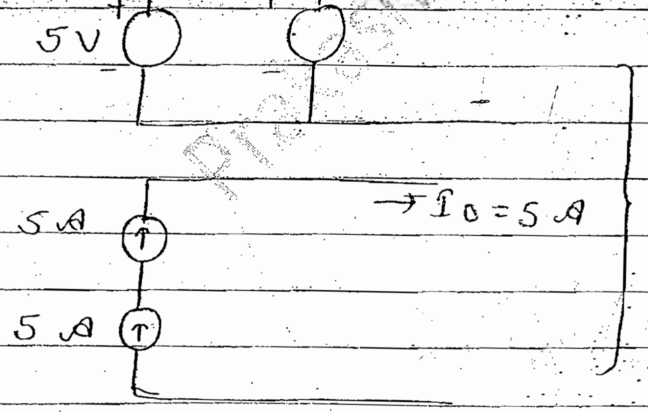


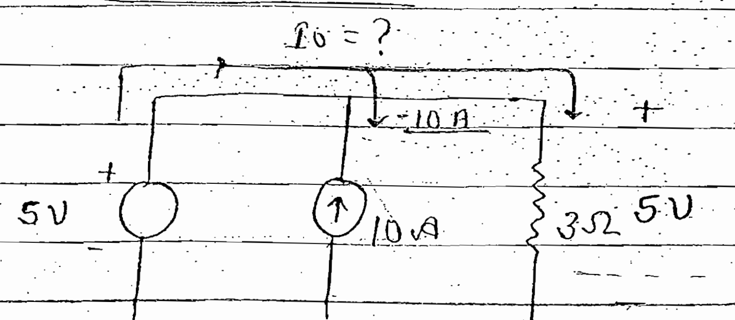
A Series combin of a voltage Source and a current source -------
is equivalent to a Constt current

The current through voltage source is purely arbitrary and depends only upon externally connected current source and externally connected element.

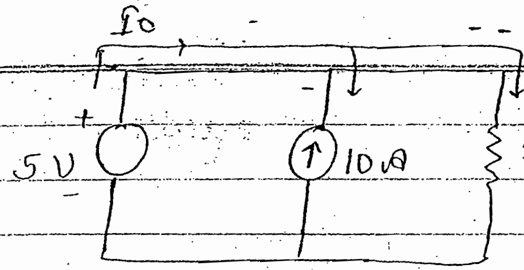
We cannot write a KCL equan at particular node with which a voltage source has been connected since current through it is arbitrary and is unknown







$I_0 = \left(-10 + 5/3\right) A$

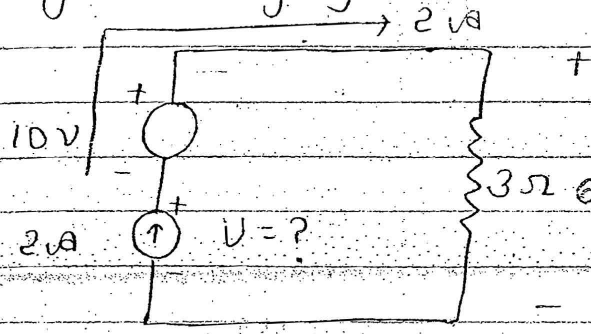


$I_o = (-10+5/7)$ A

$\longrightarrow = 5/7$ amp

by changing R_L I_o can be changed.

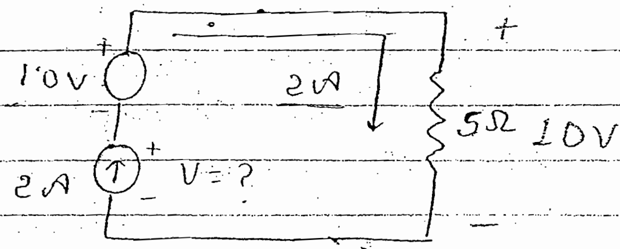
7.



KVL

$-V-10+6=0$

$\boxed{V=-4V}$

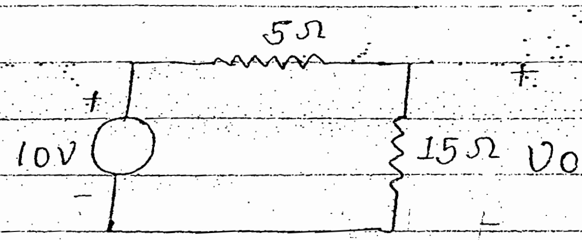


KVL:

$-V-10+10=0$

i.e $\boxed{V=0}$

8.

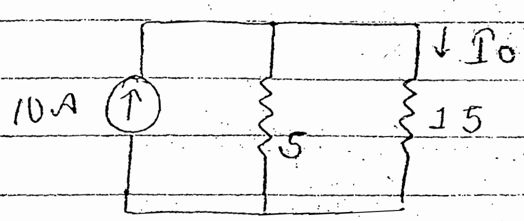


$V = IR$

$\boxed{V \propto R}$

So $V_o = \underbrace{\left(\frac{15}{5+15}\right) \times 10 \text{ ---- } V}_{\text{Voltage dividing Theorem}}$

9.

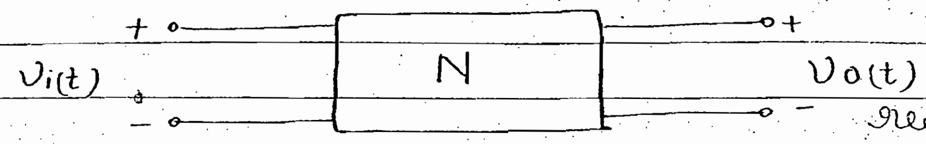


$V = IR$

$\boxed{I = \frac{1}{R}}$

So $I_o = \underbrace{\left(\frac{5}{5+15}\right) \times 10 \text{ ---- } A}_{\text{Current dividing Theorem}}$

Impulse Response of a network



Excitation → $(V_i(S))$, $V_i(\omega)$

$\underbrace{(V_i(S))}$ non sinusoidal

response → $(V_o(S))$, $V_o(\omega)$

$\underbrace{(V_o(S))}$ non sinusoidal

Transfer func^n

$$H(S) = V_o(S)/V_i(S)$$

$$V_o(S) = H(S) \cdot V_i(S)$$

If $v_i(t) = \delta(t)$ --- unit impulse func^n

$V_i(S) = 1$

$\therefore V_o(S) = H(S) \cdot 1$

$\therefore V_o(t) = L^{-1}[H(S)]$

$= h(t)$ --- impulse response of n/w

The Transfer func^n of a network represents the ratio of output voltage to the I/P voltage each in the s domain assuming all the initial condi^{n}s in various element contained in the n/w to be zero.

The Transfer func^n helps to calculate the corresponding response of the n/w for the specified excitation. For a given n/w the T.F. always have a fixed value.

The impulse response of the network represents the O/P voltage in the time domain when the excitation is unit impulse function.

The impulse response of the network is always a fixed value and also represents the inverse laplace transform of the transfer func^n of the n/w.

Ex: If $v_i(t) = e^{-t}u(t)$ then $v_o(t) = e^{-2t}u(t)$ find $v_o(t)$ if $v_i(t) = e^{-3t}u(t)$

Case I

$$v_i(t) = e^{-t}u(t)$$

$$V_i(s) = \frac{1}{s+1}$$

$$v_o(t) = e^{-2t}u(t)$$

$$V_o(s) = 1/s+2$$

$$\frac{V_o(s)}{V_i(s)} = \boxed{H(s) = \frac{s+1}{s+2}} = 1 - \frac{1}{s+2}$$

$$h(t) = \delta(t) - e^{-2t}u(t)$$

Impulse response of n/w

Case IInd

$$v_i(t) = e^{-3t}u(t)$$

$$V_i(s) = \frac{1}{s+3}$$

$$V_o(s) = H(s)\,V_i(s)$$

$$= \frac{s+1}{(s+2)(s+3)} \quad \text{(by partial frac}^n\text{)}$$

$$= \frac{-1}{(s+2)} + \frac{2}{(s+3)}$$

29/10/2014

- Laplace Transform of some Basic function

1. $\delta(t) \longrightarrow 1$
2. $u(t) \longrightarrow 1/s$
3. $r(t) \longrightarrow 1/s^2$
4. $f(t-T) \longrightarrow F(s)\,e^{-sT}$
5. $e^{-at} \longrightarrow \dfrac{1}{s+a}$
6. $\sin bt \longrightarrow \dfrac{b}{s^2+b^2}$
7. $\cos bt \longrightarrow \dfrac{s}{s^2+b^2}$
8. $\dfrac{d f(t)}{dt} \longrightarrow sF(s) - f(0^+)$; $f(0^+) = f(t)\Big|_{t=0^+}$
9. $\int_0^t f(t)\,dt \longrightarrow \dfrac{F(s)}{s} + \dfrac{f^{-1}(0^+)}{s}$; $f^{-1}(0^+) = \left(\int_0^t f(t)\,dt\right)\Big|_{t=0^+}$
10. $e^{-at}\sin bt \longrightarrow \dfrac{b}{(s+a)^2+b^2}$
11. $e^{-at}\cos bt \longrightarrow \dfrac{s+a}{(s+a)^2+b^2}$
12. $t\,e^{-at} \longrightarrow \dfrac{1}{(s+a)^2}$
13. $t^n\,u(t) \longrightarrow \dfrac{|\underline{n}}{s^{n+1}}$

- Analysis of any electrical N/w using KVL & KCL eqn

KVL

Voltage fall ⇒ +ve
Voltage Rise ⇒ −ve

Convention.

No. of KVL equation = $b-(n-1)$
where b = No. of Branches
n = No. of Nodes.

What to measure:-
(1) Branche voltage (2) Branche current
(3) Node voltage (4) loop current.

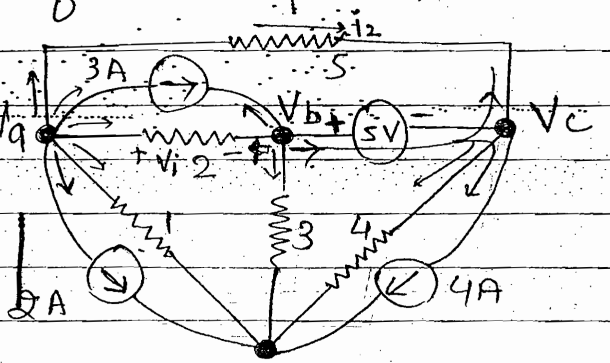
KCL

•—→— ⇒ +ve
•—←— ⇒ −ve

Convention.

No. of KCL equation = $n-1$

Ex:-



To find V_a, V_b, V_c

$V_b - V_c = 5$
$V_c = V_b - 5$

$\underbrace{V_c = V_b - 5}$

dependent or Redundant or Super-Node

- Wherever a voltage source is present b/w two Node one Node become independent and other become dependent. And this dependent get redundant put $V_c = V_b - 5$

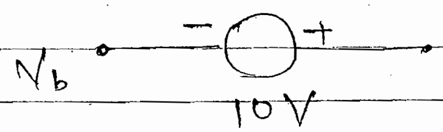
$[V_a] \Rightarrow 2 + \dfrac{V_a - 0}{1} + \dfrac{V_a - V_b}{2} + 3 + \dfrac{V_a - V_c}{5} = 0$ — (i)

$[V_b] \Rightarrow -3 + \dfrac{V_b - V_a}{2} + \dfrac{V_b - 0}{3} + \left[\dfrac{V_c - 0}{4} + 4 + \dfrac{V_c - V_a}{5}\right] = 0$ — (2)

put $V_c = V_b - 5$

find V_a, V_b, $V_c = V_b - 5$.

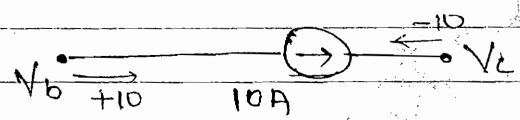
<u>case 1</u> :-



$V_c = V_b + 10$ {bcoz we don't know the value of current b/w V_b & V_c

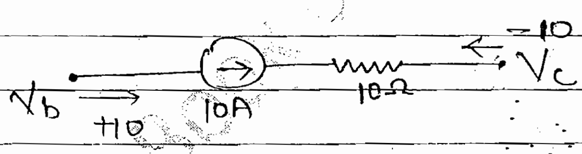
2 KCL eqn :- V_a, V_b {replace all $V_c = V_b + 10$}

<u>Case 2</u> : -



3 KCL eqn :- V_a, V_b, V_c

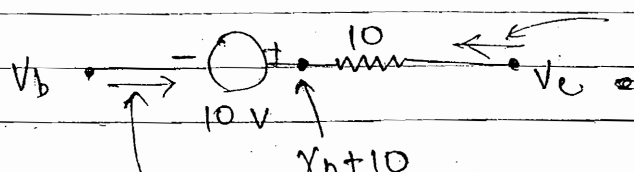
<u>Case 3</u> :



- In this case magnitude of current will remain same.
- V_b and V_c voltage value remain same. No change due to value of series resistance.
- Voltage across the dependent source is arbitrary. and voltage change due to series resistance is adjusted by dep. source. So V_b, V_c remain same

3 KCL eqn V_a, V_b, V_c --- same as case 2

<u>case 4</u> : -

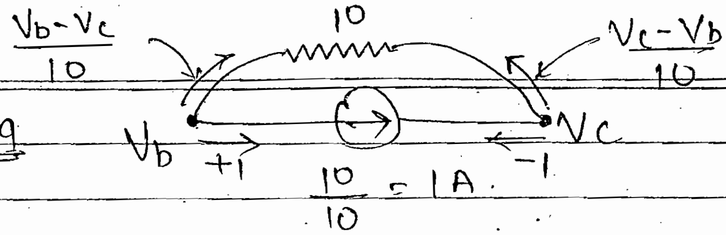


$\dfrac{V_c - (V_b + 10)}{10}$

<u>3 KCL eqn</u> :- V_a, V_b, V_c

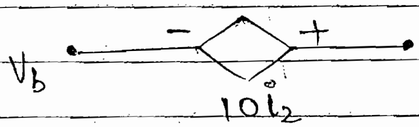
$-\left[\dfrac{V_c - (V_b + 10)}{10}\right]$

Case 4a



3 KCL eqn :- V_a, V_b, V_c

case 5 :-



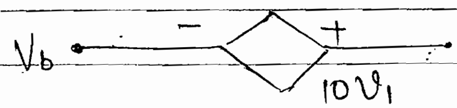
$V_c = V_b + 10i_2$

$= V_b + 10\left[\frac{V_a - V_c}{5}\right]$

$3V_c = V_b + 2V_a$

2 KCL eqn :- V_b, V_a

case 6 :-



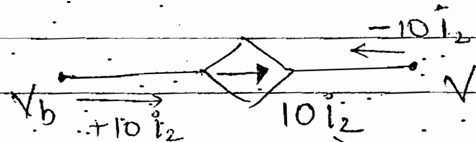
$V_c = V_b + 10V_1$

$= V_b + 10[V_a - V_b]$

$= 10V_a - 9V_b$

2 KCL eqn V_a, V_b

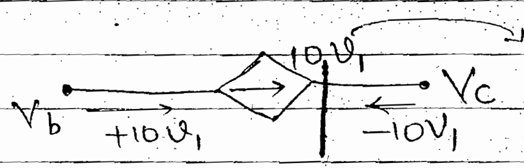
case 7 :-



$10\left[\frac{V_a - V_c}{5}\right] = 2[V_a - V_c]$

3 KCL eqn V_a, V_b, V_c

case 8 :-

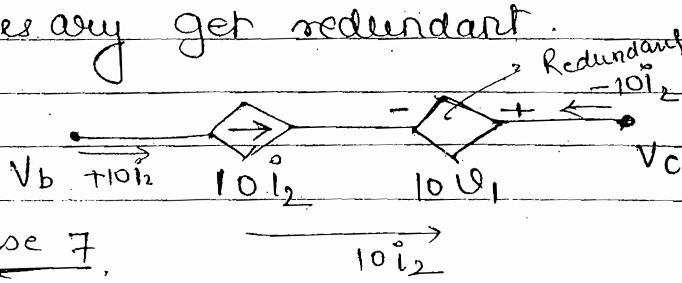


$10[V_a - V_b]$

3 KCL eqn V_a, V_b, V_c

- Whenever any voltage dependent or independent source is present b/w two Nodes in KCL, one node complesary get redundant.

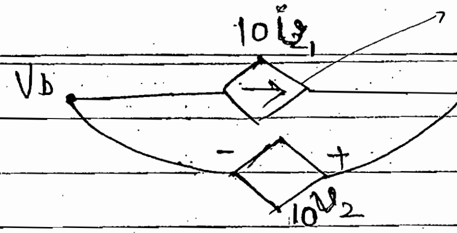
case 9 :-



3 KCL eqn V_a, V_b, V_c

Same as case 7.

Redundant

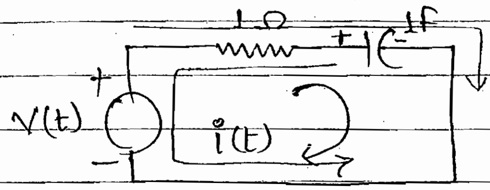


Case 10 — 2 KCL :- V_a, V_b

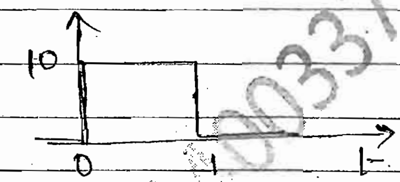
$V_c = V_b + 10 i_2$

Same as case 5.

Ex:- Series RC ckt :-

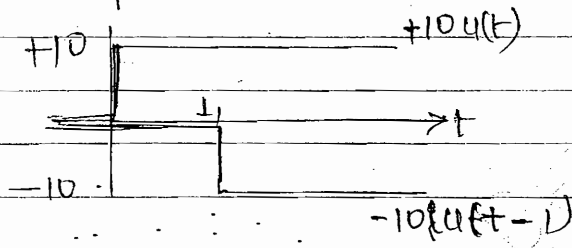


i/p



To find $i(t)$ for $t > 0$

Solution Assume : Zero initial condition



$$v(t) = 10u(t) + (-10u(t-1))$$

$$V(s) = \frac{10}{s} - \frac{10e^{-s}}{s} = \frac{10}{s}[1-e^{-s}]$$

By KVL :- $v(t) = Ri(t) + \frac{1}{C}\int_0^t i\, d(t)$

$$v(t) = i(t) + 1\int_0^t i\, d(t)$$

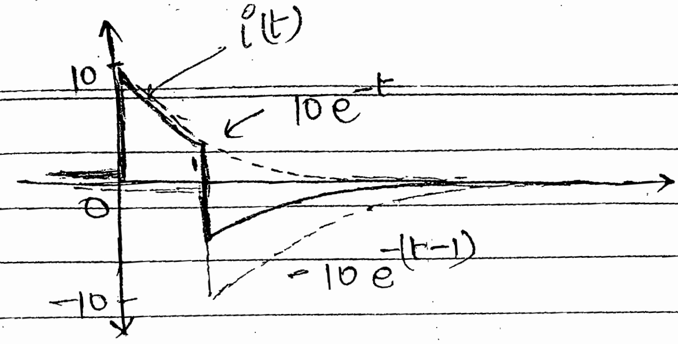
$$V(s) = I(s) + \frac{I(s)}{s} + \frac{\cancel{i^{-1}(0^+)}}{s}$$

$$I(s) = \frac{10(1-e^{-s})\cancel{s}}{(s+1)\cancel{s}} \qquad \left\{ = \left. \int_0^t i(t)\,dt \right|_{t=0^+} = Q(t)\Big|_{t=0^+} = Q(0^+) = 0 \right.$$

when No charge, No voltage

$$I(s) = \frac{10}{(s+1)}(1-e^{-s}) = \frac{10}{s+1} - \frac{10e^{-s}}{s+1}$$

$$i(t) = \underbrace{10e^{-t}u(t)}_{①} - \underbrace{10e^{-(t-1)}u(t-1)}_{②}$$



graphical Representatn.

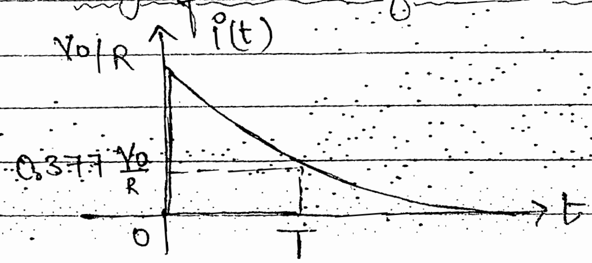
- Reversal of current is possible because capacitor will get discharge after charged.
- Initially capacitor having No charge (No voltage), so capacitor will act as s.c.

If $v(t) = V_0 u(t)$

Then $i(t) = \frac{V_0}{R} e^{-t/T} u(t)$ $\quad T = RC$

Time constant

- Significance of Time Constant :-



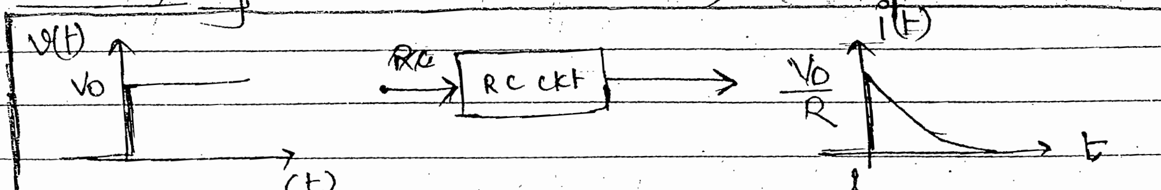
at $t = T$

$i(t) = \frac{V_0}{R} e^{-1}$

$\cong 0.377 \frac{V_0}{R}$

Summary:- (series RC ckt)

Imp



$i(t) = \frac{V_0}{R} e^{-t/T} u(t)$

$T = RC$ Time constant.

Main Points:-

(1) Initially at $t=0$, any capacitor act as a short ckt, so that, the current through RC N/w is Maximum.

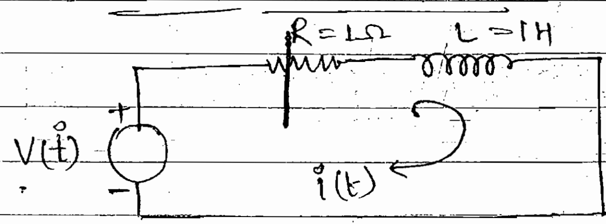
(2) At $t=\infty$, or steady state condition the capacitor acts as open ckt, so that current through RC N/w becomes zero.

(3) For Pulse excitation Reversal of current takes place in the RC N/w

(4) For step excitation the current through RC N/w decreases expoentially with a time constant T. This time constant controls the rate at which the current through RC N/w decreases and dependen only upon numerical values of R and C components value.

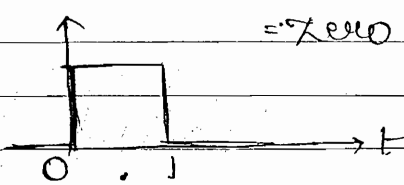
(5) The time constant of the RC N/w Represents the time at which the current through RC N/w is decreased to 37.7% of its initial value.

Ex:- Series RC ckt:



To find:- $i(t)$ for $t>0$

assume:- Initial condition = zero



Solution

KVL:-

$$\Rightarrow V(t) = R\, i(t) + L \frac{di(t)}{dt}$$

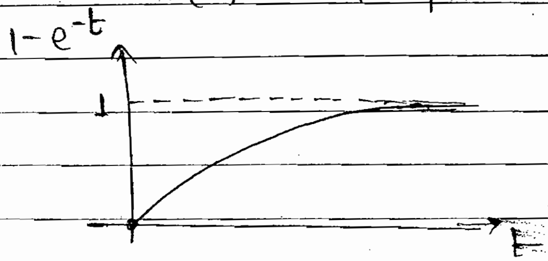
$$\Rightarrow V(s) = I(s) + s\,I(s) - \cancel{i(0^+)} = \frac{10}{s}\left(1-e^{-s}\right)$$

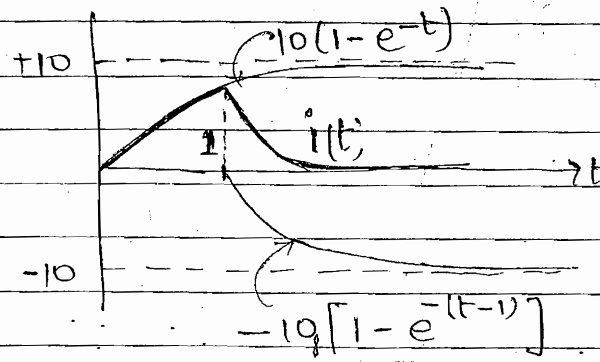
$$I(s) = \frac{10}{s(s+1)}(1-e^{-s})$$

$$= 10\left[\frac{1}{s}-\frac{1}{s+1}\right](1-e^{-s})$$

$$= 10\left[\frac{1}{s}-\frac{1}{s+1}\right] - 10\left[\frac{1}{s}-\frac{1}{s+1}\right]e^{-s}$$

$$i(t) = 10\left[1-e^{-t}\right]u(t) - 10\left[1-e^{-(t-1)}\right]u(t-1)$$





$[1-e^{-t}]$ ⇒ at $t=0$, $1-1=0$

⇒ at $t=\infty$, 1

→ RC N/W ⇒ Reversal of Current

→ RL N/W ⇒ No. Reversal of Current.

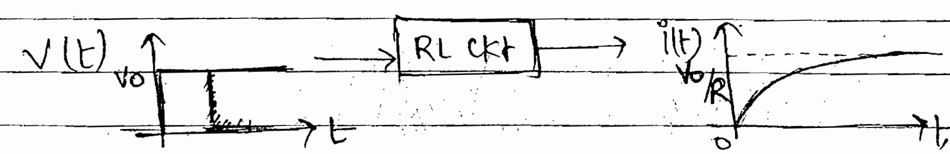
If $v(t) = V_0 . u(t)$

then $i(t) = \frac{V_0}{R}(1-e^{-t/T})u(t)$ $\qquad T = \frac{L}{R}$

Time constant.

Summary:- (series RL N/w)

Imp



$$i(t) = \frac{V_0}{R}(1-e^{-t/T})\,u(t)$$

$T = L/R$ Time constant

Main Points:

(1) Initially at $t=0$, the inductor act as open ckt so that the current through series RL N/w is zero

(2) For step excitation the current through the inductor increases exponentially with some time constant T. This time constant T depends only upon the Numerical values of R and L components and controls the rate at which current through N/w is previous exponentially.

(3) For pulse excitation in reversal of current takes place in a series RC N/w whereas No such reversal takes place in a series RL N/w.

- Initial & Final Values of A function:–

$$f(t) \begin{cases} v(t) \\ i(t) \end{cases}$$

Initial Value:

$f(0^+) = f(t)\Big|_{t=0^+}$ —— Initial condition.

$$\boxed{f(0^+) = \lim_{s \to \infty} s\, F(s)}$$ Initial value Theorem.

Final Value:– —— steady state (s.s) value.

$f(\infty) = f(t)\Big|_{t=\infty}$

$$\boxed{f(\infty) = \lim_{s \to 0} s\, F(s)}$$ Final value Theorem.

ex:– $F(s) = \dfrac{s+1}{(s+2)(s+3)}$ To find $f(0^+)$, $f(\infty)$

Solution:- $f(t) = (-e^{-2t} + 2e^{-3t})u(t)$

$f(0^+) = +1$; $f(\infty) = 0$

$s+1 = A(s+3) + B(s+2)$

put $s = -3$ $B = 2$

put $s = -2$ $B = -1$

- Cannot apply for sinusodial I/p. (oscillatory).

- Transient Response of An electrical Network.

Transient Response

$V(t)$ $i(t)$

$0 < t < \infty$

$t = 0^+$: $V(0^+)$, $i(0^+)$ — Initial condition

$t = \infty$: $V(\infty)$, $i(\infty)$ — Final conditional or steady state values.

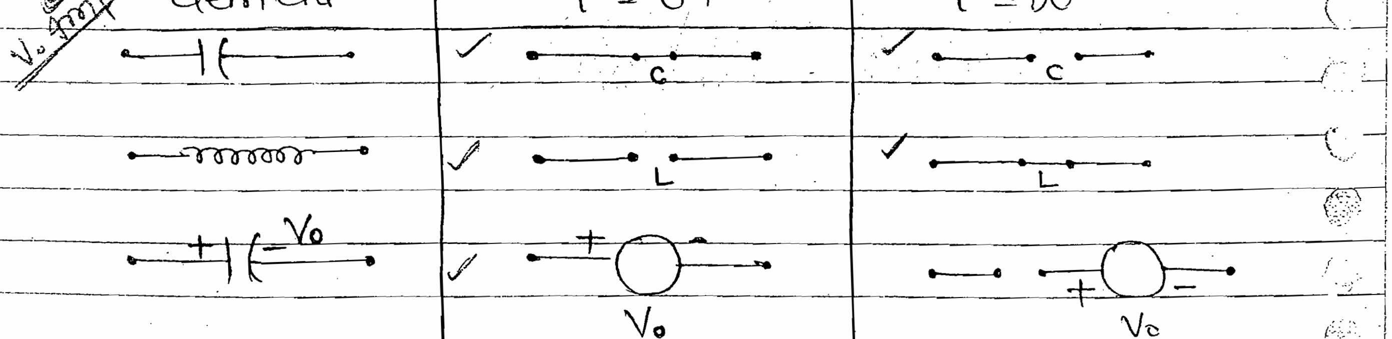
$\Rightarrow$ R (Resistor) donot store the energy.

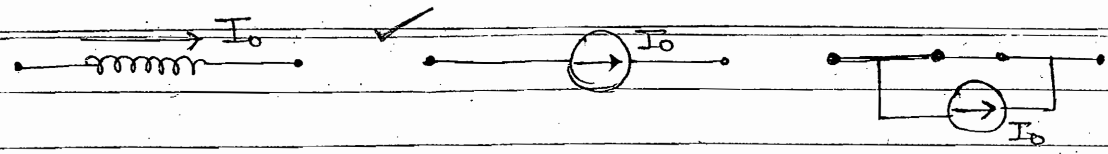
Behaviour of L & C

Case 1 In time Domain

V. Imp

Element	$t = 0^+$	$t = \infty$





(1) Initially at $t=0$ any capacitor act as a short ckt where as any inductor act as open ckt.
(2) At $t=\infty$ or in steady state condition, any capacitor acts as an open ckt where as any inductor acts as short ckt.
(3) Initially at $t=0$ any capacitor with initial voltage act as a constant voltage source whereas any inductor with initial current act as a constant current source.

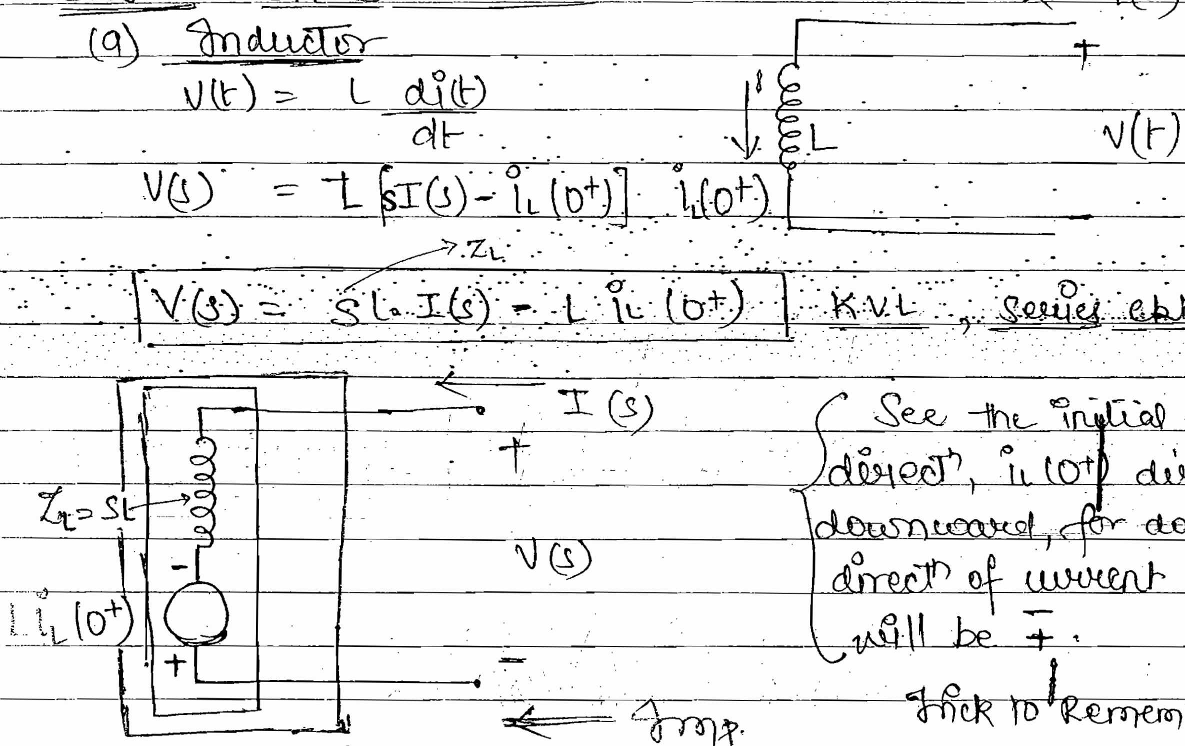
Case 2:- In S-domain

(a) Inductor

$$v(t) = L\frac{di(t)}{dt}$$

$$V(s) = L[sI(s) - i_L(0^+)] \quad i_L(0^+)$$

$$\boxed{V(s) = \underbrace{sL}_{Z_L} \cdot I(s) - L\, i_L(0^+)}$$ K.V.L., Series ckt



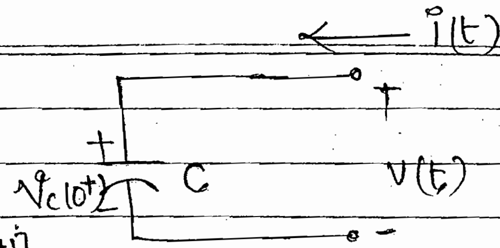
Thevenin's Model

See the initial condit^n direct^n, $i_L(0^+)$ direct^n is downward, for downward direct^n of current polarity will be $\mp$.

Trick to Remember.

(b) **Capacitor :-**

$$i(t) = C\frac{dv(t)}{dt}$$



$$I(s) = C\left[sV(s) - V_c(0^+)\right]$$

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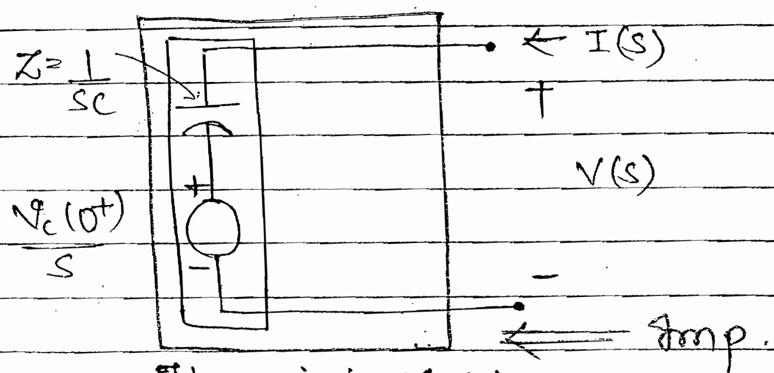
$$\boxed{V(s) = \frac{1}{Cs}I(s) + \frac{V_c(0^+)}{s}}$$ --- KVL Series ckt.

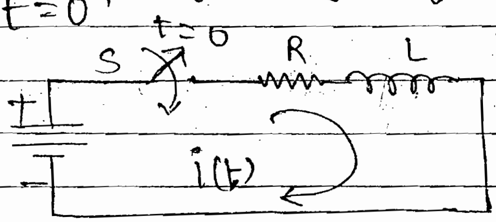


Thevenin's Model

{ according to polarity of $V_c(0^+)$ in initial Model, same to thevenin's Model, if $V_c(0^+)$ $\frac{-}{+}$
{ In Model also $\frac{V_c(0^+)}{s}$ $\frac{-}{+}$

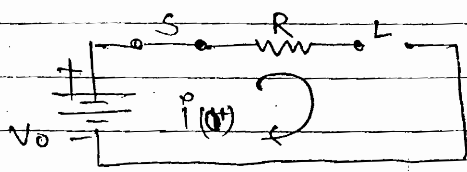
Ex:- In the circuit shown the switch is initially open. The switch is closed at $t=0$. Find the current and the rate of change of current through the N/w at $t=0^+$

Solution



To find $i(0^+)$
$\frac{di}{dt}(0^+)$

- $\underline{\text{at } t = 0^-}$
  S = open $\quad \boxed{i(0^-) = 0}$
- $\underline{\text{at } t = 0^+}$
  S = close $\quad L \Rightarrow O.C$

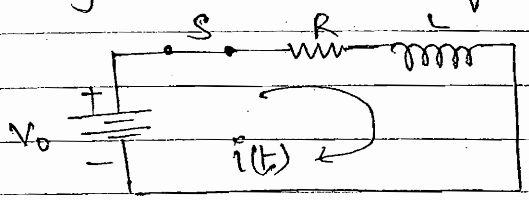


$\boxed{i(0^+) = 0}$ --- Amp.

Imppt [The current through the inductor and voltage through very Imp. $\boxed{\begin{matrix} i_L(0^-) = i_L(0^+) \\ V_c(0^-) = V_c(0^+) \end{matrix}}$

the capacitor don't change instantaneously. Therefore these values remain same just before and just after the switch has been closed.]

- General circuit for $t > 0$ [b/w 0^+ and ∞) neither behave O.C ~~and~~ nor S.C]



$$V_0 = R\,i(t) + L\frac{di(t)}{dt}$$

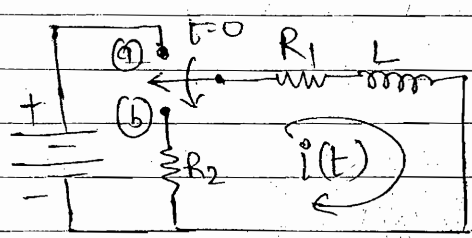
at $t = 0^+$

$$R\underbrace{i(0^+)}_{\equiv 0} + L\,di(0^+)/dt = V_0$$

$$\frac{di(0^+)}{dt} = +\frac{V_0}{L} \;\ldots\ldots\; \text{Amp/sec}$$

Ex:- In the circuit shown the switch is initially at position a for a long time till the steady state reach. The switch is moved to position b. Find an general expression of a current through the N/w for $t > 0$.

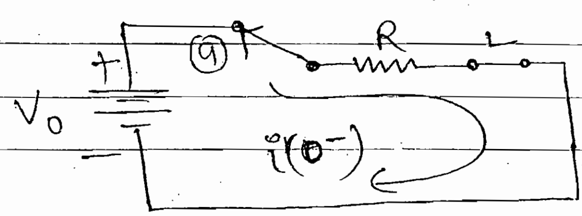
Solution: whenever long time till the steady state reach termology given consider final values



To find $i(t)$ for $t > 0$

- $t = 0^-$: switch at (a) position. [steady state condit^n/ final values]

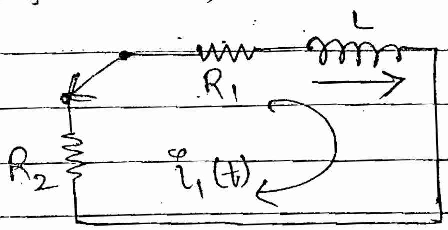
$L \Rightarrow$ S.C



$$i(0^-) = \frac{V_0}{R_1}$$

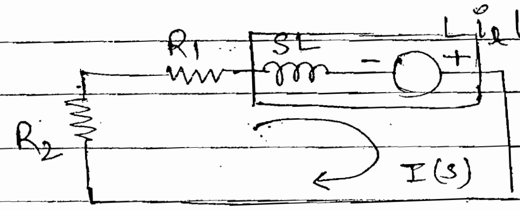
$$\boxed{i(0^-) = \frac{V_0}{R_1} = i(0^+)}$$ $\therefore$ as $i(0^-) = i(0^+)$

• $t>0$; switch at (b) position.



$i(0^+) = i_L(0^+)$
$= \frac{V_0}{R_1}$

$i_L(0^+) = V_0/R_1$



$L i_L(0^+) = \frac{L V_0}{R_1}$ $\quad \therefore L i_L(0^+) = \frac{L V_0}{R_1}$

$\Rightarrow I(s)(R_2 + R_1 + sL) - V_0 L/R_1 = 0$

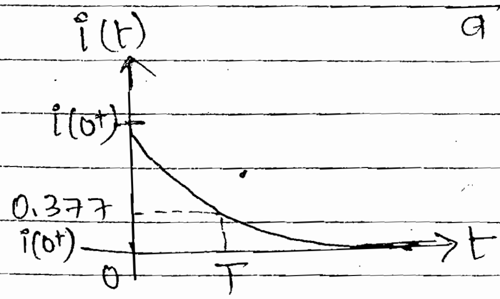
$\Rightarrow I(s) = \frac{V_0 L_1}{R} \cdot \frac{1}{L(s+a)}$; $a = \frac{R_1+R_2}{L}$

$I(s) = \frac{V_0}{R} \frac{1}{(s+a)}$

$i(t) = \frac{V_0}{R} e^{-at} \cdot u(t)$

$$\boxed{i(t) = i(0^+) e^{-at/T} u(t)}$$ $i(0^+) = \frac{V_0}{R}$

where $T = \frac{1}{a} = \frac{L}{R_1+R_2} = \frac{L}{R_{eq}}$.



at $t = T$
$i(t) = i(0^+) e^{-1} \approx 0.377\, i(0^+)$

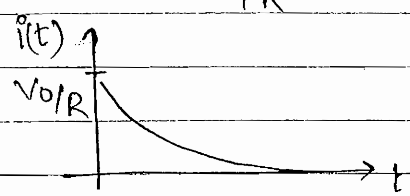
Imp pt • In an RL N/W the time constant represents the time at which the current through the N/W is decrease to 37.7% of its initial value.

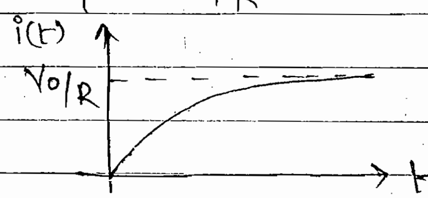
- This time constant controls the rate of exponential decay of the current. and depends purely upon the numerical values of R and L components.

- Summary :- --- series RL circuit

$i(t) = \frac{V_0}{R} e^{-t/T} u(t)$ | $i(t) = \frac{V_0}{R} (1 - e^{-t/T}) u(t)$

$T = L/R$ | $T = L/R$





- valid when IC is finite | - valid when I.C. $\equiv 0$.

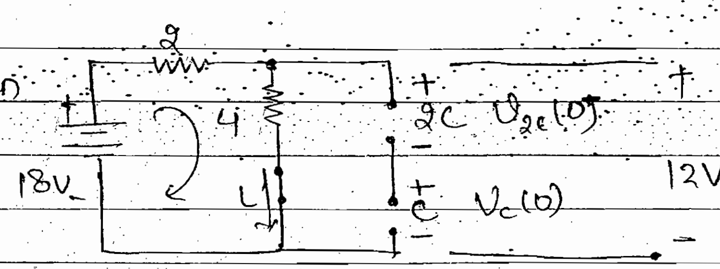
Chap2 (7)

at $t = 0^-$ Switch → open

S.S condition / Final value

L ⇒ S.C.

C ⇒ O.C.



$i(0^-) = i(0^+) = \frac{18}{6} = 3$ Amp.

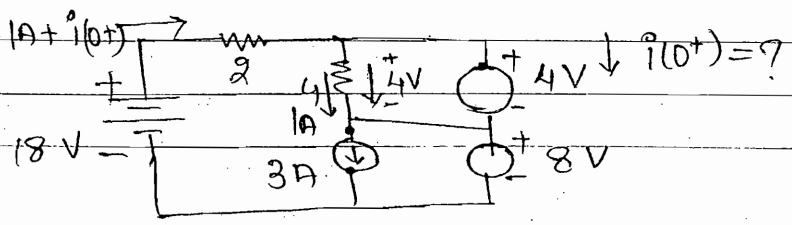
$V_{2c}(0^-) = V_{2c}(0^+) = 4$

$V_c(0^-) = V_c(0^+) = 8$

$Q = CV$ (const.)

$\boxed{V \propto \frac{1}{C}}$

at $t = 0^+$ switch → close



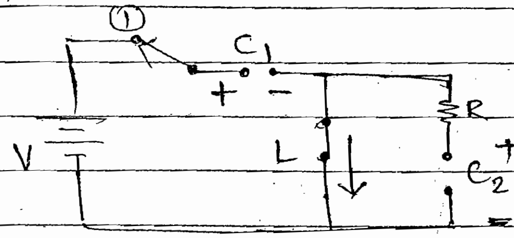
• KVL :- $-18 + 2[1 + i(0^+)] + 4 + 8 = 0$

$i(0^+) = 2$ Amp. Ans.

Q.22 at $t = 0^-$

Switch at (1). (S.S → Final Value)

$L \Rightarrow$ S.C $C \Rightarrow$ O.C



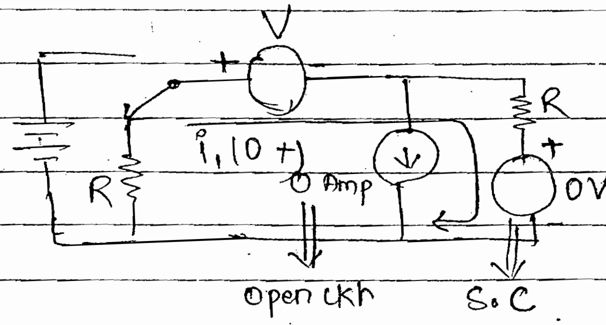
$i(0^-) = i(0^+) = 0$

$V_{C_1}(0^-) = V_{C_2}(0^+) = V$

$V_{C_2}(0^-) = V_{C_2}(0^+) = 0$

$$\boxed{i_{C_1}(0^-) \neq i_{C_1}(0^+) = 0}$$

at $t = 0^+$ switch at (2)



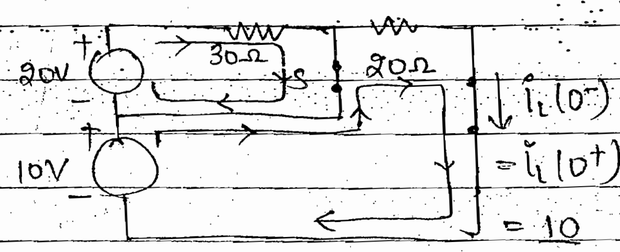
KVL eqn

$$i_1(0^+) = -\frac{V}{2R}$$

Q.25 at $t = 0^-$

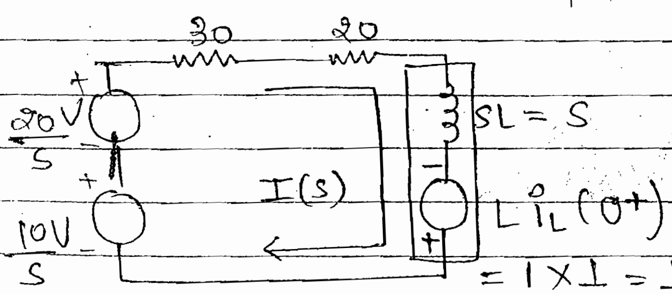
switch is closed S.S / F.C.

$L \Rightarrow$ S.C



$$i_L(0^-) = i_L(0^+) = \frac{10}{20} = \frac{1}{2} \text{ Amp.}$$

at $t > 0$ switch is open



$= 1 \times \frac{1}{2} = \frac{1}{2}$

$\Rightarrow \frac{20}{S} + \frac{10}{S} + \frac{1}{2} = I(S)[50 + S]$

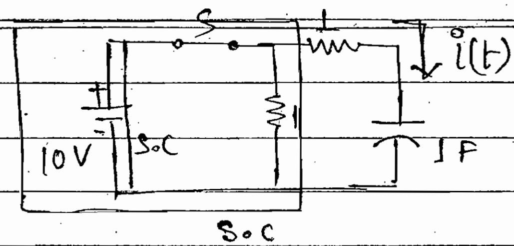
$\Rightarrow \frac{40 + 20 + S}{2S} = I(S)[50 + S]$

$\Rightarrow (60 + S)/2S = I(S)[50 + S]$

$$\Rightarrow I(S) = \frac{60 + S}{2S[50 + S]}$$

$$i(t) = 0.6 - 0.1\, e^{-50t} \text{ Ans}$$

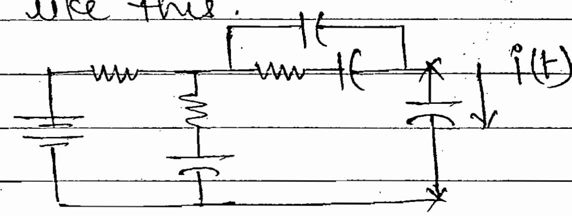
Q.19 $t>0$ switch is closed



$T = RC = 1 \times 1$

$= 1$ sec Ans.

This method of $T=RC$ only applicable for this ques. If given like this.



Not applicable above Method.

For such N/w. how to find Time constant:-

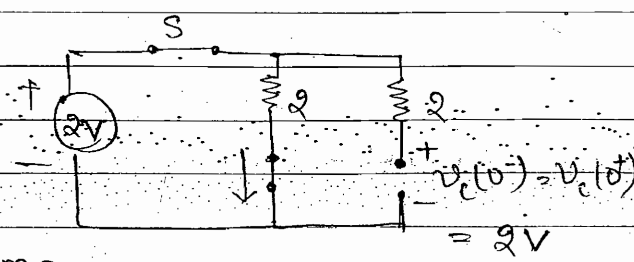
$T = R_{eq} \cdot C_{eq}$ ------ RC ckt

$= \frac{L_{eq}}{R_{eq}}$ ---------- RL ckt

Q.4 At $t=0^-$ switch-close

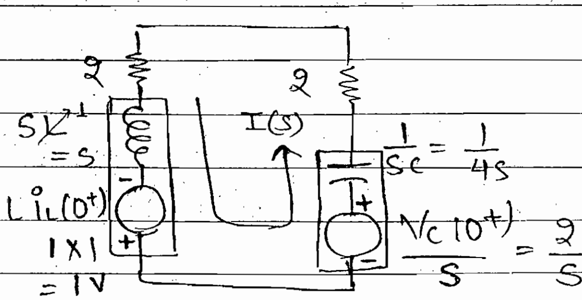
S.S / F.C

$C \Rightarrow$ O.C ; $L \Rightarrow$ S.C.



$i(0^-) = i(0^+) = 2/2 = 1$ Amp.

At $t>0$ switch-open



✓ KVL eqn in loop

$\Rightarrow 1 + \frac{2}{S} = I(S)\left[2 + S + \frac{1}{4S} + 2\right]$

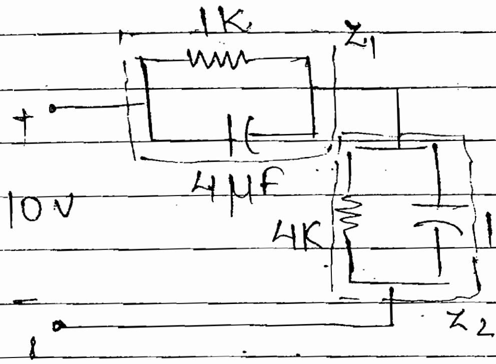
$I(S) = \dfrac{\frac{S+2}{\cancel{S}}}{\frac{8S + 4S^2 + 1 + 8S}{4\cancel{S}}}$

$= \frac{(S+2)4}{4S^2 + 16S + 1} = \frac{\cancel{(S+2)}\cancel{4}}{\cancel{4}(S+2)^{\cancel{2}}} = \frac{1}{S+2}$ & $i(t) = e^{-2t}$

$\Rightarrow i(t) = e^{-2t} = e^{-t/T} \quad \therefore T = \frac{1}{2}$ sec Ans

Q.34 Voltage division method is not applicable Becoz product of RC in Both Z_1 and Z_2 is equal. $V_c(s) = \frac{Z_2 V_s}{Z_1 + Z_2}$ X

If Z_1 and Z_2 having different values then only Voltage division method is applicable



given $V_c(0^+) = 0$

- Only for RC ckt

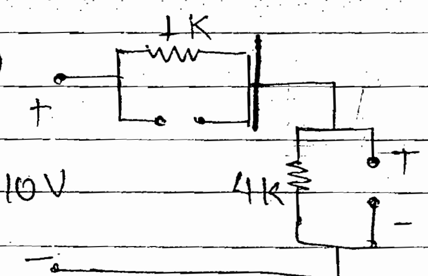
$$V_c(t) = V_c(\infty) - [V_c(\infty) - V_c(0^+)]e^{-t/T}$$

- Only for RL ckt

$$i_L(t) = i_L(\infty) - [i_L(\infty) - i_L(0^+)]e^{-t/T}$$

Imp

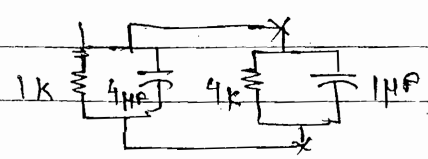
$V_c(\infty)$



$V_c(\infty) = 10 \times \frac{4}{4+1} = 8V$

$$V_c(t) = \underbrace{V_c(\infty)}_{=8} - [V_c(\infty) - \underbrace{V_c(0^+)}_{=0}]e^{-t/T}$$

$T = R_{eq} C_{eq} \Big|_{10V} \Rightarrow sc$



$R_{eq} = 1 \| 4 = 0.8K$

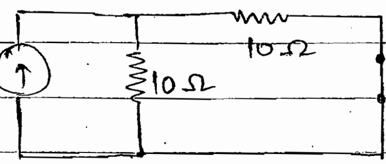
$C_{eq} = 1 + 4 = 5$ μF

$$T = R_{eq} C_{eq} = 0.8 \times 5 = 0.004 \text{ sec.}$$

$$V_C(t) = 8 - [8 - 0]e^{-t/0.004} = 8[1 - e^{-t/0.004}] \text{ Ans}$$

Q.45 at $t = 0^-$, $S \to$ open

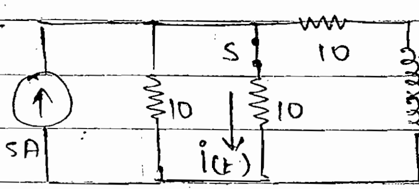
S.S / F.C $L \Rightarrow$ S.C.



$\downarrow i(0^-) = i(0^+)$
$= \frac{1.5 \times 10}{20}$
$= 0.75A$

at $t > 0$ $S \to$ close.

$$i_L(t) = \underbrace{i_L(\infty)}_{0.5A} - [i_L(\infty) - \underbrace{i_L(0^+)}_{0.75A}]e^{-t/T}$$



$\downarrow i_L(0^+) = 0.75A$
$L = 15mH$

$$= 0.5 - [0.5 - 0.75]e^{-t/T}$$

at $t = \infty$ $L \Rightarrow$ S.C, R comes in ||

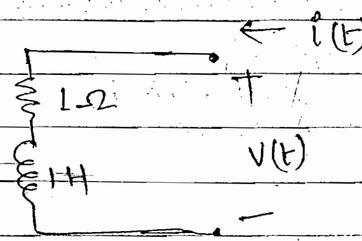
so $\frac{1.5}{3} = 0.5A$.

$$T = \frac{L^{\,15mH}}{R_{eq}}\Big|_{1.5A = O.C} = \frac{15mA}{15\,\Omega} = 1 \text{ msec}$$

$\therefore R_{eq} = (10 \| 10) + 10 = 15\Omega$

$$i_L(t) = \frac{V(t)}{10} = \frac{1}{10}\left[10\, i_L(t) + L\frac{di_L(t)}{dt}\right]$$ Ans (a)

Q. 23.



To find $i(t)$

When the supply is sinusodial
Never apply laplace

$$V(t) = 10\sqrt{2} \cos(\underset{\omega=1}{t} + 10^\circ) + 10\sqrt{5} \cdot \cos(\underset{\omega=2}{2t} + 10^\circ)$$

$$Z = R + j\omega L = 1 + j\omega = 1 + 1j = \sqrt{2}\ \angle 45^\circ \quad \ldots \omega = 1$$
$$= 1 + 2j = \sqrt{5}\ \angle\theta \quad \ldots \omega = 2$$

$$\theta = \tan^{-1}\left(\frac{2}{1}\right) = 63.4$$

$$i(t) = \frac{10\cancel{\sqrt{2}}}{\cancel{\sqrt{2}}\angle 45^\circ}\cos(t+10^\circ) + \frac{10\cancel{\sqrt{5}}}{\cancel{\sqrt{5}}\angle\theta}\cos(2t+10^\circ)$$

$$= 10\cos(t+10^\circ-45^\circ) + 10\cos(2t+10^\circ-\theta)$$

Q.15 $H(s) = \frac{1}{s+1}$ $\quad V_i(t) = \cos t \equiv V_i \cos\omega t$ $\quad V_o(t) = ?$

Solutn $V_o(t) = V_o \cos\omega t = V_o \cos t$

$s = j\omega \rightarrow 1$, $s = j$

$$H(s) = \frac{1}{s+1} = \frac{1}{j\omega+1} = \frac{1}{j+1} = \frac{1}{\sqrt{2}\angle 45^\circ}$$

$$= \frac{V_o}{V_i}, \quad V_i = 1$$

$$V_o(t) = \frac{1}{\sqrt{2}\angle 45^\circ}\cos t = \frac{1}{\sqrt{2}}\cos(t-45^\circ) \text{ Ans.}$$

Q.21 $v(t) = 10(t-0.01)e^{-100t}$

$$V(s) = \frac{10}{(s+100)^2} - \frac{0.1}{(s+100)} = \frac{10 + 0.1(s+100)}{(s+100)^2}$$

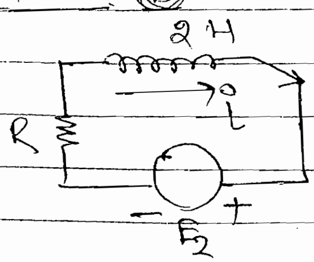
Denominator $= (s+100)^2$

$= s^2 + (100)^2 + 200s \quad | = s^2 + 2\delta\omega_n s + \omega_n^2$, $\omega_n^2 = 100$

$\delta = 1$, critically damped $\quad 2\delta\omega_n = 200$

$$\delta = \frac{200}{2\times 100} = 1$$

Q.20 (B)



$$i(t) = i(\infty) - [i(\infty) - i(0^-)]e^{-t/T}$$

$$= 4 - [4-(-8)]e^{-t/T}$$

$$= 4 - 12e^{-t/T}$$

$$\frac{di(t)}{dt} = +\frac{1}{T}\cdot 12 e^{-t/T}$$

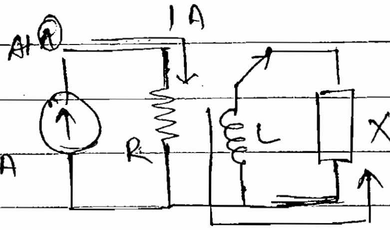
$$\frac{di(t)}{dt} = \frac{-12e^{-t/T}}{T}$$

$=$

$$\frac{di(0)}{dt} = 3 = \frac{1}{T} \cdot 12 \cdot 1 \qquad \boxed{T = 4}$$

$$T = \frac{L}{R} \; ; \; R = \frac{2}{4} = \frac{1}{2}\,\Omega \quad \text{Ans}$$

Q.13 At (A)

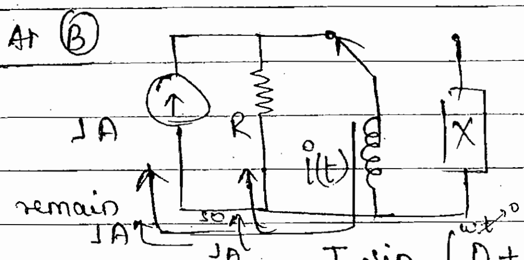


$i(t) = I \sin(\omega t + 30^\circ)$

leading

so X = capacitor.

At (B)



$I \sin(\overset{\omega t}{0} + 30^\circ)$

$= I/2$

$P = I^2 R$

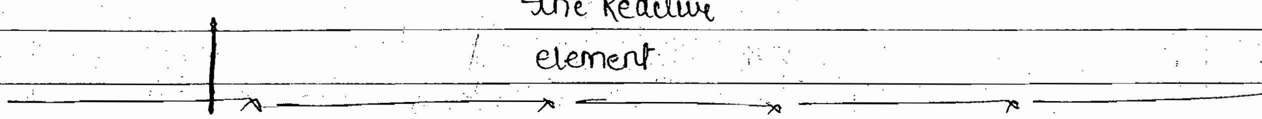
const. same   same

$\frac{1}{2} I = 1A + 1A$

$I = 4A$ Ans

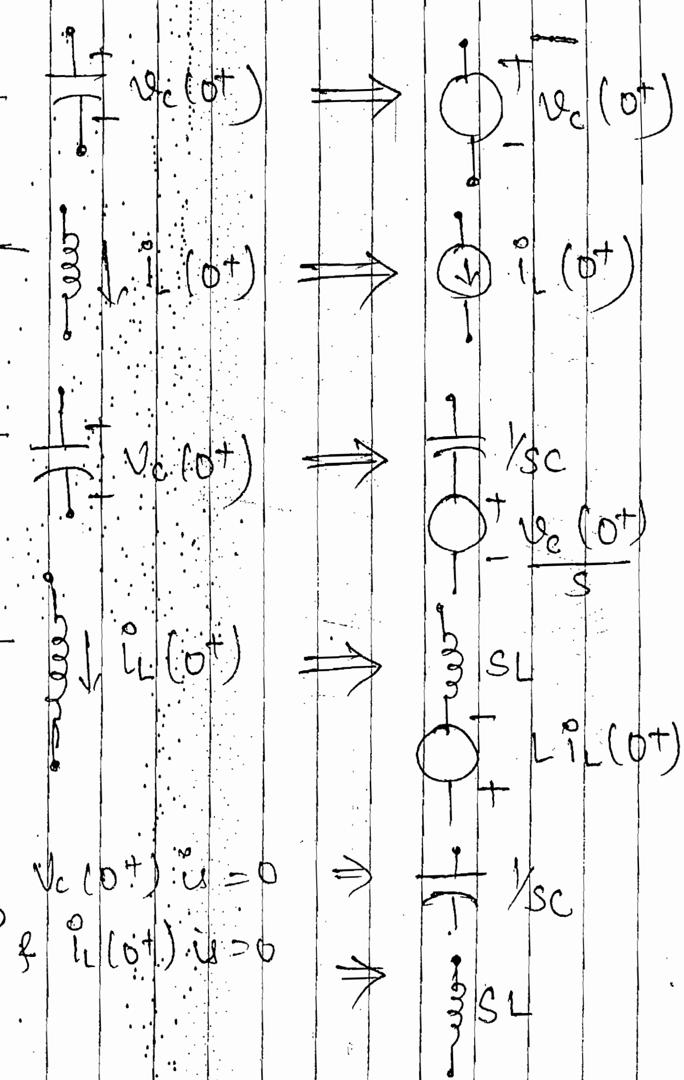
• Summary (Transient Response)   1st NOV 2014

Transient Response $\left[\begin{array}{l} v_c(0^+) \\ i_L(0^+) \end{array}\right\}$ Find I.C in all the Reactive element   (Next page)



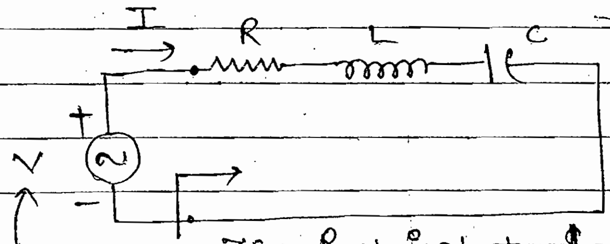
Summary (Transient Response)

- Transient Response
  - $v_c(0^+)$
  - $i_L(0^+)$
    - $v(0^+)$; $i(0^+)$ (NOT use laplace)
    - $v(t)$; $i(t)$ for $t>0$ (for general eqn use laplace)



Series RLC ckt

--- (series Resonance)



$Z_{in} = R + j\omega L + \frac{1}{j\omega C} = R + j\left(\omega L - \frac{1}{\omega C}\right)$

Sinusodial voltage source (fixed amplitude & variable freq.)

difference
- In previous topic we study the behaviour of element when t is varying
- In this topic we study the behaviour of element when freq. is varying

	L	C	
t = t	L	C	--- Transient response
$t = 0^+$	O.C	S.C	--- fixed freq. & variable time
$t = \infty$	S.C	O.C	--- behaviour of ckt
$\omega = \omega$	$j\omega L$	$\frac{1}{j\omega C}$	--- Resonance
$\omega = 0^+$	S.C	O.C	--- fixed time instant & variable freq.
$\omega = \infty$	O.C	S.C	behaviour of ckt

- at $\omega = \omega_0$; j term = 0

(*) $Z_{in} = R + j(0)$ --- minimum

$= Z_0$

$I_0 = \frac{V}{Z_{in}}\Big|_{\omega=\omega_0} = \frac{V}{Z_0} = \frac{V}{R}$ --- maximum (I_0)

at $\omega = \omega_0$

$|V_L| = |V_C|$ & are out of phase by 180°.

at $\omega = \omega_0$

$$\omega_0 L - \frac{1}{\omega_0 C} = 0$$

$$\boxed{\begin{aligned} \omega_0 &= \frac{1}{\sqrt{LC}} \\ f_0 &= \frac{1}{2\pi\sqrt{LC}} \end{aligned}}$$ ---rad/sec ---Hz

$$Z_{in} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

- $\omega = \omega_0$; Resistive
- $\omega > \omega_0$; Inductive (for this, voltage across inductor is max)
- $\omega < \omega_0$; capacitive (for this, voltage across capacitor is Max)

frequency Response :- $|I|$ Vs ω

$$|I| = \frac{V}{|Z_{in}|} = \frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$\Rightarrow$ ckt having RLC in series act as Band Pass filter.

$$I_0 = \frac{V}{R} \text{ ---- max}$$

Imp Features :-

(1) At resonance the j-term of I/p impedance of the circuit is zero.

(2) At ω_0 the N/w is resistive, I/p voltage and I/p current are in phase so that the power factor is unity

(3) At resonance I/p impedance is minimum, so that current drawn by the N/w from the I/p voltage source is maximum and therefore any series RLC N/w is behaves as Band pass filter

(4) At resonance voltage across L & C are in magnitude But out of phase by 180°, so that the net voltage across LC combination is zero

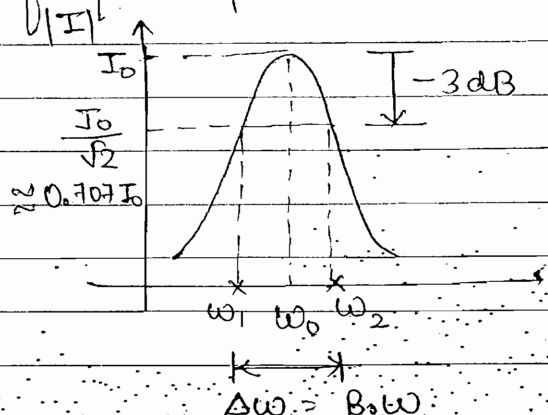
(5) The ckt behaves differently at different frequencies of operation.

for voltage and current gain ⟶ 20log.
for power gain ⟶ 10log

(a) at ω_0 N/w is Resistive and has p.f = unity
(b) above ω_0 N/w is inductive and has p.f = lagging
(c) Below ω_0 N/w is capacitive and has p.f = leading

(d) The frequency of Resonance Represent the rate at which electrical energy stored in the capacitor is transformed to The magnetic energy in the inductor and vice-versa. Transformation of these energy at a particular frequency represent phenomenon of of resonance

- freq. Response



ω_1, ω_2 — cut off frequency
... half pow freq
... 3 dB freq.

$Q_0 \rightarrow$ also voltage amplificatn factor.

at $\omega_2 \rightarrow$ ckt behaviour is inductive
$\omega_1 \rightarrow$ — — capacitive

- Quality Factor

$$Q_0 = \frac{|V_L|}{V} = \frac{|V_C|}{V} \quad ; \ (\gg 1)$$

$V_L = Q_0 V$; $V_C = Q_0 V$; V_L & V_C always greater than V

$$Q_0 = \frac{\omega_0}{\Delta\omega} = \frac{\omega_0}{\omega_2 - \omega_1}$$

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$

$$\Delta\omega = \omega_2 - \omega_1$$

$$\begin{cases} \omega_2 \cong \omega_0 + \frac{1}{2}\Delta\omega \\ \omega_1 \cong \omega_0 - \frac{1}{2}\Delta\omega \end{cases}$$

- **Quality Factor :-**

$$Q_0 = \frac{|V_L|}{V} = \frac{|V_C|}{V}$$

∴ $|V_L| = |I_0 j\omega_0 L| = I_0 \omega_0 L$; $|V_C| = \left| I_0 \cdot \frac{1}{j\omega C} \right| = \frac{I_0}{\omega_0 C}$

∴ $V = I_0 R$

$Q_0 = \frac{I_0 \omega_0 L}{I_0 R}$; $Q_0 = \frac{I_0}{I_0 R \cdot \omega_0 C}$

$$\boxed{Q_0 = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC} = \frac{1}{R}\sqrt{\frac{L}{C}}}$$

$\omega_0 = \frac{1}{\sqrt{LC}}$

- **Bandwidth :- ($\Delta\omega$)**

$\Delta\omega = \frac{\omega_0}{Q_0} = \frac{1/\sqrt{LC}}{\frac{1}{R}\sqrt{\frac{L}{C}}}$; $Q_0 = \frac{\omega_0}{\Delta\omega}$

$$\boxed{\Delta\omega = \frac{R}{L} \text{ rad/sec} \; ; \; \Delta f = \frac{R}{2\pi L} \text{ Hz}}$$

$Q_0 \uparrow \rightarrow \Delta\omega \downarrow$ — $R \downarrow$; $L \uparrow$ .. non applicable

∴ only $L\uparrow$ applicable when req. at high cost and large size

∴ So we generally kept low value of R.

- Imp. Points :-

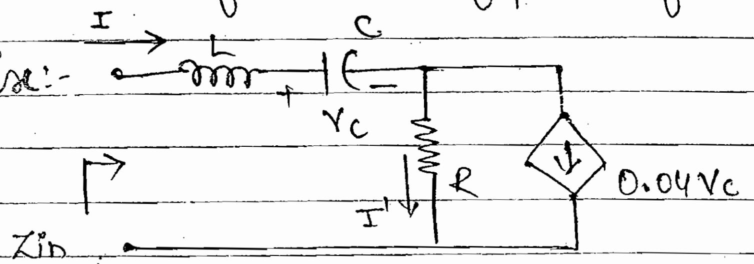
(1) Quality factor represent the ability to select a particular desire frequency, thereby rejecting all other unwanted freq. components

(2) For any tunned N/w the quality factor must be as high as possible.

(3) The bandwidth of a N/w represents the range of freq. for which the current drawn by the N/w is decreased to 0.707 of its maximum value. For any tuned N/w the B.W. must be low to keep high quality factor.

(4) The frequency of Resonance Represents the geometric mean of two half power frequency.

Ex:-



$R = 20\,\Omega$

$L = 0.1\,H$

$C = 0.1\,F$

To find:- Z_{in} ; ω_0 ; $\Delta\omega$; Q ; ω_1 ; ω_2

Solution

$$I' = I - 0.04V_C$$

$$V = j\omega L \cdot I + \frac{1}{j\omega C} \cdot I + (I - 0.04V_C)R$$

$$V = j\omega L \cdot I + \frac{I}{j\omega C} + \left(I - 0.04 \cdot \frac{1}{j\omega C} \cdot I\right)R$$

$$V = I j\omega L + \frac{I}{j\omega C} + I\left(1 - \frac{0.04}{j\omega C}\right)R$$

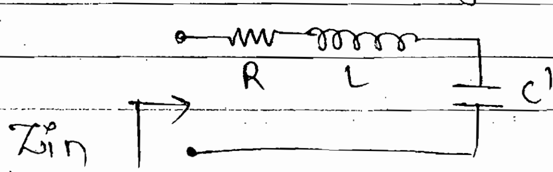
$$\frac{V}{I} = Z_{in} = R + j\omega L + \underbrace{\frac{1}{j\omega C} - \frac{0.04}{j\omega C}}$$

$$= \frac{1}{j\omega C}[1 - 0.04]$$

$$\Downarrow$$

$$\frac{1}{j\omega \dfrac{C}{1-0.04}} = \frac{1}{j\omega C'}$$

$$Z_{in} = R + j\omega L + \frac{1}{j\omega C'} \quad \ldots\ldots \text{ series RLC ckt}$$



1. $\omega_0 = \dfrac{1}{\sqrt{LC'}}$ $= 4470$ rad/se

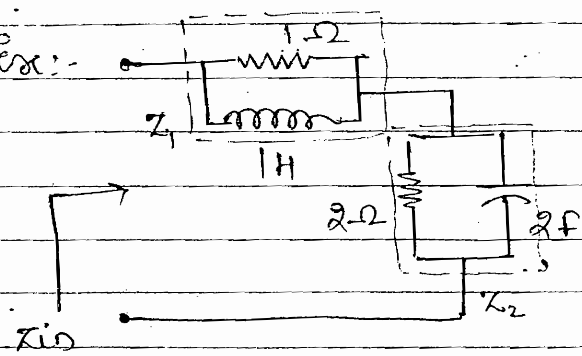
2. $Q_0 = \dfrac{\omega_0 L}{R} = \dfrac{1}{\omega_0 RC'} = \dfrac{1}{R}\sqrt{\dfrac{L}{C'}}$ $= 22.4$

3. $\Delta\omega = \dfrac{\omega_0}{Q_0} = \dfrac{R}{L} = 199.55$ rad/sec

4. $\omega_2 \simeq \omega_0 + \dfrac{1}{2}\Delta\omega = 4569.775$

$\omega_1 \simeq \omega_0 + \frac{1}{2}\Delta\omega = 4370.225$

Ex:-



To find ω_0

$Z_{in} = Z_1 + Z_2$ $\quad X_L = j\omega L$

where $Z_1 = \dfrac{R \times j\omega L}{R + j\omega L} = \dfrac{j\omega}{1 + j\omega}$

$Z_2 = \dfrac{R \times \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \dfrac{R}{1 + j\omega RC}$

$Z_1 = \dfrac{j\omega}{1+j\omega} \times \dfrac{1-j\omega}{1-j\omega} = \cancel{\dfrac{j\omega}{1+\omega^2}}^{\,0} + \dfrac{\omega^2}{1+\omega^2}$

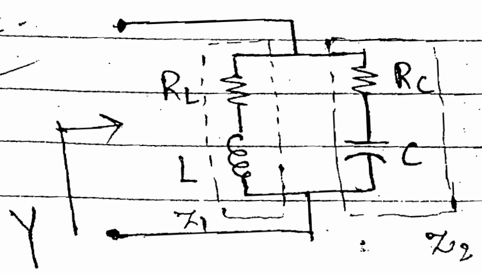
$Z_2 = \dfrac{2}{1+j\omega 4} \times \dfrac{1-4j\omega}{1-4j\omega} = \dfrac{2}{1+16\omega^2} - \cancel{\dfrac{j\omega 8}{1+16\omega^2}}^{\,0}$

at $\omega = \omega_0$ j terms $= 0$ so;

$\dfrac{\omega_0}{1+\omega_0^2} - \dfrac{j\omega_0 8}{1+16\omega_0^2} = 0$

$\omega_0 = \sqrt{\dfrac{7}{8}}$ -- rad/sec

Ex:- ~~Imp~~



To find ω_0

$Y_{in} = Y_1 + Y_2$

$X_L = j\omega L \qquad X_C = \frac{1}{\omega C}$

$$Y_{in} = \frac{1}{Z_1} + \frac{1}{Z_2} = \frac{1}{R_L + jX_L} + \frac{1}{R_C - jX_C}$$

$$= \frac{R - jX_L}{R_L^2 + X_L^2} + \frac{R_C + jX_C}{R_C^2 + X_C^2}$$

at $\omega = \omega_0$; j terms $= 0$

$$\Rightarrow \frac{-jX_L}{R_L^2 + X_L^2} + \frac{jX_C}{R_C^2 + X_C^2} = 0$$

$$\Rightarrow -X_L R_C^2 - X_L X_C^2 + X_C R_L^2 + X_C X_L^2 = 0$$

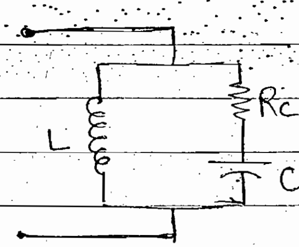
$$\Rightarrow -\omega_0 L R_C^2 - \omega_0 L \frac{1}{\omega_0^2 C^2} + \frac{1}{\omega_0 C} R_L^2 + \frac{1}{\omega_0 C} \omega_0^2 L^2 = 0$$

$$\boxed{\omega_0 = \frac{1}{\sqrt{LC}} \sqrt{\frac{R_L^2 - L/C}{R_C^2 - L/C}}}$$ freq. of Resonance.

- $Z_{in} = (\quad) + j\underbrace{(\quad)}_{=0 \text{ at } \omega = \omega_0}$

$Y_{in} = (\quad) + j\underbrace{(\quad)}_{=0 \text{ at } \omega = \omega_0}$

<u>case 1</u> :- $R_L = 0$



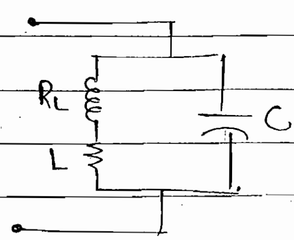
$$\omega_0 = \frac{1}{\sqrt{LC}} \sqrt{\frac{L/C}{L/C - R_C^2}}$$

To make the result +ve

$$\frac{L}{C} - R_C^2 > 0$$

$$\boxed{R_C < \sqrt{\frac{L}{C}}}$$ ---- condition of Resonance

<u>case 2</u> :- $R_C = 0$

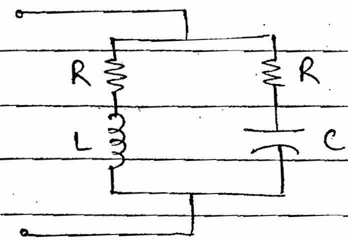


$$\omega_0 = \frac{1}{\sqrt{LC}} \sqrt{1 - \frac{R_L^2 C}{L}}$$

For tue Resuet $1 - \frac{R_L^2 C}{L} > 0$

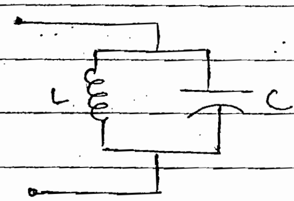
$$\boxed{R_L < \sqrt{\frac{L}{C}}}$$ ... condition for Resonance

<u>case 3</u> :- $R_L = R_C = R$



$$\boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$$

<u>case 4</u> : $R_L = R_C = 0$



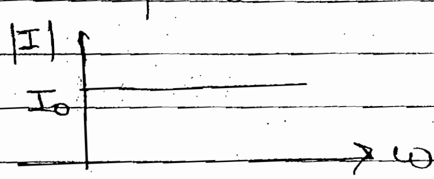
$$\boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$$

<u>NOTE</u> :- The frequeny of Resonance does not depend upon the type of connection b/w L and C. Therefore the frequency of Resonance Remain same wheather L & C are connected in series or in parallel

Imp <u>case 5</u> :- $R_L = R_C = \sqrt{\frac{L}{C}}$

Current is independent to any value of frequency. I/p voltage and I/p current are in phase.

$$Y_{in} = (\text{Real}) + j(0)$$

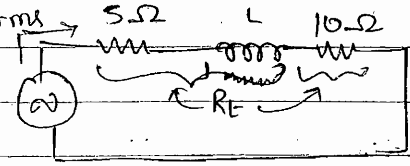


NOTE : For the specified value of RL & RC, the imaginary part of the I/p Impedance is alway zero and have only the real part. Therefore irrespective of frequency of operation Input voltage and I/p current are alway in phase.

↳ Therefore the current drawn By the N/w from I/p voltage source is alway constant irrespective of the frequency of operation. No specific frequency of resonance occurs and the given ckt is resonating at all frequencies for the specified value of RL and RC

Chapter-7

(1) $V_{max} = 50V$



$V_{rms} = \frac{50}{\sqrt{2}} = 35.35V$

$P_5 = 10W = I_{rms}^2 \, 5 \Rightarrow I_{rms} = \sqrt{2}$

$p.f = \cos\theta = \frac{R_t}{|Z|} = \frac{I_{rms}. R_t}{I_{rms} |Z| \equiv V_{rms}}$

$= \frac{\sqrt{2} \times (5+10)}{35.35} = \frac{\sqrt{2} \times 15}{35.35} = 0.6.$

(2) $v = 200 \sin(2000t + 50^\circ)$

$= 200 \sin(2000t + 90^\circ \underbrace{- 90 + 50^\circ}_{-40})$

$= 200 \cos(2000t - 40^\circ)$

$i = 4 \cos(2000t + 13.2^\circ)$

leading ⇒ capacitive.

i leads v

↳ capacitive → R, C elements
→ R, L, C where $X_C > X_L$

Q.8 $1/Z(s) = Y(s) = \frac{s^2 + 0.5s + 100}{5s} = \frac{s}{5} + 0.5 + \frac{20}{s}$

$\equiv Y_1 + Y_2 + Y_3$

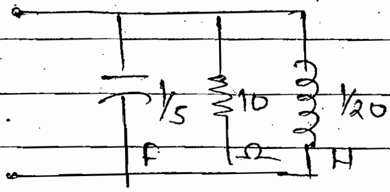
$\equiv SC_1 + \frac{1}{R_2} + \frac{1}{SL_3}$

Bcoz. it is admittance

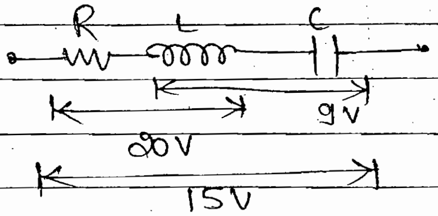
$C_1 = \frac{1}{5} F$,

$R_2 = 10\,\Omega$

$L_3 = 1/20\,H$.



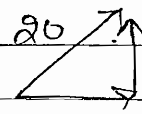
Q.11



$V_R + V_L = 20$

wrong Bcoz R, L are not in phase, having the phase shift of 90°

$\sqrt{V_R^2 + V_L^2} = 20$ — ①



$V_R = 400 - V_L^2$

$= 400 - (9 + V_C)^2$

$V_L - V_C = 9$ — ②

$V_L = 9 + V_C$

$\sqrt{V_R^2 + (V_L - V_C)^2} = 15$ — ③

$\boxed{V_C = 7V}$ Ans

✓ why we don't write $V_C - V_L$.

$Z = R + jX_L - jX_C = R + j(X_L - X_C)$

$V = ZI$

$= IR + j(IX_L - IX_C) = V_R + j(V_L - V_C)$

Q.13 Series RLC ckt

Given $R = 20\,\Omega$ $\theta = -45°$

$|V_L| = 2|V_C|$

$IX_L = 2IX_C$ $\quad \boxed{X_L = 2X_C}$

$I = \frac{V}{Z} = \frac{V}{R + j(X_L - X_C)}$

$\theta = \cancel{-}\tan^{-1}\frac{X_L - X_C}{R} = \cancel{-}45°$

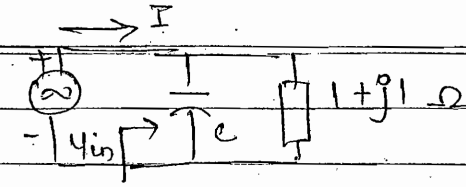
$\frac{X_L - X_C}{R} = 1$

$X_L - \frac{1}{2}X_L = R$ $\quad X_L = 2R = 2 \times 20$

$= 40\,\Omega$ Ans

Q.19 $V_s = \sin 2t$

$\omega_0 = 2$



V_s; I --- in phase

$\Rightarrow$ ckt is in resonance at $\omega_0 = 2$

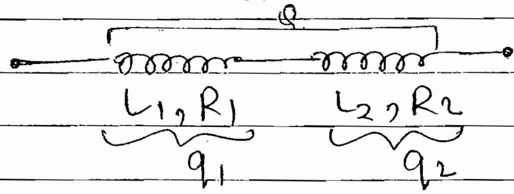
$$Y_{in} = Y_1 + Y_2 = j\omega_0 C + \frac{1}{1+j1} \times \frac{(1-j1)}{(1-j1)}$$

$$= j2C + \frac{1-j1}{1+1} = j2C + \frac{1}{2} - j\frac{1}{2}$$

at resonance the j Terms become zero.

$$2C - \frac{1}{2} = 0 \quad \Rightarrow C = \frac{1}{4} \text{ F Ans}$$

Q.21



Resonance condition will not occur bcoz capacitor is not present.

$$q_1 = \frac{L_1\omega}{R_1}$$

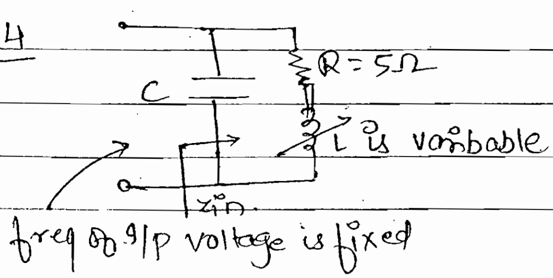
$$q_2 = \frac{\omega L_2}{R_2}$$

Here ω = freq. of operation
= Supply freq.
= self resonating freq.
freq. at which resonance conditⁿ occurs

$$Q = \frac{\omega L_{eq}}{R_{eq}} = \frac{1}{R_1+R_2}\left[\underbrace{\frac{\omega L_1}{R_1}}_{q_1} \times R_1 + \underbrace{\frac{\omega L_2}{R_2}}_{q_2} \times R_2\right]$$

$$= \frac{1}{R_1+R_2}\left[q_1 R_1 + q_2 R_2\right]$$

Q.14



$$Z_{in} = \frac{(-jX_C) \times (R + jX_L)}{-jX_C + R + jX_L}$$

$$= \frac{(-jX_C)(R+jX_L)}{R+j(X_L-X_C)} \times \frac{R-j(X_L-X_C)}{R-j(X_L-X_C)}$$

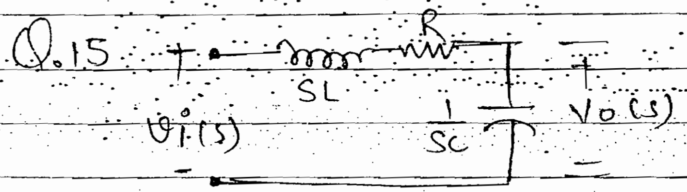
$$= (\text{Real}) + j\underbrace{(\text{Imaginary})}_{=0 \text{ at resonance.}}$$

$$X_L = \frac{1}{2}\left[X_C \pm \underbrace{\sqrt{X_C^2 - 4R^2}}\right]$$

$X_C^2 < 4R^2$	$X_C^2 = 4R^2$	$X_C^2 > 4R^2$
X_L = complex	one Resonance	for Two Resonance
∴ Not possible	occurs.	
Because $X_L \rightarrow$ Real in jX_L	possible	

$$X_C > 2R; \Rightarrow X_C > 2 \times 5; \boxed{X_C > 10\,\Omega}$$ Ans

Q.15



$$\frac{V_o(s)}{V_i(s)} = H(s) = \frac{1/SC}{SL + R + 1/SC} = \frac{1}{S^2LC + SRC + 1}$$

$$= \frac{1/LC}{S^2 + \frac{R}{L}S + \frac{1}{LC}}$$

$$\text{denominator} = S^2 + S\frac{R}{L} + \frac{1}{LC}$$

$$= S^2 + 2\delta\omega_n S + \omega_n^2 \qquad \omega_n = \frac{1}{\sqrt{LC}}$$

$$2\delta\omega_n = \frac{R}{L}; \quad \delta = \frac{R}{2}\sqrt{\frac{C}{L}}$$

RLC are not in parallel Bcoz:- $V_{in} = V_o$.

$H(s) = T.F = \frac{V_o}{V_{in}} = 1$

Q.15. $\delta = \frac{R}{2}\sqrt{\frac{C}{L}}$

for No oscillation, $\delta \geq 1$

$\frac{R}{2}\sqrt{\frac{C}{L}} \geq 1$

$R \geq 2\sqrt{\frac{L}{C}}$

Q.12 $2\delta\omega_n = \frac{R}{L} = \Delta\omega$

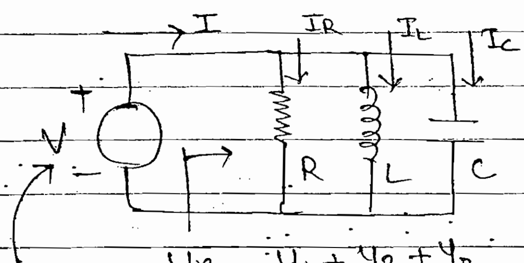
$\omega_n = \frac{1}{\sqrt{LC}} = \omega_0$

$Q_0 = \frac{\omega_0}{\Delta\omega}$

$H(s) = \frac{10^6}{s^2 + \underbrace{20}_{\Delta\omega}s + \underbrace{10^6}_{\omega_0}}$

$Q_0 = \frac{10^3}{20} = 50$ Ans.

• Parallel R-L-C ckt ---- Parallel Resonance.



$y_{in} = y_1 + y_2 + y_3$

Sinusoidal source (Variable frequency)

$y_{in} = \frac{1}{R} + \frac{1}{j\omega L} + j\omega C$

$= \frac{1}{R} + j\left[\omega C - \frac{1}{\omega L}\right]$

at $\omega = \omega_0$ j term $= 0$

$\omega_0 C - \frac{1}{\omega_0 L} = 0$

$\boxed{\omega_0 = \frac{1}{\sqrt{LC}} \text{ --- rad/sec} \qquad f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ --- Hz}}$

at $\omega = \omega_0$

$Y_{in} = \frac{1}{R}$ ...... minimum

$Z_0 = R$ ..... maximum

$I_0 = V.Y_0$ ...... minimum

at $\omega = \omega_0$

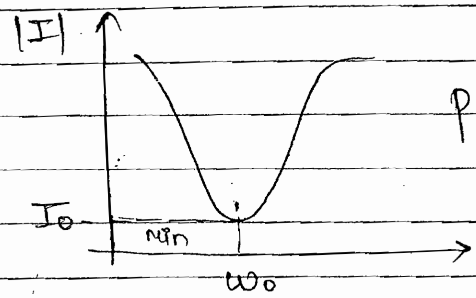
$|I_L| = |I_C|$

& are out of phase By 180° so that the net current drawn By L & C component is zero.

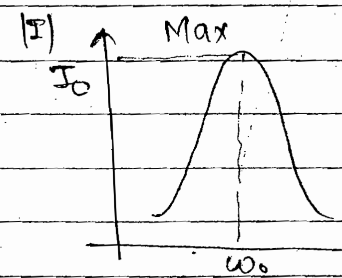
$$Y_{in} = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$

→ $\omega = \omega_0$; Resistive
→ $\omega > \omega_0$; capacitive
→ $\omega < \omega_0$; inductive.

frequency Response



parallel RLC ckt
BSF / BRF



Series RLC ckt BPF

BSF → Band stop filter
BRF → Band Rejection filter

Imp

- **Features :-**

(1) At resonance the imaginary part of the I/p Admittance is zero.

(2) At resonance the N/w has only resistive part, I/p admittance is minimum, I/p voltage and current are in phase so that the p.f. is unity

(3) At resonance current drawn by the N/w has minimum value so that any parallel RLC N/w behaves as a Band stop filter or a Band Reject filter.

(4) At resonance the current through the inductor and capacitor are equal in magnitude and are phase shifted by 180°. So that the net current through LC combination has zero value.

(5) The N/w Behaves differently at various frequencies

of operation.

(a) at ω_0 the N/w is resistive ; p.f = unity
(b) above ω_0 the N/w is capacitive ; pf = leading
(c) below ω_0 the N/w is inductive ; pf = lagging

(6) The quality factor represents the current amplificatn factor and for any tunned N/w the quality factor must have a high value. Therefore, at a specific frequency of resonance it must have low B.W.

- Quality Factor:-

$$Q_0 = \frac{|I_L|}{I_0} = \frac{|I_C|}{I_0}$$

$$|I_L| = \left|\frac{V}{j\omega_0 L}\right| = \frac{V}{\omega_0 L} \; ; \; |I_C| = \left|\frac{V}{1/j\omega_0 C}\right| = \omega_0 C V$$

$$I_0 = \frac{V}{R}$$

$$Q_0 = \frac{V}{\omega_0 L} \cdot \frac{R}{V} \; ; \; Q_0 = \frac{\omega_0 C V}{V/R}$$

$$\boxed{Q_0 = \frac{R}{\omega_0 L} = \omega_0 C R = R\sqrt{\frac{C}{L}}}$$

General Results:-

① $\Delta\omega = \omega_2 - \omega_1$
② $Q_0 = \omega_0 / \Delta\omega$
③ $\omega_0 = \sqrt{\omega_1 \omega_2}$
④ $\omega_2 \simeq \omega_0 + \frac{1}{2}\Delta\omega$
⑤ $\omega_1 \simeq \omega_0 - \frac{1}{2}\Delta\omega$

- Bandwidth :-

$$\Delta\omega = \frac{\omega_0}{Q_0} = \frac{1/\sqrt{LC}}{R\sqrt{C/L}} \qquad \therefore Q_0 = \frac{\omega_0}{\Delta\omega}$$

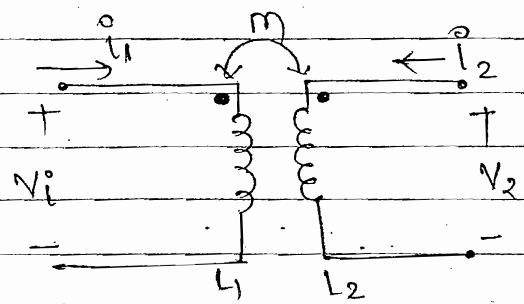
$$\boxed{\Delta\omega = \frac{1}{RC}}$$

$Q_0 \uparrow \rightarrow \Delta\omega \downarrow$ — $\rightarrow R\uparrow$ ... not advisable (due to of losses)
$\rightarrow C\uparrow$

✓ $\boxed{\Delta\omega = \frac{1}{RC} \text{ parallel RLC ckt} \; ; \; \Delta\omega = \frac{R}{L} \text{ ; series RLC ckt}}$

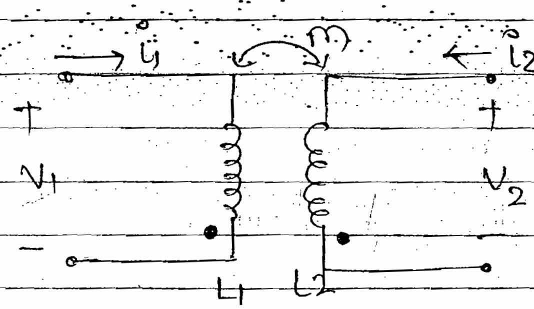
⊙ EFFECT OF MUTUAL INDUCTANCE
--- (coupled ckt)

Case 1:



Current is entering to DOT

Case 2:



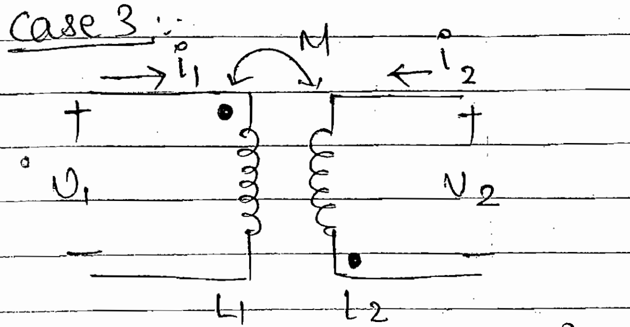
Current is leaving to DOT

→ When Both current entering or leaving the dot Mutual inductance takes +ve

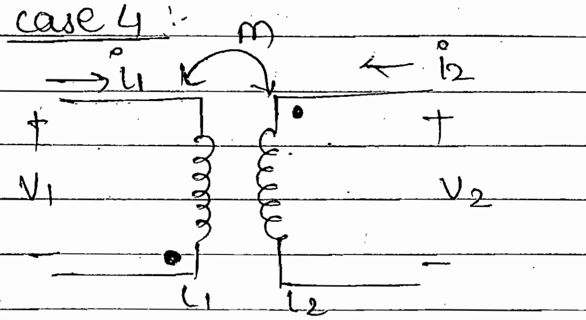
• $\boxed{\text{Effect of M is +ve} \quad E_{ind} \text{ is additive}}$ case 1, 2

Physical Meaning of dot → sense of wdg (same) in above cases either clockwise or anticlockwise.

Case 3 :-



Case 4 :-



current entering to dot | current leaving to dot | current leaving to dot | current entering to dot

Physical Meaning :- In above cases (3,4) if one wdg is clockwise other wdg will anti-clockwise

- $\boxed{\text{Effect of Mutual is -ve Eind is Subtractive}}$ case 3,4

$$\left.\begin{aligned} v_1(t) &= L_1\frac{di_1(t)}{dt} \pm M\frac{di_2(t)}{dt} \\ v_2(t) &= L_2\frac{di_2(t)}{dt} \pm M\frac{di_1(t)}{dt} \end{aligned}\right\}$$ ---- in time domain ---- for non-sinusoidal excitation

$$\left.\begin{aligned} V_1(s) &= sL_1I_1(s) \pm sM\,I_2(s) \\ V_2(s) &= sL_2I_2(s) \pm sM\,I_1(s) \end{aligned}\right\}$$ -- in s-domain assuming zero I.C.s

$$\begin{bmatrix} V_1(s) \\ V_2(s) \end{bmatrix} = \begin{bmatrix} sL_1 & \pm sM \\ \pm sM & sL_2 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j\omega L_1 & \pm j\omega M \\ \pm j\omega M & j\omega L_2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

above Matrix is ---- for sinusoidal excitation
--- using phasors $s = j\omega$

NOTE:- (1) Effect of Mutual inductance is consider when ever two inductors are placed physically closed to each other

(2) When the current flows in first conductor, their is some magnetic field therefore some magnetic flux. Part of this magnetic flux will link with second inductor. Due to rate of change of magnetic flux and due to faraday's law of electromagnetic induction some emf is induce B/w the Two Terminal of the second inductor. The polarity of this induce
(3) voltage may be +ve or −ve and depends upon the dot convention therefore depends upon the relative sense of windg of the two inductor.
↳ Then all the KVL equations are modify.

• Factors affecting the mutual Inductance
1. M
2. K ... Coefficient of coupling [show closeness of two inductor]

$$\boxed{K = \frac{M}{\sqrt{L_1 L_2}}} \; ; \; \boxed{M = K\sqrt{L_1 L_2}}$$

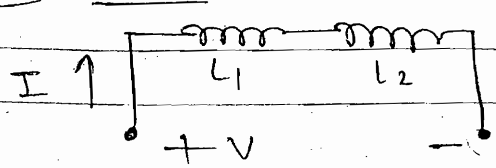
Case 1: $K = 1$; $M = \sqrt{L_1 L_2}$ ... critical coupling
such ckt ... critically coupled ckt

Case 2: $K < 1$; $M < \sqrt{L_1 L_2}$ ... loose coupling
such ckt ... under coupled ckt

Case 3: TURNS RATIO :-

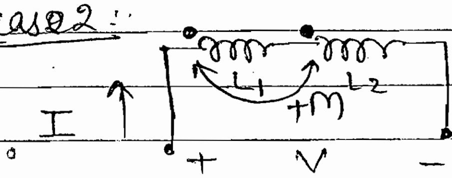
$$\boxed{\frac{V_2}{V_1} = \frac{n_2}{n_1}} \qquad \boxed{\frac{I_2}{I_1} = \frac{n_1}{n_2}}$$

Ex: case 1. No mutual



$$\boxed{L_{eq} = L_1 + L_2}$$

case 2 :

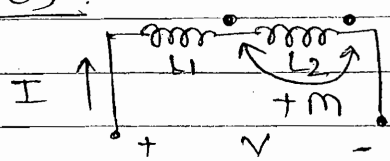


$$V = j\omega L_1 I + j\omega L_2 I + j\omega M I + j\omega M I$$

$$\frac{V}{I} = Z_{eq} = j\omega (L_1 + L_2 + 2M)$$

$$\boxed{L_{eq} = L_1 + L_2 + 2M}$$

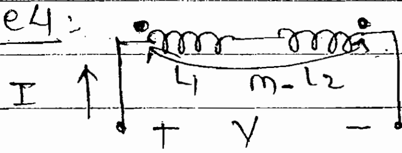
case 3 :



$$\boxed{L_{eq} = L_1 + L_2 + 2M}$$

→ Net inductance ↑ By factor 2M

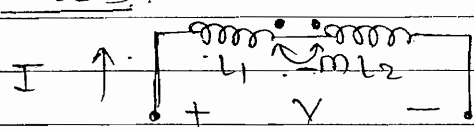
case 4 :



$$V = j\omega L_1 I + j\omega L_2 I - j\omega M I - j\omega M I$$

$$\boxed{L_{eq} = L_1 + L_2 - 2M}$$

Case 5 :

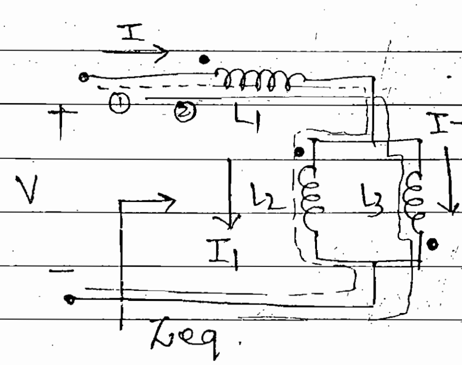


$$\boxed{L_{eq} = L_1 + L_2 - 2M}$$

→ Net inductance ↓ By factor 2M

Find $\frac{V}{I} = Z_{eq} = j\omega L_{eq}$ To find: L_{eq}

Ex :



$L_1 = L_2 = L_3 = 10H$

$M_{12} = M_{21} = 1H$

$M_{13} = M_{31} = 2H$

$M_{23} = M_{32} = 3H$

To find : L_{eq}

Solution : $Z_{eq} = \frac{V}{I} = j\omega L_{eq}$

KVL loop ① :

$$V = j\omega L_1 I + j\omega L_2 I_1 + j\omega M_{21} I_1 - j\omega M_{31}(I - I_1) + j\omega M_{12} I - j\omega M_{32}(I - I_1)$$

$M_{21} \rightarrow$ Read like:- Mutual inductance in inductor 2 due to inductor 1

Due to inductor L_1, so leave L_1 and write according to L_2 & L_3.

$$V = j\omega(L_1 - M_{31} + M_{12} - M_{32})I + j\omega(L_2 + M_{21} + M_{31} + M_{32})I_1 \quad \text{---(1)}$$

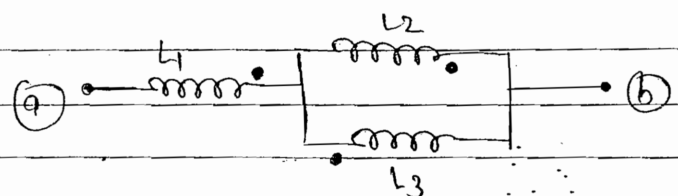
KVL loop (2) :-

$$V = j\omega L_1 I + j\omega L_3 (I - I_1) + j\omega M_{21} I_1 - j\omega M_{31}(I - I_1) - j\omega M_{13} I - j\omega M_{23} I_1$$

$$V = j\omega I(L_1 + L_3 - M_{31} - M_{13}) + j\omega I_1(-L_3 + M_{21} + M_{31} - M_{23})$$

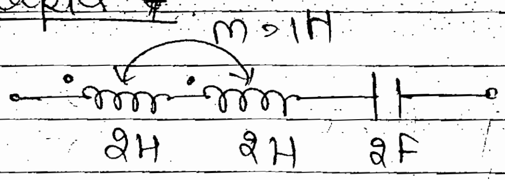
Eliminate I_1 find $\frac{V}{I} = Z_{eq} = j\omega L_{eq}$; $L_{eq} = ?$

Ex



Chapter-7

(6)



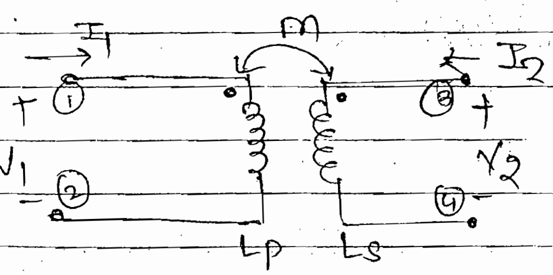
$$f_0 = \frac{1}{2\pi\sqrt{L_{eq}C}} \text{ Hz}$$

$C \leftarrow 2F$

$L_{eq} \leftarrow L_1 + L_2 + 2M = 2 + 2 + 2 = 6$

$$f_0 = \frac{1}{2 \times 3.14\sqrt{12}} = \frac{1}{4\pi\sqrt{3}} \text{ Hz} \quad \text{Ans}$$

Chap. 1 :
Q. 15



$$K = \frac{M}{\sqrt{L_p L_s}}$$

$\sqrt{L_p L_s} \leftarrow 2H$

case 1 :- 3-4 ⇒ O.C   case 2 :- 3-4 ⇒ S.C

So $I_2 = 0$

$\frac{V_1}{I_1} = Z_{eq} = j\omega L_{eq} = j\omega 4$ —①

$V_1 = j\omega L_p I_1 + j\omega M \cancel{I_2}^{0}$

$= j\omega L_p I_1 + 0$ —②

$\frac{V_1}{I_1} \Rightarrow j\omega L_p = j\omega 4$

$\boxed{L_p = 4\,H}$

so $V_2 = 0$

$V_1 = j\omega L_p . I_1 + j\omega M I_2$ —ⓐ

$V_2 = 0 = j\omega L_s I_2 + j m\omega I_1$ —ⓑ

$\frac{V_1}{I_1} = j\omega L_{eq} = Z_{eq} = j\omega 3$ —ⓒ

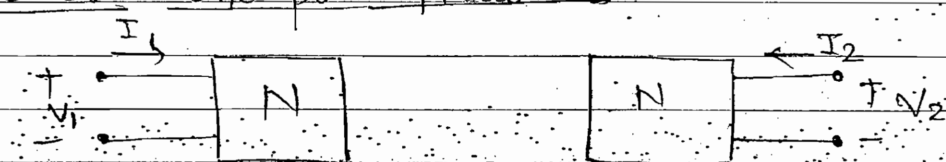
$K = \frac{M}{\sqrt{L_p L_s}} = 0.5$ ($M \leftarrow \sqrt{2}$, $L_p \leftarrow 4$, $L_s \leftarrow 2$)

∴ ($K \ll 1$) under-coupled ckt

• Network Functions :-

used to represent any type of Network Mathematically

case 1 : One port Networks :



- $Z_{11} = \frac{V_1}{I_1}$ ---- DRIVING POINT INPUT IMPEDANCE FUNCTION.
- $Y_{11} = \frac{I_1}{V_1}$ --- DRIVING POINT INPUT ADMITTANCE FUNCTION

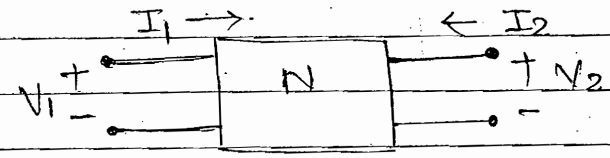
→ for series element Impedance finding is convinient
→ for parallel element admittance finding is convinent

- $Z_{22} = \frac{V_2}{I_2}$ --- DRIVING POINT OUTPUT IMPEDANCE FUNCTION
- $Y_{22} = \frac{I_2}{V_2}$ --- DRIVING POINT OUTPUT ADMITTANCE FUNCTION

- $Z_{11}, Z_{22}, Y_{11}, Y_{22} \rightarrow$ Driving point immittance functn

Impedance → (im) ; Admittance → (mittance)

Case 2 : Two port Networks



1) Driving point immittance function :-

$$Z_{11} = \frac{V_1}{I_1} \quad ; \quad Z_{22} = \frac{V_2}{I_2}$$

$$Y_{11} = \frac{I_1}{V_1} \quad ; \quad Y_{22} = \frac{I_2}{V_2}$$

2) Transfer Impedance function (Ratio) :-

$$Z_{12} = \frac{V_1}{I_2} \quad ; \quad Z_{21} = \frac{V_2}{I_1}$$

3) Transfer admittance Ratio :-

$$Y_{12} = \frac{I_1}{V_2} \quad ; \quad Y_{21} = \frac{I_2}{V_1}$$

4) Transfer Voltage Ratio :-

$$G_{12} = \frac{V_1}{V_2} \quad ; \quad G_{21} = \frac{V_2}{V_1}$$

5) Transfer Current Ratio :-

$$\alpha_{12} = \frac{I_1}{I_2} \quad ; \quad \alpha_{21} = \frac{I_2}{I_1}$$

Network Function

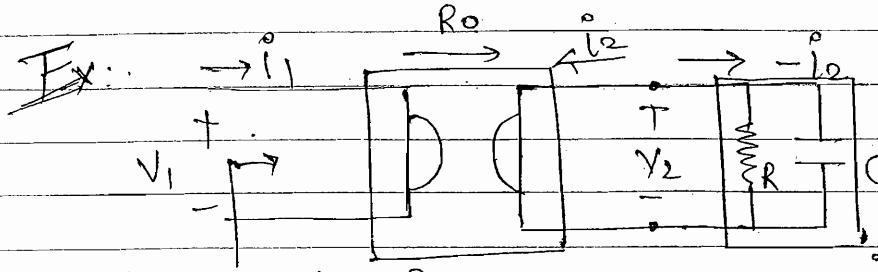
$F(s)$

$\underbrace{Z_{11}, Z_{22}, Y_{11}, Y_{22}}_{\text{Driving pt immittance function}}$

$\underbrace{Z_{12}, Z_{21}, Y_{12}, Y_{21}, G_{12}, G_{21}, \alpha_{12}, \alpha_{21}}_{\text{Transfer functn}}$

NOTE :- (1) for the driving point immittance function all the poles and zeros lie in left half of the complex s-plane. For Transfer function all the poles still lie in LHP (left half plane) whereas some of the zeros may also lie in RHP

Ex:



$Z_{in} = Z_{11} = \frac{V_1}{I_1} = ?$

$Z = \frac{R \times \frac{1}{sC}}{R + \frac{1}{sC}} = \frac{R}{1+RsC}$

$V_1 = R_o i_2$ —① ; $V_2 = -R_o i_1$ —②

$V_2 = -i_2 Z = -i_2 \frac{R}{1+RsC}$ —③

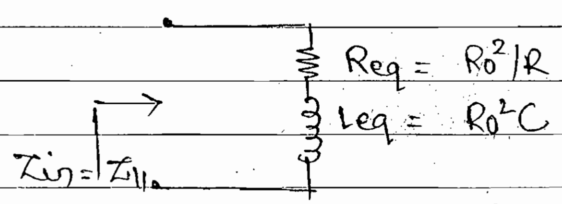
eq^n ② & ③ $\quad -R_o i_1 = -i_2 \frac{R}{1+Rsc} \qquad i_2 = \frac{R_o i_1 (1+RsC)}{R}$

$V_1 = \frac{R_o^2 (1+RsC)\, i_1}{R}$

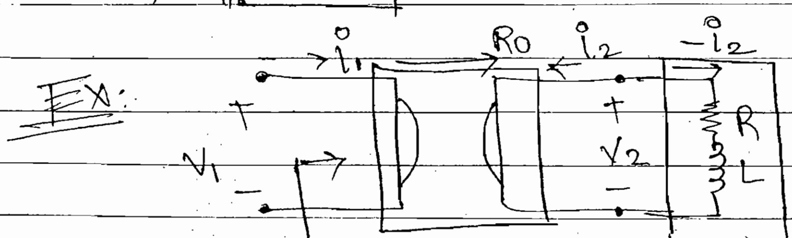
$Z_{in} = \frac{V_1}{I_1} = \frac{R_o^2 (1+Rsc)}{R} = \underbrace{\frac{R_o^2}{R}}_{Z_1} + \underbrace{\frac{R_o^2 sC}{R}}_{Z_2}$ due to S (inductor)

$Z_{in} = Z_1 + Z_2$

$= R_{eq} + S L_{eq}$ $\qquad \boxed{R_{eq} = \frac{R_o^2}{R} \; ; \; L_{eq} = R_o^2 C}$



Ex:



$Z = R + SL$

$Z_{in} = Z_{11} = \frac{V_1}{i_1}$

Solution:- $v_1 = R_0 i_2$ —(1) ; $v_2 = -R_0 i_1$ —(2)

$v_2 = -i_2 (R + SL)$ —(3)

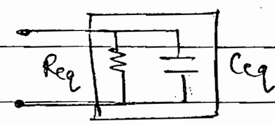
$\therefore i_2 = \frac{R_0}{R+SL} i_1 \qquad v_1 = \frac{R_0^2}{R+SL} i_1$

[$\because i_2 = -\frac{v_2}{R+SL}$ $\quad$ $\therefore v_2 = -R_0 i_1$] $\qquad$ [$\because v = R_0 i_2$ $\quad$ $\therefore i_2 = \frac{R_0 i_1}{R+SL}$]

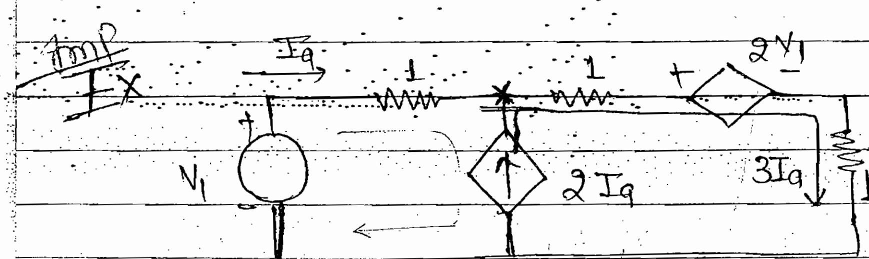
$$\boxed{\frac{v_1}{i_1} = \frac{R_0^2}{R+SL}} = \frac{1}{Y_{11}} = \frac{1}{Y_{in}}$$

$$Y_{in} = \frac{R+SL}{R_0^2} = \underbrace{\frac{R}{R_0^2}}_{Y_1} + \underbrace{\frac{SL}{R_0^2}}_{Y_2}$$

$$Y_{in} = Y_1 + Y_2 = \frac{1}{R_{eq}} + C_{eq}$$



Imp Ex



To find $G_{21} = \frac{V_2}{V_1}$

Cannot apply KVL eqn across X box, voltage across $2I_a$ we donot known (arbitary voltage)

So KVL

$$+V_1 + 1\times I_a + 1\times 3I_a + 2V_1 + 1\times 3I_a = 0$$

$$+V_1 = -7 I_a \quad \text{—(1)}$$

$$V_2 = 1\times 3 I_a \quad \text{—(2)}$$

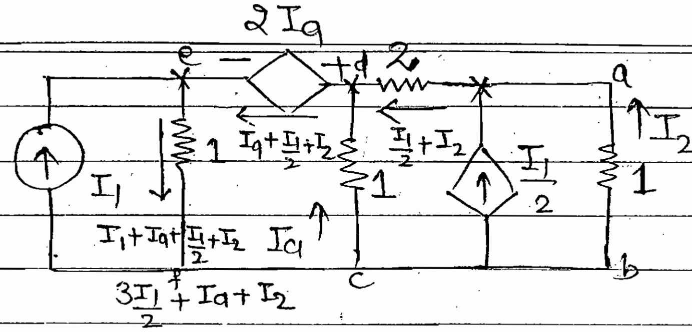
$$\frac{V_2}{V_1} = G_{21} = -\frac{3}{7} \quad \text{Ans}$$

Select that loop which does not contains any current source

Chap-5

Q.3 Conversion:-

To find $\alpha_{21} = \frac{I_2}{I_1}$



KVL loop (1) badcb

$\Rightarrow I_2 + 2\left(\frac{I_1}{2}+I_2\right) - I_q = 0$

$\Rightarrow I_q = I_2 + 2(I_1/2 + I_2) = 3I_2 + I_1$ —(1)

KVL loop (2) defcd

$\Rightarrow I_q + 2I_q + 1.(I_1 + I_q + I_1/2 + I_2) = 0$

$\Rightarrow 4I_q + 3/2\, I_1 + I_2 = 0$ —(2)

put (1) in (2). $4[3I_2 + I_1] + \frac{3}{2} I_1 + I_2 = 0$

$\Rightarrow 12I_2 + 4I_1 + 3/2\, I_1 + I_2 = 0$

$\Rightarrow 13\, I_2 + 11/2\, I_1 = 0$

$\alpha_{21} = \frac{I_2}{I_1} = -\frac{11}{26}$ Ans

Q For any N/w in particular N/w function is specified as follows. Find the pole zero pattern and explain the physical significance of poles and zero.

Solution $F(s) = \frac{(s+1)(s+3)}{s(s+2)(s+4)} = \frac{\text{Numerator poly}}{\text{Denominator poly}}$

Z_{11}, Z_{22} — Z_{12}, Z_{21}

Y_{11}, Y_{22} — Y_{12}, Y_{21}

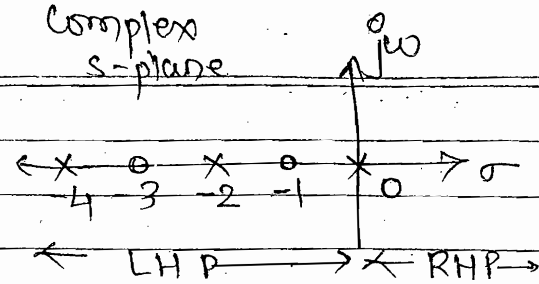
G_{12}, G_{21}

α_{12}, α_{21}

- For Zeros :- Num poly = 0 ; $F(s) = 0$; $s = -1, -3$
- For poles :- Deno. poly = 0 ; $F(s) = \infty$; $s = 0, -2, -4$

----- Zeros & poles ------ singularities ------ Natural freq (Bcz. Img. part is zero)	---- Complex frequencies (In general)

2nd NOV 2014



- Pole → control shape of response
- Coefficient → — " — amplitude — " —
(zero)

$$f(t) = \left(A + Be^{-2t} + Ce^{-4t}\right) u(t)$$

Coefficient of partial fractⁿ expansion
— Residues of poles.

- Some Imp. points :-

(1) The zeros represent the complex frequencies at which the numerator polynomial of the N/w functⁿ become zero. Therefore at these frequencies the N/w functⁿ become zero. The zeros control the magnitude of the response of a given N/w.

(2) The poles represent the complex frequencies where the denominator poly. of a N/w functⁿ become zero. Therefore at these freq. the N/w functⁿ Become ∞. The poles control the nature or shape of response of a given N/w.

⇒ Therefore these control overall stability of any given N/w.

(3) For a stable N/w → (a) NO pole should exist in RHP
(b) NO multiple poles should exist along the jω axis including the origin.

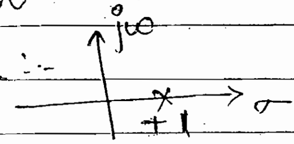
(4) In any N/w functⁿ The no. of zeros and NO. of poles are alway equal.

(5) The locatⁿ of poles and zeros control the type of elements contain in any given N/w.

(6) For Immittance functⁿ all the poles and zeros lie in LHP. For transfer functⁿ all the poles lie in LHP whereas some of the zeros may also lie in RHP.

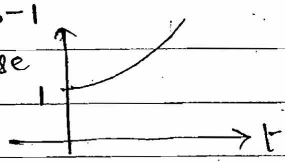
• Effect of location of poles on the Response of N/w.

case 1 :-



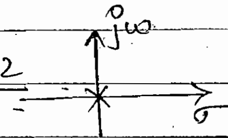
$\frac{1}{S-1} \longrightarrow e^{+t}u(t)$

Response

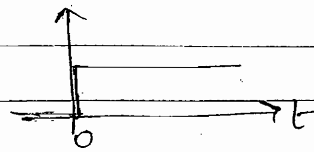


∴ unstable N/w.

case 2

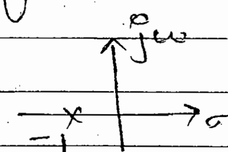


$\frac{1}{s} \longrightarrow u(t)$

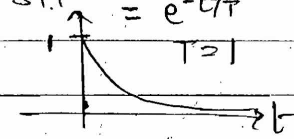


∴ Ideal/lossless or stable sys.

case 3

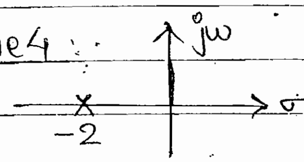


$\frac{1}{S+1} \longrightarrow e^{-t}u(t)$
$= e^{-t/T}$
$T=1$

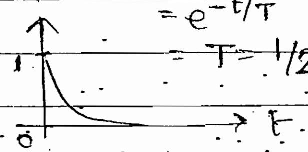


$\boxed{\text{RC/RL CKt}}$
stable

case 4 :-



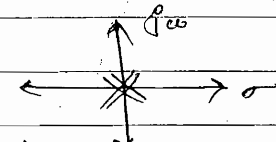
$\frac{1}{S+2} \longrightarrow e^{-2t}u(t)$
$= e^{-t/T}$
$\Rightarrow T = 1/2$



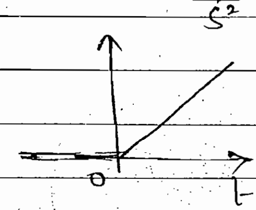
→ decay at faster Rate

• This N/w is More stable than case 3 N/w
• pole shift to L.H.S. (stable)

case 5 :-



$\frac{1}{s^2} \longrightarrow t\,u(t).$



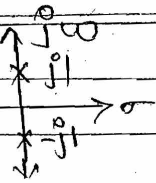
∴ unstable N/w

• For stable N/w :- (1) The pole should not exist in RHP (case 1)
(2) Multiple poles should not exist at the origin (case 5)

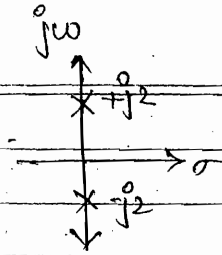
• When any pole lies along the -ve real axis the response is expotential decaying and the rate of decay depends upon the exact locatn of the pole along the -ve real axis.

• When the pole is shifted away from the origin along the -ve real axis, the rate of decay of the expotential response increases, the time constant ↓, and the N/w will become more stable.

case 6:



case 7:

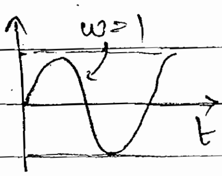


$\frac{1}{(s-2j)(s+2j)}$

$\frac{1}{(s-j1)(s+j1)} = \frac{1}{s^2+1} = \frac{a}{s^2+a^2}$ $\Rightarrow \frac{1}{s^2+4^2} \Rightarrow \frac{1}{2} \cdot \frac{2}{s^2+2^2}$

$\Rightarrow \sin t$ $= \frac{1}{2} \sin 2t$

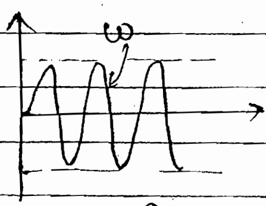
$\omega = 1$ rad/sec $\omega = 2$ rd/sec



R → Not present

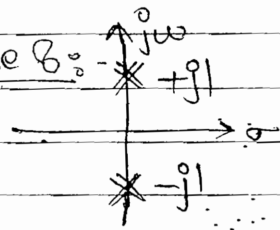
$\boxed{\text{LC ckt}}$

$s = 0 \pm j1$



• freq of Response ↑

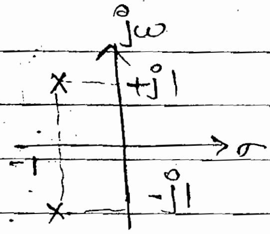
case 8:



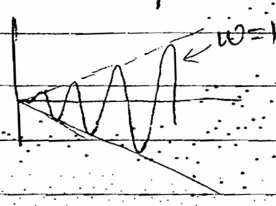
$= \frac{1}{(s^2+1)^2}$

$= t \sin t$

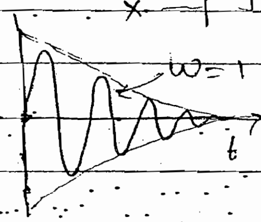
case 9



$\boxed{\text{RLC ckt}}$



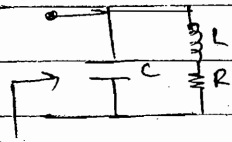
unstable N/w.



- When a pair of poles lie along the jω axis then the response is oscillatory in Nature whose freq. is controlled By the exact locatⁿ of poles along the jω axis.
- The the poles are shifted away from the origin along the jω axis then the freq. of sinusodial response ↑.
- When the poles are complex conjugate in nature having both real and imaginary part in LHP. Then the response is oscillatory in nature whose amplitude ↓ exponentially.
- For stable N/w Multiple pole must not exist along the jω axis.

Chap. 5

Q.18



Given $Z(0) = 1$; zero $\rightarrow -1 = s$

poles : $s_1, s_2 \rightarrow \frac{1}{2} \pm j\sqrt{3}/2$

To find R, L, C.

$$Z(s) = \frac{(R + sL)(1/Cs)}{\left(\frac{1}{Cs}\right) + (sL + R)} = \frac{R + sL}{1 + s^2LC + sCR}$$

$$= \frac{L\,(s + R/L)}{LC\,[s^2 + sR/L + 1/LC]}$$

$$Z(0) = \frac{R + 0}{1 + 0 + 0} = 1 \quad \boxed{\Rightarrow R = 1\,\Omega}$$

✓ zero :- $s = \overset{\text{compare}}{-R/L = -1}$ $\quad \boxed{L = 1\,H}$

✓ poles :- $s_1, s_2 = \frac{-R/L \pm \sqrt{R^2/L^2 - 4 \times 1/LC}}{2} = \frac{1}{2} \pm j\frac{\sqrt{3}}{2}$

$$= -\frac{R}{2L} \pm j\sqrt{1/LC - (R/2L)^2}$$

$$= -1/2 \pm j\,\sqrt{3}/2$$

$\Big] \rightarrow$ compare.

$$\frac{1}{LC} - \left(\frac{R}{L2}\right)^2 = \frac{3}{4}$$

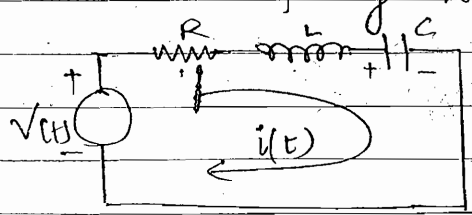
$$1/C = \frac{3}{4} + \left(\frac{1}{2}\right)^2 = \frac{3 + 1}{4} = \frac{4}{4} = 1$$

$$\boxed{C = 1\,F} \text{ Ans.}$$

• Effect of variation of R in a series RLC circuit :-

$s^2 + 2\delta\omega_n s + \omega_n^2$

$\rightarrow$ $\delta \rightarrow$ vary only when R is varying in ckt :-



$$V(t) = Ri(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int_0^t i\,dt$$

$$\frac{dV(t)}{dt} = R\frac{di(t)}{dt} + L\frac{d^2i(t)}{dt^2} + \frac{1}{C}i(t)$$

Characteristic eqn in s-domain.

$$Rs\,I(s) + Ls^2\,I(s) + \frac{1}{C}I(s) = 0$$

$\Rightarrow Rs + Ls^2 + \frac{1}{c} = 0$

$\Rightarrow s^2 + \frac{R}{L}s + \frac{1}{Lc} = 0$ compare with $[s^2 + 2\delta\omega_n s + \omega_n^2]$

$\Rightarrow 2\delta\omega_n = \frac{R}{L}$; $\delta = \frac{R}{2}\sqrt{\frac{C}{L}}$; $\omega_n = \frac{1}{\sqrt{LC}}$

damping coefficient — natural freq.

$$S_1, S_2 = \frac{-2\delta\omega_n \pm \sqrt{(2\delta\omega_n)^2 - 4\omega_n^2}}{2}$$

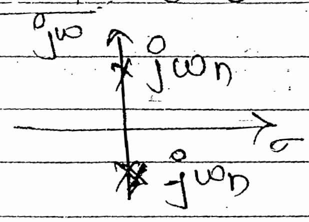
$$\boxed{S_1, S_2 = -\delta\omega_n \pm j\omega_n\sqrt{1-\delta^2}}$$

$\delta = \frac{R}{2}\sqrt{\frac{C}{L}}$; $\omega_n = \frac{1}{\sqrt{LC}}$; $\omega_d = \omega_n\sqrt{1-\delta^2}$

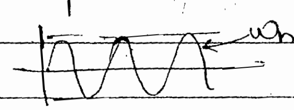
POINTS :-

- Char. eqn represent the denominator poly. of any funct^n.
- The roots of this eqn represent the poles of the N/w and control overall stability of N/w.
- This eqn control the transient response of a given N/w for zero excitation.

Case 1. $\delta = 0$; $\boxed{R = 0}$; $S_1, S_2 = \pm j\omega_n$ - roots on imag axis.



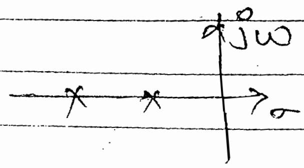
Response oscillatory with constant magnitude

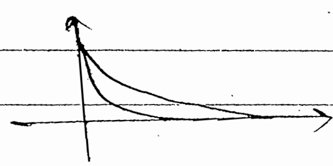


undamped Response

Case 2:- ~~...~~ $\delta > 1$; $\delta = \frac{R}{2}\sqrt{\frac{C}{L}}$ $\boxed{R > 2\sqrt{\frac{L}{C}}}$

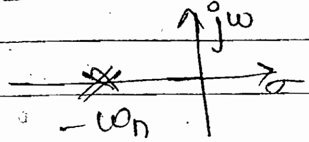
S_1, S_2 :- ~~$-a \pm jb$~~ - unequal Roots → ~~imag~~ axis lie along $-a \pm b$ — -ve real axis



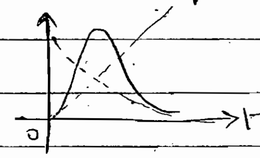


over damped Response

<u>case3</u> $\delta = 1$; $s_1, s_2 = -\omega_n$ .. multiple Roots along -ve Real axis.



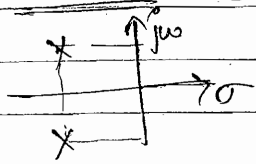
$$\frac{1}{(s+\omega_n)^2} \; ; \quad = t\,e^{-\omega_n t}$$

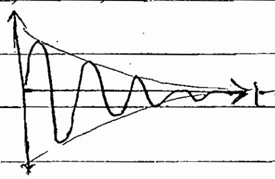


<u>critically damped</u>

$$\boxed{R = 2\sqrt{\frac{L}{C}}}$$

<u>case4</u> $\delta < 1$; $s_1, s_2 = -a' \pm jb'$ roots are complex conjugate





$\boxed{\text{under damped Response}}$

- For NO Oscillation condition $\delta \geq 1$

- 2 PORT N/W PARAMETERS

--- another way to represent the any 2-port N/w Mathematically

x, y

$y = mx + c$

dependent variable — independent variable

<u>case1:-</u> Z or impedance, or O.C. parameters :-

$$\boxed{\begin{aligned} V_1 &= Z_{11} I_1 + Z_{12} I_2 \\ V_2 &= Z_{21} I_1 + Z_{22} I_2 \end{aligned}}$$

--- series

--- series

$I_2 = 0$	$I_1 = 0$
$Z_{11} = \frac{V_1}{I_1}$ Ω	$Z_{12} = \frac{V_1}{I_2}$ Ω
$Z_{21} = \frac{V_2}{I_1}$ Ω	$Z_{22} = \frac{V_2}{I_1}$ Ω

→ Network function

- $Z_{11}, Z_{12}, Z_{21}, Z_{22}$ → Solve without any condition. (O.C, S.C)
- $z_{11}, z_{12}, z_{21}, z_{22}$ → Solve without pre condition (O.C, S.C).

→ Network parameters.

(1) The Network functions are calculated without any precondition whereas N/w parameter are calculated either by S.C or O.C the i/p or o/p port.

(2) A single N/w funct^n is sufficient to represent a particular N/w whereas all the four parameters are required simultaneously to represent the same N/w.

case 2 :- Y parameter or S.C or admittance parameter.

$$\boxed{\begin{aligned} I_1 &= y_{11} V_1 + y_{12} V_2 \\ I_2 &= y_{21} V_1 + y_{22} V_2 \end{aligned}}$$

— shunt
— shunt

$V_2 = 0$	$V_1 = 0$
$y_{11} = \frac{I_1}{V_1}$ S	$y_{12} = \frac{I_1}{V_2}$ S
$y_{21} = \frac{I_2}{V_1}$ S	$y_{22} = \frac{I_2}{V_2}$ S

i/p and o/p are short circuited

S → Simenon

case 3: h or hybrid Parameter :-

$$\boxed{\begin{aligned} V_1 &= h_{11} I_1 + h_{12} V_2 \\ I_2 &= h_{21} I_1 + h_{22} V_2 \end{aligned}}$$

— series
— parallel.

o/p → S.C i/p → O.C

$V_2 = 0$	$I_1 = 0$
$h_{11} = \frac{V_1}{I_1}$ Ω	$h_{12} = \frac{V_1}{V_2}$
$h_{21} = \frac{I_2}{I_1}$	$h_{22} = \frac{I_2}{V_2}$ S

Biasing → only for D.C.
Model → a.c.

Case 4 g parameter:

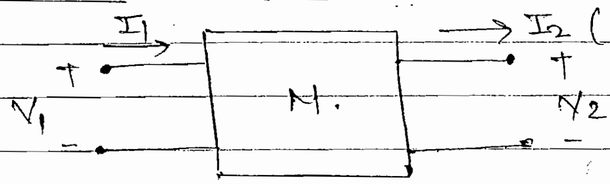
$$\boxed{\begin{aligned} I_1 &= g_{11} V_1 + g_{12} I_2 \\ V_2 &= g_{21} V_1 + g_{22} I_2 \end{aligned}}$$

— parallel
— series.

$I_2 = 0$	$V_1 = 0$
$g_{11} = \frac{I_1}{V_1}$ S	$g_{12} = \frac{I_1}{I_2}$
$g_{21} = \frac{V_2}{V_1}$	$g_{22} = \frac{V_2}{I_1}$ Ω

I/p → S.C O/p → O.C.

Imp Case 5: Transmission or T or ABCD parameters:-



$$\begin{aligned} V_1 &= A V_2 + B I_2 \\ I_1 &= C V_2 + D I_2 \end{aligned}$$

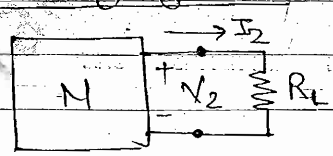
$I_2 = 0$	$V_2 = 0$
$A = \frac{V_1}{V_2}$	$B = \frac{V_1}{I_2}$ Ω
$C = \frac{I_1}{V_2}$ S	$D = \frac{I_1}{I_2}$

→ Change in Imp. will change only C, B parameters.
ex: Impe double ⇒ B → double, C → ½ Bcoz admittance

Interview question

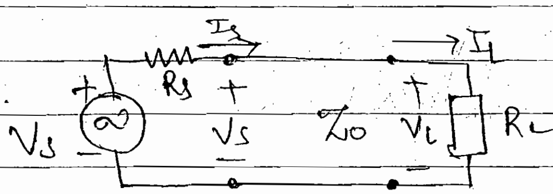
Direction of I_2:-

1.



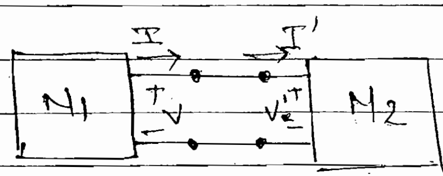
$V_2 = + I_2 R_L$

2.



Transmission line

3.



Cascading the N/ws
(direct of should in same for simplicity)

Imp

• Conditions for a Network to Be Reciprocal & Symmetrical:-

RECIPROCAL	SYMMETRICAL
• Z-parameter $z_{12} = z_{21}$	$z_{11} = z_{22}$
• Y-parameter $y_{12} = y_{21}$	$y_{11} = y_{22}$
• $\begin{vmatrix} A & B \\ C & D \end{vmatrix} = 1$; $\Delta = 1$ $AD - BC = 1$	$A = D$
• $h_{12} = -h_{21}$	$\begin{vmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{vmatrix} = 1$; $\Delta_h = 1$ $h_{11}h_{22} - h_{12}h_{21} = 1$
• $g_{12} = -g_{21}$	$\begin{vmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{vmatrix} =$; $\Delta g = 1$ $g_{11}g_{22} - g_{12}g_{21} = 1$

Q. Calculate the parameters which are inverse of given set of parameter:-

Given : $[z]$ To find : $[y]$

Solution

$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}^{-1}$$

$$= \frac{1}{\Delta_z} \begin{bmatrix} z_{22} & -z_{21} \\ -z_{12} & z_{11} \end{bmatrix}$$

So $y_{11} = \frac{z_{22}}{\Delta_z}$ $y_{12} = \frac{-z_{21}}{\Delta_z}$ $y_{21} = \frac{-z_{12}}{\Delta_z}$ $y_{22} = \frac{z_{11}}{\Delta_z}$

$$\Delta_z = z_{11} \times z_{22} - z_{12}z_{21}$$

Q Calculate the parameters By given set of parameter which are not inverse of it.

Given [h] To find :- [z]

$h_{11} = 2 \quad h_{12} = -1 \quad h_{21} = 3 \quad h_{22} = +2$

Solution:-

h. para.
$$V_1 = h_{11} I_1 + h_{12} V_2 \quad ; \quad V_1 = 2 I_1 - V_2$$
$$I_2 = h_{21} I_1 + h_{22} V_2 \quad ; \quad I_2 = 3 I_1 + 2 V_2$$

Substitute.

$$\Rightarrow V_2 = \frac{I_2 - 3I_1}{2} \quad \text{--- (1)}$$

z. param.
$$V_1 = z_{11} I_1 + z_{12} I_2$$
$$V_2 = z_{21} I_1 + z_{22} I_2$$

$$V_1 = \underbrace{\left(\frac{7}{2}\right)}_{z_{11}} I_1 + \underbrace{\left(\frac{-1}{2}\right)}_{z_{12}} I_2$$

from. eq^n (1) $$V_2 = \underbrace{\left(\frac{-3}{2}\right)}_{z_{21}} I_1 + \underbrace{\frac{1}{2}}_{z_{22}} I_2$$

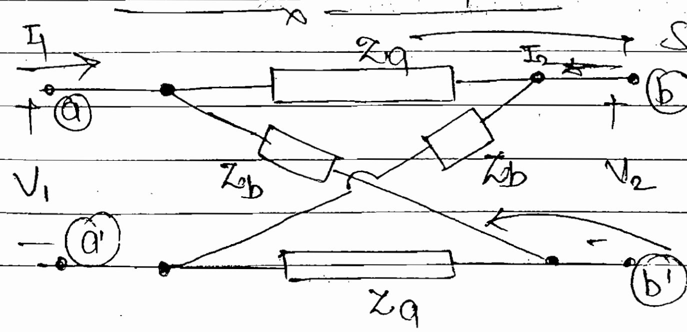
$$[Z] = \begin{bmatrix} 7/2 & -1/2 \\ -3/2 & 1/2 \end{bmatrix} \; ; \Omega$$

Compare $[Z_{11}, Z_{22}]$, $Z_{11} \neq Z_{22}$, $\frac{7}{2} \neq \frac{1}{2}$ ... not symmetrical

Compare $[z_{12}, z_{21}]$, $z_{12} \neq z_{21}$, $-\frac{3}{2} \neq \frac{1}{2}$ ... not reciprocal.

$\Longrightarrow$ By checking the symmetrical and reciprocal property from given set parameter — It is equal to calculated set of parameter [No need to solve for it like this]

Ex: LATTICE N/W



Straight arm Impedance. (always symm.)

Cross-arm Impedance (always reciprocal)

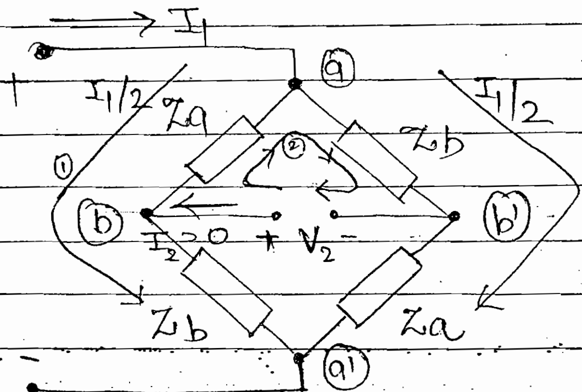
To find :- 1) [Z]   2. Z_a, Z_b in terms of [Z]

Solution: $I_2 = 0$ $Z_{11} = \frac{V_1}{I_1}$; $Z_{21} = \frac{V_2}{I_1}$

Imp. point:

1. A lattice Network is said to be symmetrical if the Two straight arm Impedances are equal i.e.
   $Z_{11} = Z_{22}$ ...... symmetrical type N/w. (N)

2. A lattice N/W is said to be reciprocal if the Two cross arm Impedances are equal. i.e
   $Z_{12} = Z_{21}$ ...... reciprocal type N/w.



KVL equation :

(1)

$$V_1 = \frac{I_1}{2}[Z_a + Z_b] \quad \text{---(1)}$$

(2) $-V_2 - \frac{I_1}{2} Z_a + Z_b \frac{I_1}{2} = 0$

$$V_2 = \frac{I_1}{2}[Z_b - Z_a] \quad \text{---(2)}$$

from eq^n (1) $\boxed{Z_{11} = \frac{V_1}{I_1} = \frac{Z_b + Z_a}{2}}$ $= Z_{22}$

from eq^n (2) $\boxed{Z_{21} = \frac{V_2}{I_1} = \frac{[Z_b - Z_a]}{2}}$ $= Z_{12}$

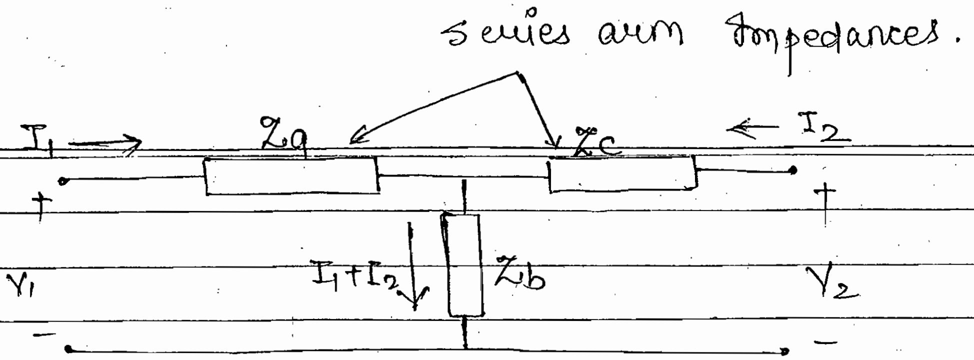
$Z_a + Z_b = 2Z_{11}$

$Z_b - Z_a = 2Z_{12}$

Add ± $\quad 2Z_b = 2Z_{11} + 2Z_{12} \Rightarrow Z_b = Z_{11} + Z_{12}$

Subtract : $\quad 2Z_a = 2Z_{11} - 2Z_{12} \Rightarrow Z_a = Z_{11} - Z_{12}$

Ex :-



To find [z] :- T OR Y OR Star N/w

Solution $V_1 = \underbrace{[Z_a + Z_b]}_{z_{11}} I_1 + \underbrace{[Z_b]}_{z_{12}} I_2$

$V_2 = \underbrace{[Z_b]}_{z_{21}} I_1 + \underbrace{[Z_c + Z_b]}_{z_{22}} I_2$

- N/w is non-symmetrical $(z_{11} \neq z_{22})$ but reciprocal $[z_{12} = z_{21}]$

Imp pt - Any general T N/w is always a reciprocal N/w but may not be symmetrical in nature. Such N/w may be symmetrical if the two series arm impedances Z_a and Z_c are equal.

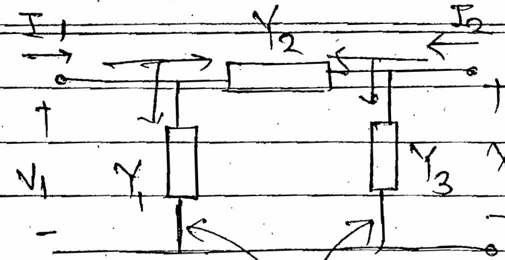
$$[Z] = \begin{bmatrix} Z_a + Z_b & Z_b \\ Z_b & Z_b + Z_c \end{bmatrix} ; \; \Omega$$

$$\Delta_z = (Z_a + Z_b)(Z_b + Z_c) - Z_b^2$$
$$= Z_a Z_b + Z_a Z_c + Z_b Z_c$$
$$= \Sigma Z_a Z_b = \Sigma Z_b Z_c = \Sigma Z_c Z_a$$

- $y_{11} = \dfrac{z_{22}}{\Delta_z} = \dfrac{Z_b + Z_c}{\Sigma Z_a Z_b}$, • $y_{22} = \dfrac{z_{11}}{\Delta_z} = \dfrac{Z_a + Z_b}{\Sigma Z_a Z_b}$

- $y_{12} = \dfrac{-z_{12}}{\Delta_z} = y_{21} = \dfrac{-Z_b}{\Sigma Z_a Z_b}$

Ex Pie π OR Δ Network



Shunt arm admittances

To find [y]

$$I_1 = V_1 Y_1 + (V_1 - V_2) Y_2$$

$$= V_1 \underbrace{(Y_1 + Y_2)}_{y_{11}} + V_2 \underbrace{(-Y_2)}_{y_{12}}$$

$$I_2 = (V_2 - V_1) Y_2 + V_2 Y_3 = V_2 \underbrace{[Y_2 + Y_3]}_{y_{22}} + \underbrace{(-Y_2)}_{y_{22}} V_1$$

$$[y] = \begin{bmatrix} Y_1 + Y_2 & -Y_2 \\ -Y_2 & Y_2 + Y_3 \end{bmatrix}$$

Imp pt

- Any general π N/W is always a reciprocal N/W But may not be symmetrical in nature. Such N/W may be symmetrical in nature if the two shunt arm admittance Y_1 & Y_3 are equal.

$$\Delta y = (Y_1 + Y_2)(Y_2 + Y_3) - Y_2^2$$

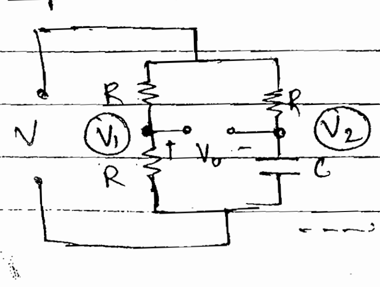
$$= Y_1 Y_2 + Y_1 Y_3 + Y_2 Y_3$$

$$= \Sigma Y_1 Y_2 = \Sigma Y_1 Y_3 = \Sigma Y_2 Y_3$$

- $z_{11} = \dfrac{y_{22}}{\Delta y} = \dfrac{Y_2 + Y_3}{\Sigma Y_1 Y_2}$ • $z_{22} = \dfrac{Y_1 + Y_2}{\Delta y_1 Y_2} = \dfrac{y_{11}}{\Delta y}$
- $z_{12} = z_{21} = -\dfrac{y_{12}}{\Delta y} = \dfrac{Y_2}{\Sigma Y_1 Y_2}$

Chapter-5

Q.4



$$V_0 = V_1 - V_2$$

$$V_1 = V \cdot \frac{R}{R+R} = 10 \cdot \frac{\cancel{R}}{2\cancel{R}} = 5$$

All pass filter

$$V_2 = 10 \times \frac{1/j\omega C}{R + \frac{1}{j\omega C}}$$

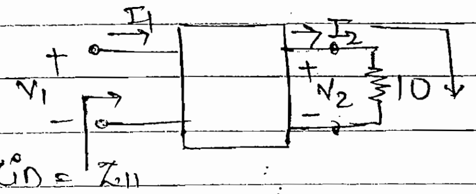
$$V_0 = 5 - \frac{1}{1+jR\omega C} \times 10 \times \frac{\cancel{(1+j\omega RC)}}{\cancel{(1-j\omega RC)}} = 5 - \frac{10}{1+j\omega RC}$$

$$= \cancel{50} - \frac{\cancel{10}\ \cancel{(1-j\omega RC)}}{\cancel{1+R^2\omega^2C^2}}$$

$$= \frac{5+5j\omega RC-10}{(1+j\omega RC)} = \frac{5\,[-1+j\omega RC)}{(1+j\omega RC)}$$

$|V_0| = 5\ V$ Ans.

Q.5



$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$$

$Z_{in} = Z_{11}$

Solution $V_1 = A V_2 + B I_2 = V_2 + 2I_2$ —(1)

$I_1 = C V_2 + D I_2 = V_2 + 3I_2$ —(2)

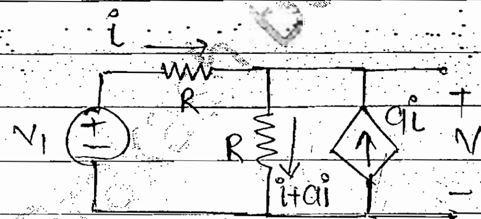
$V_2 = 10 I_2$ —(3)

Put eqn (3) in (1) & (2) $\quad V_1 = 12 I_2$

$I_1 = 13 I_2$

$$Z_{in} = \frac{V_1}{I_1} = \frac{12}{13}\ \Omega \quad \text{Ans.}$$

Q.10



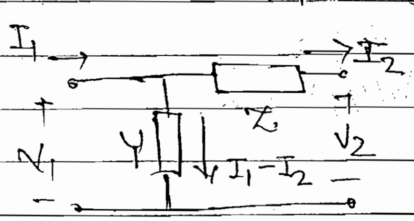
$V_1 = iR + R[i + ai]$

$= iR + iR[1+a]$ —(1)

$V_2 = [1+a]\,iR$ —(2)

$$\frac{V_2}{V_1} = \frac{[1+a]\,i\cancel{R}}{i\cancel{R}\,[2+a]} = \frac{1+a}{2+a} \quad \text{Ans.}$$

Q.21



$I_1 - I_2 = V_1 Y$ —(1)

$V_1 = I_2 Z + V_2$ —(2)

eqn (2) $\quad V_1 = \underbrace{[1]}_{A} V_2 + \underbrace{[Z]}_{B} I_2$

eqn (1). $\quad I_1 - I_2 = [I_2 Z + V_2]\,Y, \quad I_1 = V_2 \underbrace{[Y]}_{C} + I_2 \underbrace{[ZY+1]}_{D}$

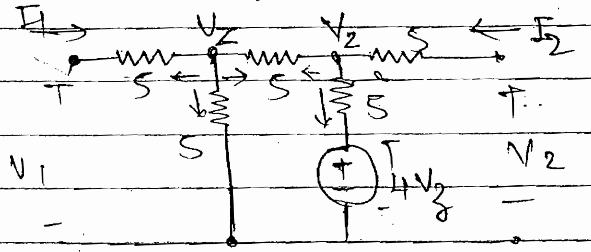
$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & Z \\ y & 1+yZ \end{bmatrix}$$ A

$A \neq D$, ~~$B \neq C$~~ , Neither symm, ~~Nor~~ &

$AD - BC = 1 = 1(1+yz) - yz = 1$ (reciprocal N/w)

Q.29 $z_{11} = \frac{V_1}{I_1}$, $I_2 = 0$, $z_{11} = \frac{V_1}{I_1}$

Both z_{11} and z_{11} are having equal value bcoz load terminal are open



Node V_z

$$\frac{V_z - V_1}{5} + \frac{V_z}{5} + \frac{V_z - V_2}{5} = 0$$

$$3V_z = \cancel{20V_2}\ V_1 + V_2 \quad \text{— (1)}$$

Node V_2

$$\frac{V_2 - V_z}{5} + \frac{V_2 - 4V_z}{5} = 0$$

$$\Rightarrow 2V_2 - 5V_z = 0 \quad \text{— (2)} \qquad V_2 = \frac{5V_2}{2}$$

I/p side $\Rightarrow \frac{V_1 - V_z}{5} = I_1$ — (3)

$$3V_z = V_1 + \frac{5V_z}{2} \Rightarrow V_1 = \frac{3V_2 - 5V_2}{2} = \frac{V_2}{2}$$

Put in eqn (3) $\frac{\frac{V_2}{2} - V_2}{5} = I_1 \Rightarrow \frac{-V_2}{10} = I_1$

$$I_1 = -\frac{2V_1}{10}, \quad \frac{V_1}{I_1} = \frac{10}{-2} = -5 \quad \text{Ans}$$

Q.31 $$Z_{11} = \frac{SL \times (R_1+R_2)}{SL + (R_1+R_2)} = \frac{L(R_1+R_2)}{L} \frac{S}{S+a}$$

$$a = \frac{R_1+R_2}{L}$$

Coefficient of S must Be 1.

6

$$Z_{22} = \frac{R_2 \times (R_1 + sL)}{R_2 + (R_1 + sL)} = \frac{s + R_1/L}{s + \frac{R_1 + R_2}{L}} \cdot \frac{R_2 L}{L} = R_2 \left(\frac{s+b}{s+a} \right)$$

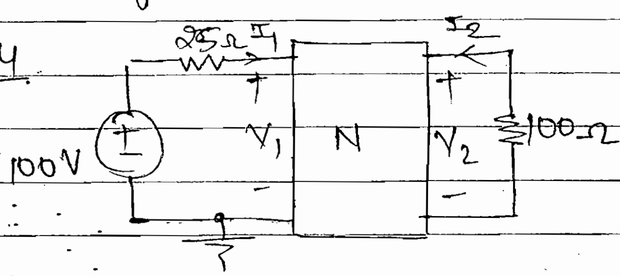
Origin → near should Be zero } so $s+2$ → zero
→ away should Be pole } so $s+5$ → pole

- How the S.C and O.C are related to pole and zero in o/p port.

ex: $Z_{22} = \frac{V_2}{I_2}$

If $V_2 = 0$ $Z_{22} = 0$ (zero) That mean o/p port is S.C
If $I_2 = 0$ $Z_{22} = \infty$ (pole) — " — o/p port is O.C.

Q.34



To find $G_{21} = \frac{V_2}{V_1}$

Solution: $I_1 = 0.1\ V_1 - 0.01\ V_2$ — (1)

$I_2 = 0.01 V_1 + 0.1 \cdot V_2$ — (2)

$V_2 = -100\ I_2$ — (3)

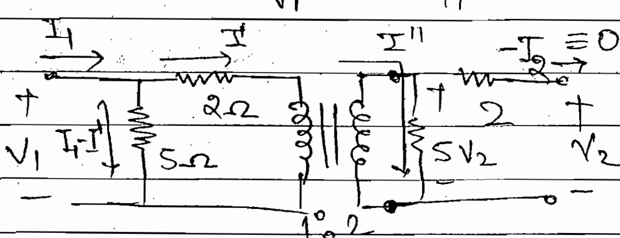
$V_1 = 25\ I_1 + V$ — (4)

eq^N(2) $V_2 = -100\ I_2 = -100 [0.01 V_1 + 0.1 V_2]$
$= -1 V_1 - 10 V_2$

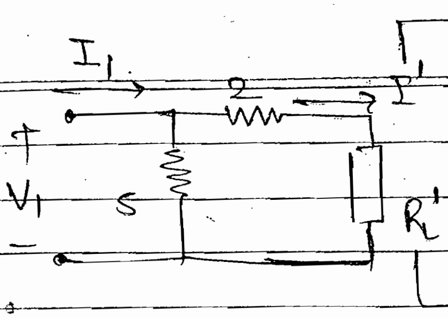
$11 V_2 = -1 V_1$ $\therefore$ $G_{21} = \frac{V_2}{V_1} = \frac{-1}{11}$ Ans

Q.36. $V_1 = A V_2 + B I_2$

$A = \frac{V_1}{V_2} \Big|_{I_2 = 0}$



Transferring secondary to primary side



$\rightarrow \left(\frac{n_2}{n_1}\right) I'' = \frac{2}{1} \times \frac{V_2}{5} = \frac{2V_2}{5}$

$V' = \left(\frac{n_1}{n_2}\right) \times V_2 = \left(\frac{1}{2}\right) \times V_2 = \frac{V_2}{2}$

$\rightarrow \left(\frac{n_1}{n_2}\right)^2 R_L = \left(\frac{1}{2}\right)^2 \times 5 = \frac{5}{4}\,\Omega$

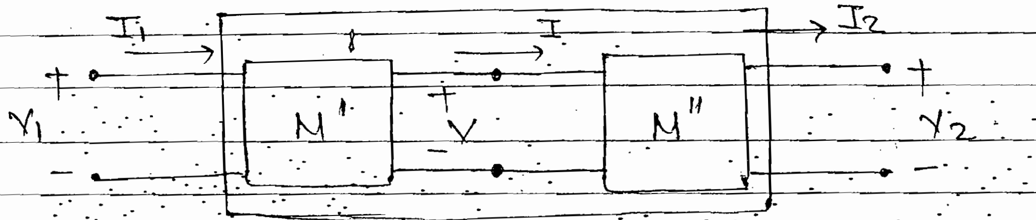
$V_1 = 2I' + V'$

$= 2 \times \frac{2}{5} V_2 + \frac{V_2}{2}$; $\frac{V_1}{V_2} = \frac{13}{10}$ A Ans

Imp

- Inter-Connection of Various 2-port Network :-

- case 1 : Cascaded Networks



N' : $\begin{cases} V_1 = A'V + B'I \\ I_1 = C'V + D'I \end{cases}$ N'' : $\begin{cases} V = A''V_2 + B''I_2 \\ I = C''V_2 + D''I_2 \end{cases}$

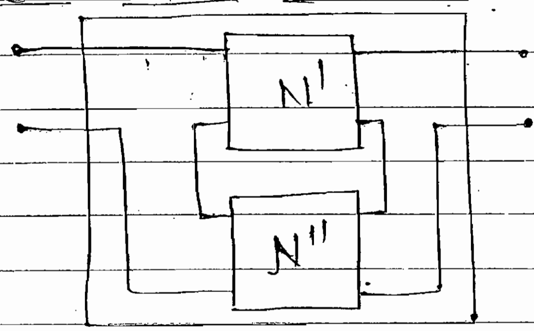
Overall N/w :- $V_1 = A V_2 + B I_2$
$I_1 = C V_2 + D I_2$

Substitute the V, I from N'' into V, I in N' we get the V_1, I_1 in terms of V_2, I_2.

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{bmatrix} A'' & B'' \\ C'' & D'' \end{bmatrix}$$

NOTE: When two N/w are connected in cascade than the ABCD parameters of the individual N/w are multipled to obtain overall ABCD parameters of the resulting N/w.

- Case 2: Series Network:

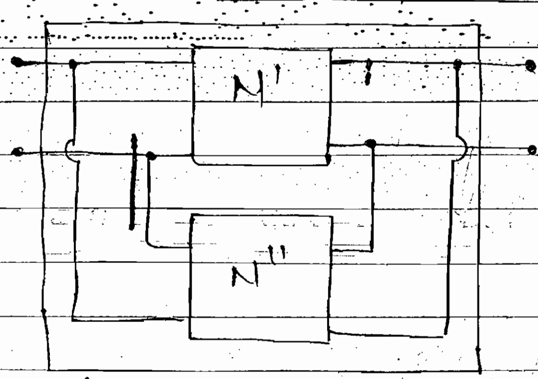


NOTE:- When two N/w. are connected in series at the I/p and in series at O/p Then the respective Z parameter of the two N/w are added to obtain overall Z parameters of the resultant N/w.

$$\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} = \begin{bmatrix} z'_{11} + z''_{11} & z'_{12} + z''_{12} \\ z'_{21} + z''_{21} & z'_{22} + z''_{22} \end{bmatrix}$$

$$\Rightarrow \quad [z] = [z'] + [z'']$$

- Case 3: Parallel Network:



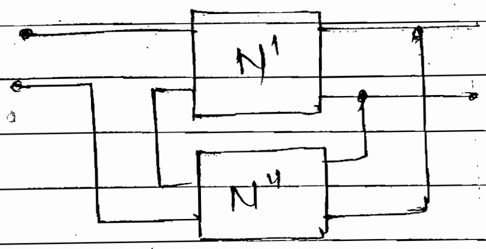
$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \begin{bmatrix} y'_{11} + y''_{11} & y'_{12} + y''_{12} \\ y'_{21} + y''_{21} & y'_{22} + y''_{22} \end{bmatrix}$$

$$[y] = [y'] + [y'']$$

NOTE:- When Two N/w are connected in parallel at the I/p and in parallel at the O/p. Then respective Y parameter of the two N/w are **added** to obtain overall Y parameter of the resulting N/w.

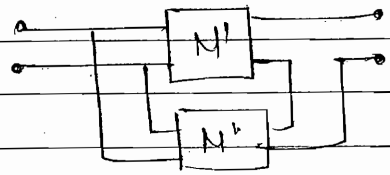
Numerically :- ABCD and Y parameter are most Imp.

case 4:- Series-Parallel Network



$$[h] = [h'] + [h'']$$

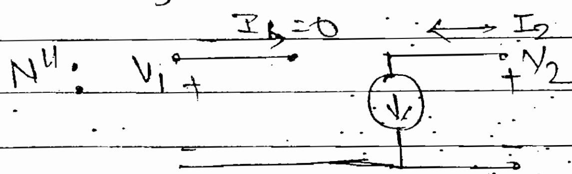
case 5:- Parallel-Series Networks



$$[g] = [g'] + [g'']$$

Q.14 N': $y_{11} = y_{22} = -y_{12} = -y_{21} = Y$

N'':



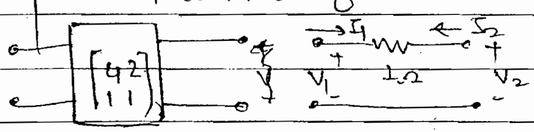
$V_2 = 0$	$V_1 = 0$
$y_{11} = \frac{\cancel{I_1}}{V_1} = 0$	$y_{12} = \frac{\cancel{I_1}}{V_2} = 0$
$y_{21} = \frac{I_2}{\cancel{V_1}}$	$y_{22} = \frac{I_2}{V_2}$
$= g\frac{\cancel{V_1}}{\cancel{V_1}}$	$= \frac{g\cancel{V_1}}{V_2} = 0$
$= g$	

N/w are connected in parallel

Overall parameter $y_{21} = y_{21}' + y_{21}''$

$$y_{21} = -Y + g$$

Q. parallel Connection



$I_1 = -I_2$ —— (1)

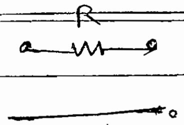
$V_1 = 1 \times I_1 + V_2$

$V_1 = I_1 + V_2$ —— (2)

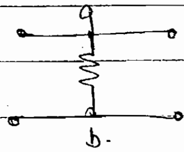
$$\Rightarrow I_1 = [1]V_1 + [-1]V_2 \; ; \; I_2 = [-1]V_1 + [1]V_2$$

$$y = y' + y'' = \begin{bmatrix} 4 & 2 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 0 & 2 \end{bmatrix} \text{ Ans}$$

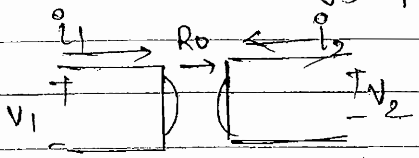
Whenever R is connected b/w I/p & o/p terminal. all parameters get modified

Ex. ①  Same previous Question. | find [Z]

As $\Delta_Z = 0$; Z parameter donot exist.

②  [y] ; [Z] when O/p port is S.C., a, b also get S.C. so $\Delta_y = 0$

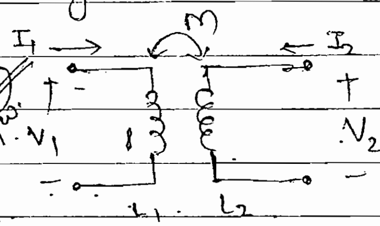
∴ [y] parameter donot exist.

③ Such N/w 

$V_1 = R_0 i_2 = \overset{z_{11}}{(0)} i_1 + \overset{z_{12}}{R_0} i_2$

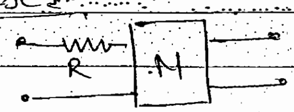
$V_2 = -R_0 i_1 = \underset{z_{21}}{-R_0} i_1 + \underset{z_{22}}{(0)} i_2$

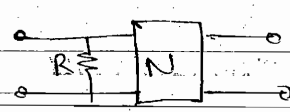
- Symmetrical N/w. $\Delta_Z = R_0^2$; $[Z] = \begin{bmatrix} 0 & R_0 \\ -R_0 & 0 \end{bmatrix}$

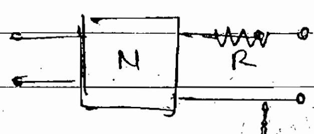
④ Such N/w. 

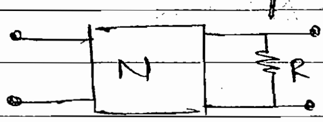
$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \underbrace{\begin{bmatrix} j\omega L_1 & \pm j\omega M \\ \pm j\omega M & j\omega L_2 \end{bmatrix}}_{[Z]} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

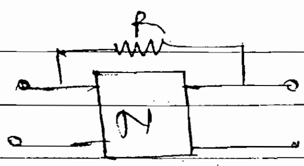
Assume Sinusoidal excitatⁿ. N/w is reciprocal. But NO. Symm.

Q.39 Imp case I  Parameter affected Z_{11} only

case 2  y_{11} "

Case 3  Z_{22} "

Case 4  y_{22} "

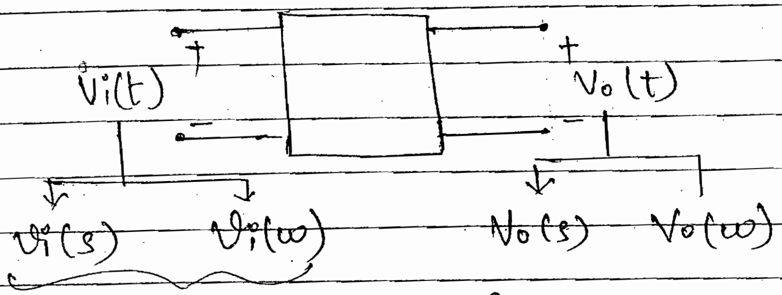
Case 5  $y_{11}, y_{21}, y_{12}, y_{22}$ all. (as connected in parallel so admittance)

Filters

5/11/2014.

freq. Selective Networks

- Amplitude → fixed, variable → frequency.

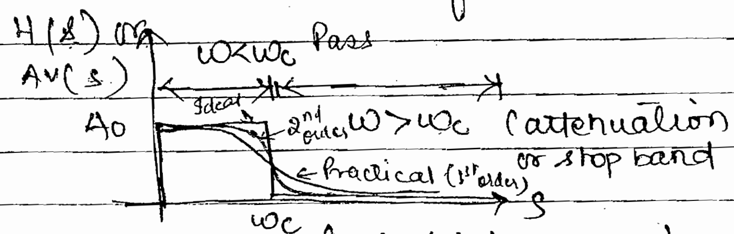


$H(s) = \frac{V_o(s)}{V_i(s)} = A_v(s)$

$H(\omega) = \frac{V_o(\omega)}{V_i(\omega)} = A_v(\omega)$

fixed amplitude & variable freq.

- case 1 : Low Pass filter (LPF)



(cut off frequency).
(depends upon RC)

- capacitor means → determine the order. $\frac{1}{Cs}$
- one capacitor → 1st order (Blue colour)
- two cap. → 2nd order (Red colour).

↑cap. or ↑order, To achieve Ideal charact.

- gain Av(s) can be adjust By using op-amp.

⇒ Characteristic of a Practical filter

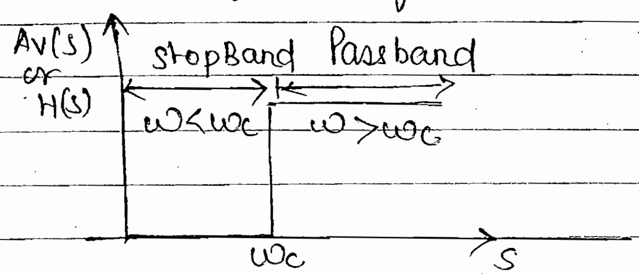
(1) The cut off frequency of any filter canbe adjusted as per our requirement by suitably selecting the numerical values of R and C components.

(2) Any ideal filter has sharp cut off characteristic. A practical filter cannot have such sharp cut off characteristic since the freq. response of such filter is limited by its time constant. Therefore any practical filter has gradual cut off characteristic. These cut off characteristic may be approached to

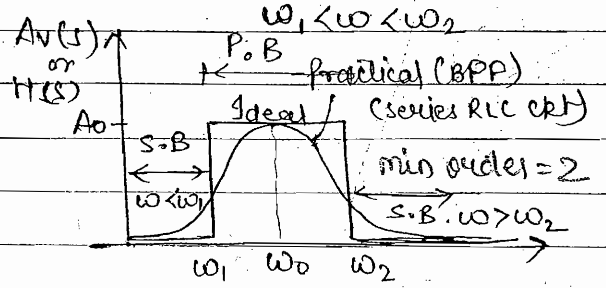
those of ideal filter By increasing the order of the filter. By increasing the no. of stages which are being cascaded.

(3) The gain during a pass band can be adjusted by including op-amp along with a RC selection. Such filter are then called <u>active filter</u>

<u>Case 2:</u> High pass filter (HPF).



<u>case 3:</u> Band Pass Filter (BPF)



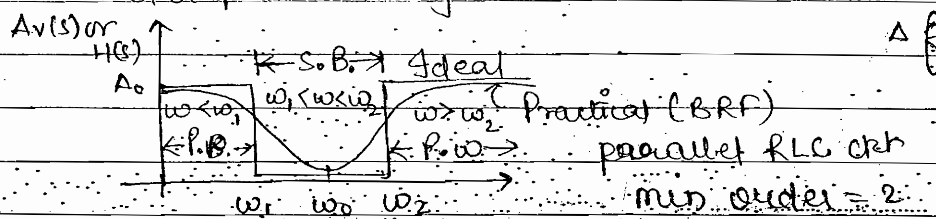
B.W. = $\Delta\omega = \omega_2 - \omega_1$

B.W = $\Delta\omega = R/L$ rad/sec

$\Delta f = R/2\pi L$ Hz.

<u>case 4:-</u> BSF / BRF

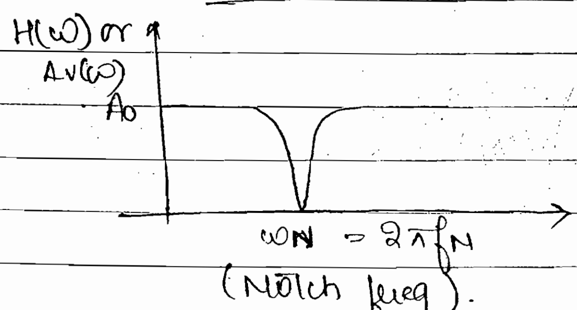
Band Stop / Band Rejection Filter.



B.W. $\Delta\omega = \omega_2 - \omega_1 = 1/RC$

$\Delta f = 1/2\pi RC$

<u>case 5:-</u> <u>Notch Filter:</u> (special case of BRF)



→ only single frequency is rejected.

→ used in I/P of audio signal.

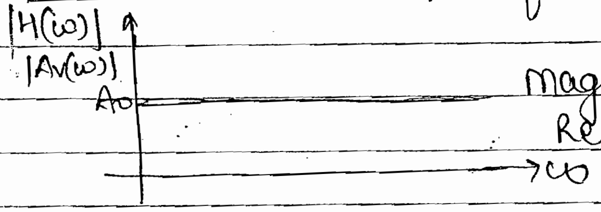
→ for Blocking 3rd harmonic.

✓ A notch filter is a further special case of a Band stop filter where a single freq. called the notch freq is not allow to enter the system.

✓ Such filter is used at the I/p of any audio system, where 50 Hz main freq is not allowed to enter in the system to minimize the noise or distortion in the system.

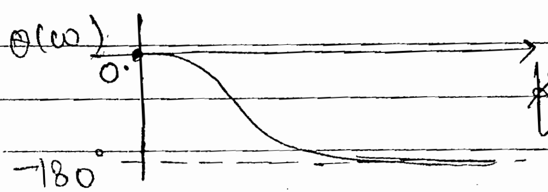
✓ Such filter is also used at the I/p of any system, to minimize 3rd harmonic distortion.

case 6 : APF (all pass filter)



Magnitude Response

• If having RHS zero → surely → APF



~~frequency~~ phase Response.

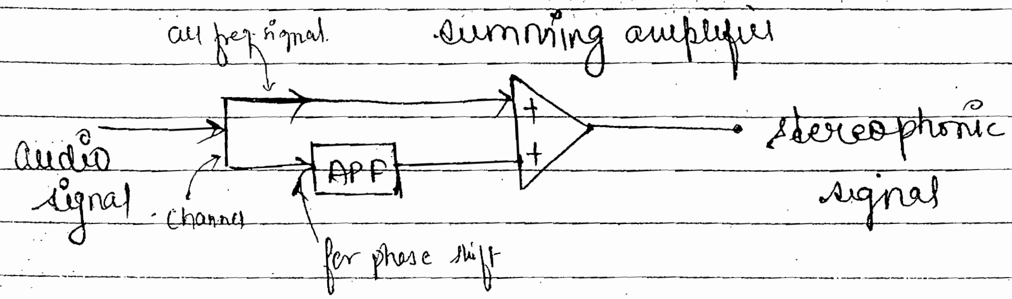
✓ Such filter passes all the freq. components with constant gain. But various freq. components are subjected to different amount of phase shift having a minimum value to 0° and maximum value of -180°.

✓ Such filter has all the zero in RHP and all the poles in LHP.

✓ Since any N/w the no. of zero and poles are equal. Then the no. of zeros in RHP or no. of poles in LHP control the order of filter.

✓

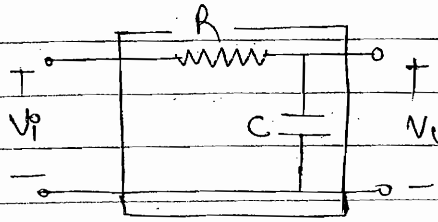
summing amplifier



- An APF can be used to provide a stereophonic music signal.

Ex:- 1st Order LPF :-

(By adjusting the value of RC (Time constant) same ckt can use as LPF and as integrator)



To find:
(1) $H(s) = T.F. = V_o(s)/V_i(s)$
(2) Cut off freq. ω_c
(3) Zero-pole pattern.
(4) freq. Resp. H(s) Vs S
(5) Amplitude Response. $|H(\omega)|$ Vs ω.

shape wise no much diff. B/w Amplitude & freq. Response.

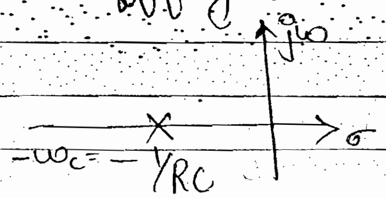
(6) Identification of filter.

Solution (1) $\frac{V_o(s)}{V_i(s)} = H(s) = \frac{1/Cs}{R + 1/Cs} = \frac{1}{(1+sRC)} = \frac{1}{\left(1+\frac{s}{\omega_c}\right)} \equiv H(s)$

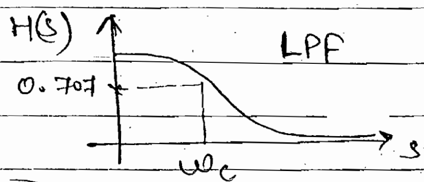
(2) $\omega_c = \frac{1}{RC}$; $f_c = \frac{1}{2\pi RC}$

- cutoff freq:- Where gain ↓ to value 0.707 or −3dB

(3) pole zero-pattern.



(4)



LPF ← freq. Resp.

{ Resemble to LPF so it called LPF

or also $V_o(s)/V_i(s)$ Vs S.

(5)

put $s = j\omega$; $H(j\omega) = \frac{1}{1 + \frac{j\omega}{RC}}$; $|H(\omega)| = \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_c}\right)^2}}$; $H(\omega)$ Vs ω

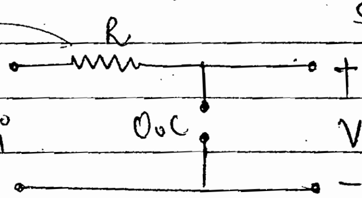
1. $\omega \ll \omega_c$; $|H(\omega)| \cong 1$
2. $\omega = \omega_c$; $|H(\omega)| = \frac{1}{\sqrt{2}} \cong 0.707$
3. $\omega \gg \omega_c$; $|H(\omega)| \cong \frac{\omega_c}{\omega}$

} method of sketching graph.

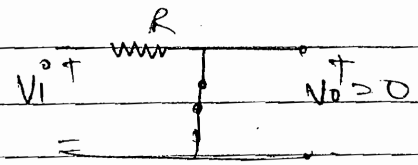
(6) Identification :-

case 1 $s = 0$; $Z = \frac{1}{sC} = O.C$

≃ No voltage drop



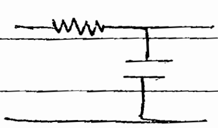
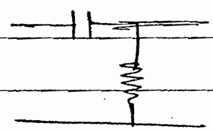
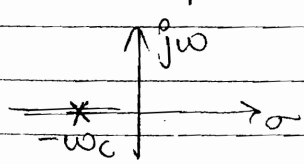
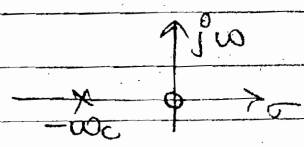
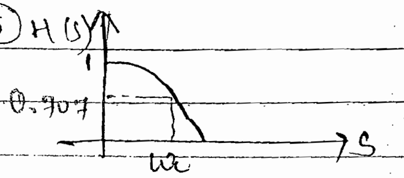
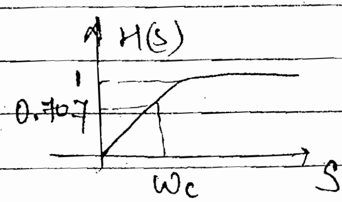
Case 2 $s = \infty$; $Z = 1/sC = S.C$



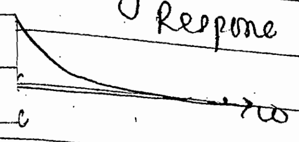
Interview

Disadvantage of this method :- Can't identify ... of sys.

Ex. 1st order HPF.

LPF	HPF	Cor...
(1)  Integrator	 differentiator	Interchange location of R & C
(2) $H(s) = \frac{1}{1+\frac{s}{\omega_c}}$	$H(s) = \frac{1}{1+\frac{1}{s/\omega_c}} = \frac{s/\omega_c}{1+s/\omega_c}$	Rep... $/\omega_c$
(3) $\omega_c = \frac{1}{RC}$	$\omega_c = \frac{1}{RC}$	
(4) 		
(5) 		

(5) magnitude Response



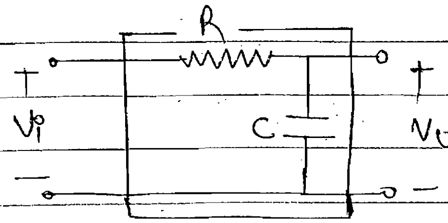
g some drop

→ s

✓ An APF can be used to provide [illegible] music signal.

Ex:- 1st Order LPF :-

(By adjusting the value of RC [illegible]) same ckt can use as LPF and as in[illegible]



To fin[illegible]

① $H(s) = V_o(s)/V_i(s)$

② Cut[illegible]

③ Zeros [illegible]

④ freq. Resp. H(S) Vs S

⑤ [illegible] $|H(\omega)|$ Vs ω.

⑥ Identificatn of filter.

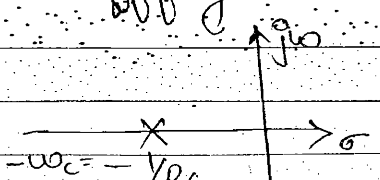
[illegible] diff. B/w Amplitude [illegible]

Solution ① $\frac{V_o(s)}{V_i(s)} = H(s) = \frac{1/CS}{R + 1/CS} = \frac{1}{\left(1 + \frac{S}{\omega_c}\right)} = H(s)$

② $\omega_c = \frac{1}{RC}$; $f_c = \frac{1}{2\pi RC}$

• cutoff freq :- Where gain is To -3dB

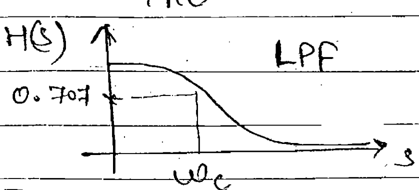
③



pole

$-\omega_c = -1/RC$

④



LPF ← freq. Resp.

LPF so it

[illegible] Vs S.

⑤

put $S = j\omega$ $\quad H(j\omega) = \frac{1}{1 + \frac{j\omega}{RC}}$

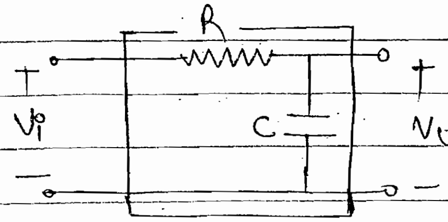
; H(ω) Vs ω

1. $\omega \ll \omega_c$; $|H(\omega)| \cong 1$
2. $\omega = \omega_c$; $|H(\omega)| = \frac{1}{\sqrt{2}}$
3. $\omega \gg \omega_c$; $|H(\omega)| \cong \frac{c}{\omega}$

✓ An APF can be used to provide a stereophonic music signal.

Ex:- 1st Order LPF :-

(By adjusting the value of RC (Time constant) same ckt can use as LPF and as integrator)



To find:

① $H(s) = T.F. = V_o(s)/V_i(s)$

② cut off freq. ω_c

③ Zero-pole pattern

④ freq. Resp. $H(s)$ Vs s

⑤ Amplitude Response. $|H(\omega)|$ Vs ω. shape wise no much diff. b/w Amplitude & freq. Response.

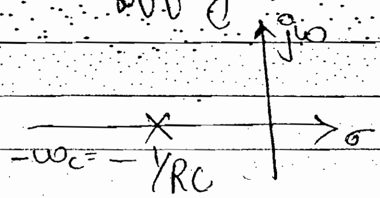
⑥ Identification of filter.

Solution ① $\dfrac{V_o(s)}{V_i(s)} = H(s) = \dfrac{1/Cs}{R + 1/Cs} = \dfrac{1}{(1+sRC)} = \dfrac{1}{\left(1+\frac{s}{\omega_c}\right)} \equiv H(s)$

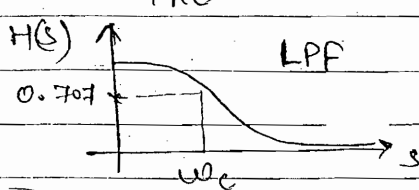
② $\omega_c = \dfrac{1}{RC}$; $f_c = \dfrac{1}{2\pi RC}$

• cut off freq:- where gain ↓ to value 0.707 or −3dB

③ pole zero-pattern.



④



← freq. Resp. {Resemble to LPF so it called "LPF"}

or also $V_o(s)/V_i(s)$ Vs S.

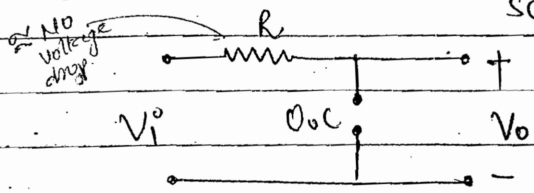
⑤ put $s = j\omega$ $H(j\omega) = \dfrac{1}{1+\frac{j\omega}{RC}}$; $|H(\omega)| = \dfrac{1}{\sqrt{1+\left(\frac{\omega}{\omega_c}\right)^2}}$; $H(\omega)$ Vs ω

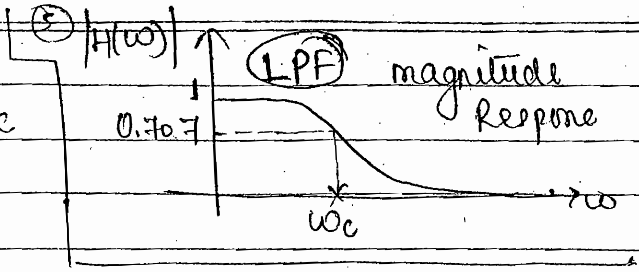
1. $\omega \ll \omega_c$; $|H(\omega)| \cong 1$
2. $\omega = \omega_c$; $|H(\omega)| = \dfrac{1}{\sqrt{2}} \cong 0.707$
3. $\omega \gg \omega_c$; $|H(\omega)| \cong \dfrac{\omega_c}{\omega}$

} method of sketching graph.

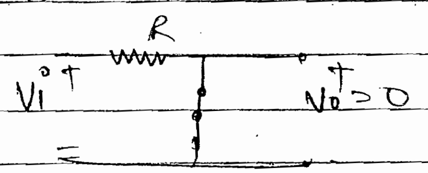
(8) Identification:

case 1 $s=0$; $Z = \frac{1}{sC} = O.C$

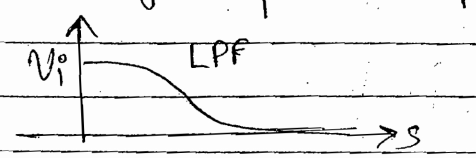


(5) 

case 2 $S = \infty$; $Z = 1/sC = S.C$



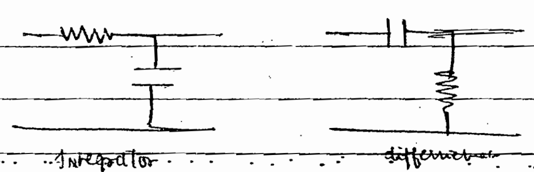
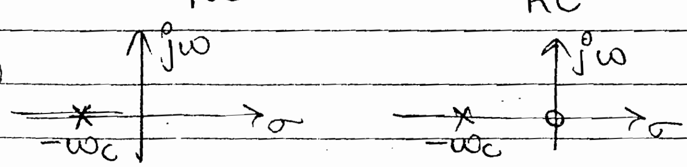
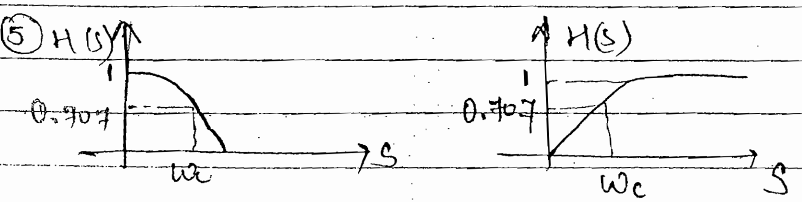
But if having some drop



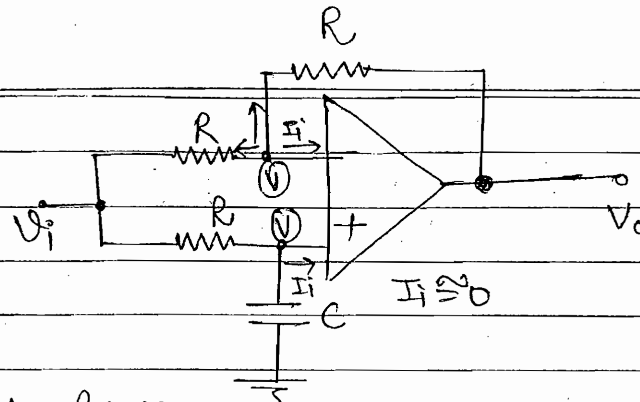
Interview

Disadvantage of this method :- Can't identify the order of sys.

Ex 1st order HPF.

	LPF	HPF	Comment
(1)	 Integrator	Differentiator	Interchange the location of R & C components.
(2)	$H(s) = \frac{1}{1+\frac{s}{\omega_c}}$	$H(s) = \frac{1}{1+\frac{1}{s/\omega_c}} = \frac{s/\omega_c}{1+s/\omega_c}$	Replace $\frac{s}{\omega_c}$ By $\frac{1}{s/\omega_c}$
(3)	$\omega_c = \frac{1}{RC}$	$\omega_c = \frac{1}{RC}$	cut off frequency.
(4)			Poles - Zero plot
(5)			freq. Response

Ex: 1st order APF



To find:

(1) T.F. $\Rightarrow H(s) = \dfrac{V_o(s)}{V_i(s)}$

(2) Pole-zero pattern.

(3) $|H(\omega)|$ Vs ω .. magnitude Response

$\theta(\omega)$ Vs ω phase response.

→ O/p of op-amp is arbitrary → So never write the KCL at Vo(o/p)

(1) The voltages at I/p. terminals of the op-amp are always equal since any ideal op amp. has infinite voltage

(2) The I/p current drawn by the op-amp is zero since any ideal op-amp has ∞ I/p Impedance.

(3) Always write KCL eqn at the I/p of the op-amp.

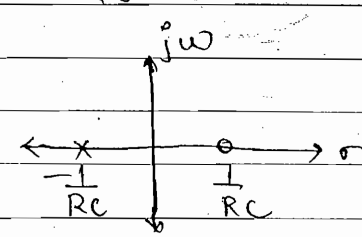
(4) Never write KCL eqn at the O/p of op-amp unless it is terminated by load impedance. Under No load condtn the O/p current of the op-amp is arbitrary and is unknown

KCL-1 $\quad \dfrac{V-V_i}{R} + \dfrac{V-V_0}{R} + 0 = 0 \quad \therefore 2V = V_i + V_0$ —— (1)

KCL-2 $\quad \dfrac{V-V_i}{R} + \dfrac{V-0}{1/sC} + 0 = 0 \qquad V = \dfrac{V_i}{1+sCR}$ —— (2)

$$\Rightarrow \frac{2V_i}{1+sCR} = V_i + V_0 \Rightarrow V_i\left(\frac{2}{1+sCR} - 1\right) = V_0$$

$$\Rightarrow \frac{V_0}{V_i} = \frac{1-sRC}{1+sRC} = H(s) = \text{T.F.}$$

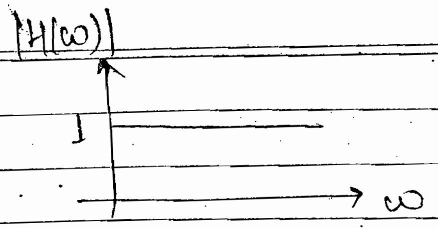


pole-zero pattern

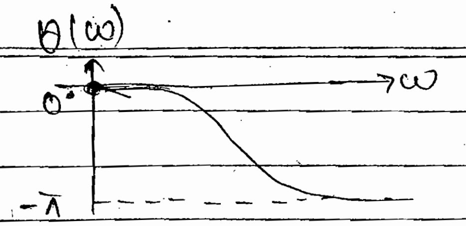
$s = j\omega.$

$$H(j\omega) = \frac{1-j\omega RC}{1+j\omega RC}$$

$|H(\omega)| = 1 \qquad \theta(\omega) = -2\tan^{-1}\dfrac{\omega RC}{1}$



Amplitude response

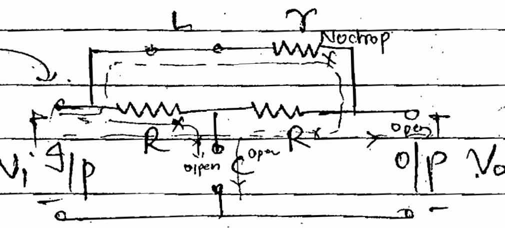


Phase response.

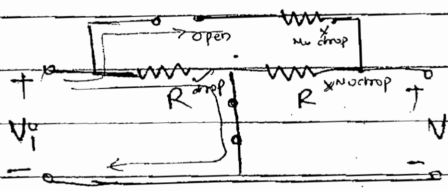
Chapter 8

Q.2 $S=0$; $L \Rightarrow$ S.C , $C \Rightarrow$ O.C

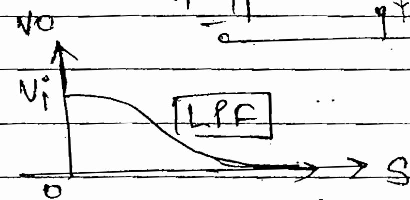
$S = \infty$; $L \Rightarrow$ O.C , $C \Rightarrow \infty$



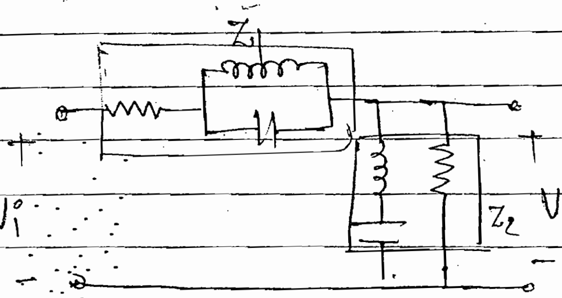
So $V_o = V_i$



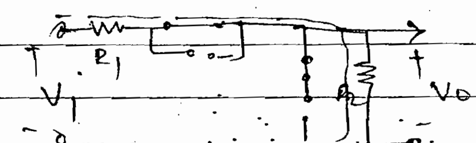
$V_o = 0$



Q.3



1. $S=0$, $L \Rightarrow$ S.C $C \Rightarrow$ O.C



$V_o = \frac{R_2 . V_1}{R_2 + R_1}$ finite

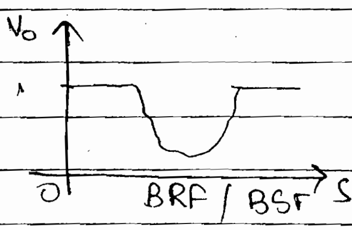
2. $S = \infty$; $L =$ O.C , $C =$ S.C

ckt is again same as case 1.

$V_o = \frac{R_2}{R_1+R_2} V_i$ ---- finite.

3. $0 < S < \infty$

$\downarrow V_o = \frac{Z_2 \downarrow V_i}{\uparrow Z_1 + Z_2 \downarrow}$



When R is in series Total impe $\uparrow$ f

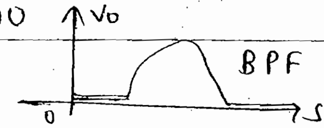
R is in parallel Total impe $\downarrow$.

Q.6 $H(S) = \frac{10S}{S^2 + 10S + 100}$

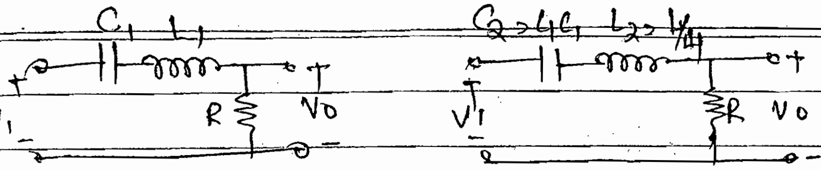
at $S=0$ $= \frac{0}{0+0+100}$

$= 0$

at $S = \infty$ $= \frac{10/S}{1 + \frac{10}{S} + \frac{100}{S^2}} = 0$



Q.9

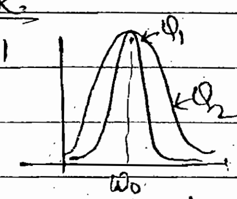


ω_0 = Remain same.

$\Delta\omega_1 = B\omega_1 = B_1$ $\qquad$ $\Delta\omega_2 = B\omega_2 = B_2$

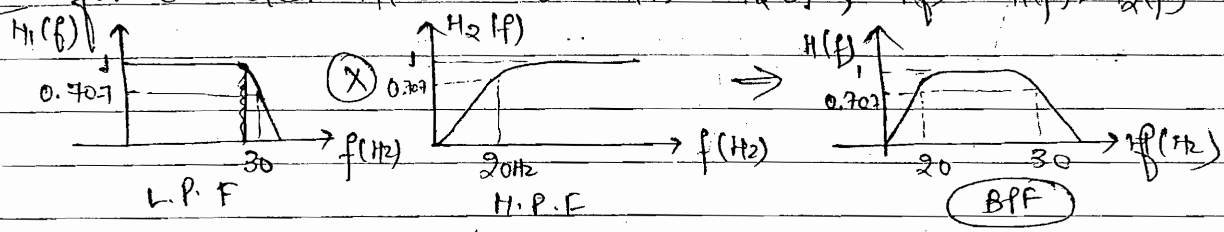
Series $\quad \Delta\omega_1 = \frac{R}{L_1}$ $\qquad$ $\Delta\omega_2 = \frac{R}{L_1/4} = \frac{4R}{L_1}$

$$B_1/B_2 = \frac{R/L_1}{4R/L_1} = \frac{1}{4}$$



- $B_1 < B_2 \Rightarrow Q_1 > Q_2$; ckt (1) Behaves as Better tuned ckt.

Q.10 for cascaded N/w. $H(s) = H_1(s) \cdot H_2(s)$; $H(f) = H_1(f) \cdot H_2(f)$



$\Delta f = 30 - 20 = 10\,Hz.$

$f_0 = \sqrt{f_1 f_2} = \sqrt{20 \times 30} = 10\sqrt{6}\ Hz$

$Q_0 = \frac{f_0}{\Delta f} = \frac{10\sqrt{6}}{10} = \sqrt{6}.$

5/11/2014

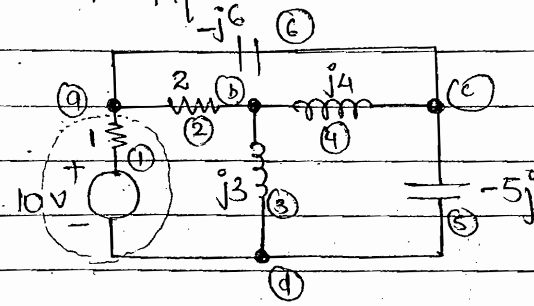
Network Topology :

.... Graph Theory

→ Another method to solve any elect. N/w.

Ex Given :- Any elect. N/w

• To find :- 1. Graph of N/w.
2. Oriented Graph
3. Tree of graph
4. Co-tree



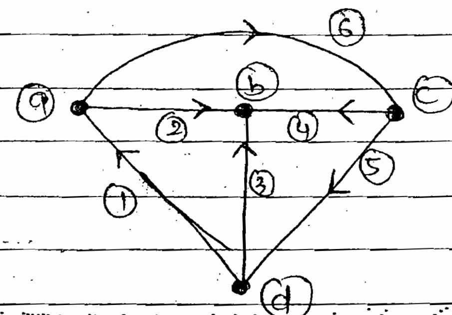
No. of Nodes = n = ~~3~~ 4

No. of Branches b = 6

Graph of Network : Graph : n : [a, b, c, d]
b : [1, 2 ..... 6]

• For fully connected graph :-

$$\text{No. of Branches} = {}^nC_2 = \frac{n(n-1)}{2}$$



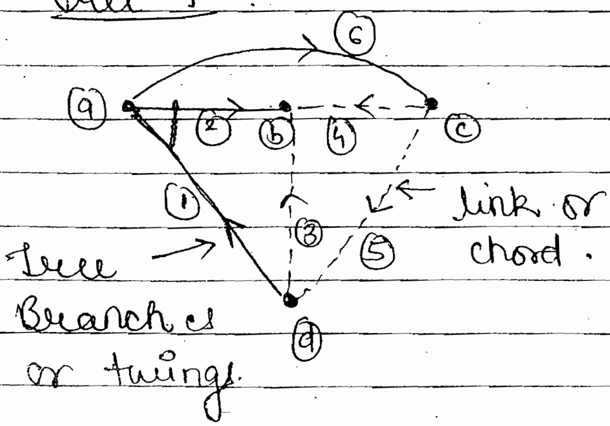
Oriented graph / fully connected graph

$V_2 = 2.i_2$, $V_3 = j3 i_3$

$V_3 = j3\ i_3$, $V_4 = j4\ i4$

$V_5 = -j5\ i5$, $V_6 = -j6\ i_6$

Tree-1 :-



Tree : [1, 2, 6] Tree Branches or twings

Co-Tree : [3, 4, 5] links / Chord.

No. of Branches connected to a node → is degree of Node

→ fully connected graph = degree of each Node is same.

- No of Branches (Tree) = $n-1$ = no. of KCL eq^n
- No. of links or chords = $b-(n-1)$ = no. of KVL eq^n.

- Degree of Node = $n-1$

  $r \equiv$ Rank of graph = $n-1$

- Co-tree → each node not having same degree - then highest degree of Node ⇒ define Rank of Tree.

$\boxed{\text{No. of Trees} = n^{(n-2)}}$ ONLY FOR FULLY CONNECTED GRAPH.

$\boxed{\text{No. of Tree} = \det[AA^T]}$ FOR GENERAL GRAPH [NOT F.C. OR F.C]

where [A] = reduced incidence Matrix

- The N/w Topology is concerned the manner in which various elements are interconnected b/w various Nodes irrespective of their values and types.
- Basic Tree of a graph:
(1) The tree contains all the nodes of the graph
(2) Their is no close path and hence an tree is circuit less.
(3) The tree of a graph is not unique.
(4) If the graph contains n nodes. Then its a tree contain $(n-1)$ Tree Branches.
(5) The Tree of the graph can be utilize to solve the given N/w using
(i) Tie-set Matrix (ii) Cut-set Matrix.
(6) Any close loop in a given graph may not contain all the nodes of the graph.
(7) In a fully connected graph each node is connected

to every node of other graph which is with a direct Branch.

(8) The degree of any node represents the no. of Branches which are meeting at the specified Node.

(9) In a fully connected graph the degree of each node is equal and represents the rank of the graph.

(10) When the graph is not fully connected, then the degree of each node is different. But the highest degree of the node represents the rank of the graph.

Main Points:

1. No. of Branches in a fully connected graph $= {}^nC_2 = \frac{n(n-1)}{2}$

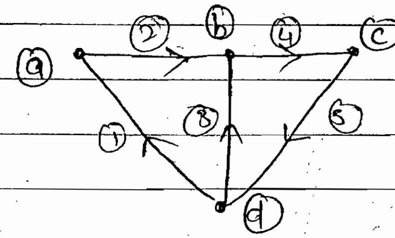
2. No. of Trees of a graph $= n^{(n-2)}$ ... ONLY for F.C graph
$= \det[AA^T]$ ... for general graph [whether FC or not]

3. No. of links / chords $= b-(n-1)$
= No. of KVL eq^n
= No. of fundamental loop
$l = b-(n-1)$ ⟸ Imp.
= no. of f- tie sets

4. No. of tree - Branches $= n-1$
= No. of KCL eq^n
= degree of each node in a F.C graph.
= Rank of graph
= Max degree of Node when graph is not F.C

= no. of nodes - pair voltages.
= no. of f- cut sets.

Ex:- Given : graph of a Network
To find : 1. Incidence Matrix
2. Reduce incidence Matrix [A]



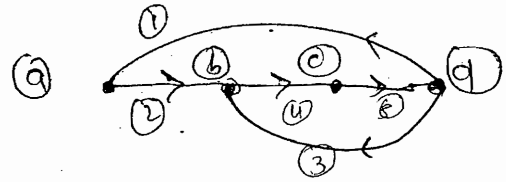
Incidence Matrix.

1. The incidence Matrix translates all the geometrical features of the graph into an algebraic expression
2. Every graph has an incidence Matrix and vice-versa
3. Each row of the matrix contains entering of +1, -1, 0
   •⟶ ⇒ +1 , •⟵ ⇒ -1 , • ⇒ 0
4. Each column of the matrix contains only one entry of +1 and only one entry of -1. so that that algebraic sum of each column is always equal to zero.
5. The determinant of the incidence matrix of any close loop is always equal to zero.
6. The size of matrix is [n x b]
   where n = No. of Nodes = No. of rows in the matrix
   b = No. of Branches = No. of column in the matrix
7. Two graphs having same incidence matrix are called isomorphic graph.

Solution:- Incidence Matrix

Nodes ↓ Branches ⟶

$$\begin{array}{cc} & \begin{array}{ccccc}(1) & (2) & (3) & (4) & (5)\end{array} \\ \begin{array}{c} a \\ \xrightarrow{\text{delete}} b \\ c \\ d \end{array} & \begin{bmatrix} -1 & +1 & 0 & 0 & 0 \\ 0 & -1 & \cancel{0}-1 & +1 & 0 \\ 0 & 0 & 0 & -1 & +1 \\ +1 & 0 & +1 & 0 & -1 \end{bmatrix} \end{array} \begin{array}{c} [n \times b] \\ [4 \times 5] \end{array}$$



⇒ Write the Matrix ⇒ row wise
⇒ Draw the graph ⇒ column wise

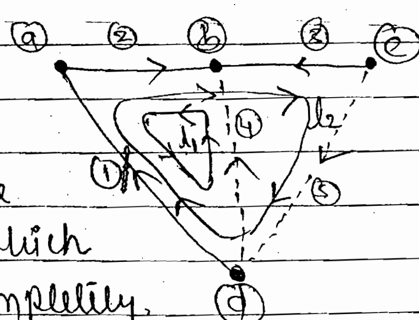
• Reduced Incidence Matrix:-
(1) A particular Node is taken as a reference Node
(2) Delete the row corresponding to the specific ref. node to obtain the reduced incidence Matrix.
(3) The size of the Matrix is $[(n-1)\times b] = [3\times 5]$
(4) This Matrix can be used to find the No. of Tree for any general graph using the result det $[AA^T]$

• Taking Ref. Node: b

Node ↓ Branches →

$$[A] = \begin{array}{c} a \\ c \\ d \end{array}\begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} -(+1) & +1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & +1 \\ +1 & 0 & +1 & 0 & -1 \end{bmatrix} \end{array} \quad [(n-1)\times b] = [3\times 5]$$

Ex: Given: Tree of a graph:
To find:
(1) f-tie set Matrix, f-loop Matrix, f Branch Matrix $[B_f]$
(2) f-cut set Matrix $[Q_f]$



• Main points of F-Tie Sets

1. Fundamental Tie Sets are the optimum of Tie sets with which a given ckt can be solved completely.
2. A fundamental T-set represents the No. of Branches required to form a close loop

(3) Each fundamental Tie set contain one and only one link or chord, remaining are the tree Branches.

(4) The No. of fundamental Tie set is

$$f\text{-Tie set} = \text{No. of links / chords}$$
$$= b-(n-1) \qquad [5-(4-1)] = 2$$

(5) The direction of fundamental Tie-set is same as the direction of link of the chord.

(6) The fundamental Tie set Matrix can be utilize to write Branch current in terms of loop current.

Solution :- f-Tie-Set :-

$l_1 : [1, 2, \underset{\text{link}}{4}]$ $\qquad$ $l_2 : [1, 2, 3, \underset{\text{link}}{5}]$

$\Rightarrow$ f-Tie-set Matrix

$$\begin{array}{c} \\ l_1 \\ l_2 \end{array}\begin{array}{c} \begin{array}{ccccc} ① & ② & ③ & ④ & ⑤ \end{array} \\ \left[\begin{array}{ccc|cc} -1 & -1 & 0 & +1 & 0 \\ +1 & +1 & -1 & 0 & +1 \end{array}\right] \end{array} = [B_f]$$

$$\underbrace{\qquad\qquad}_{B_t} \qquad \underbrace{\qquad}_{B_l \,=\, U}$$

$B_t \rightarrow$ tree
$B_l \rightarrow$ link
$U \rightarrow$ unity Matrix

$$[B_f] = [B_t \,\vdots\, U]$$

Branch Current :-

$$\boxed{i_1 = -I_1 + I_2,\ i_2 = -I_1 + I_2,\ i_3 = -I_2,\ i_4 = +I_1,\ I_5 = +I_2}$$

- Main Points of f-Cut Set :-

(1) Fundamental Cut set are optimum no. of cut set required to solve the N/W completely

(2) Each fundamental cut set represent a group of Branches which must be cut so that the graph is divided into two parts

(3) Each fundamental cut set contain one and only one tree branch, Remaining are links or chord.

(4) The direction of fundamental cut set is the same as the direction of tree branch.

(5) The no. of fundamental cut set is equal to the no. of tree Branch

f-cut set = No. of tree Branches

$= n-1$ $\quad (4-1) = 3.$

(6) The fundamental cut set Matrix can be utilize to write Branche voltage in terms of Node voltages

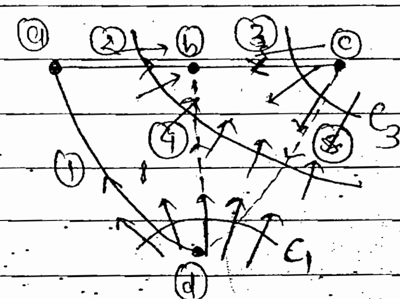
Tree Branch.

$C_1 : [\,1, 4, 5\,]$

$C_2 : [\,2, 4, 5\,]$

$C_3 : [\,3, 5\,]$

Tree Branch



f-Cut Set matrix

$$[Q_f] = \begin{array}{c|ccccc} & 1 & 2 & 3 & 4 & 5 \\ \hline C_1 & 1 & 0 & 0 & 1 & -1 \\ C_2 & 0 & 1 & 0 & 1 & -1 \\ C_3 & 0 & 0 & 1 & 0 & 1 \end{array}$$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{Q_t = U} \; \underbrace{\begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}}_{Q_l}$$

$[Q_f] = [Q_t \,\vdots\, Q_l]$

$[Q_f] = [U \,;\, Q_l]$

$$\boxed{[Q_l] = -[B_t^T] \quad , \quad [B_t] = -[Q_l^T]}$$

Branch voltage :

$$\boxed{v_1 = V_1 \;,\; v_2 = V_2 \;,\; v_3 = V_3 \;,\; v_4 = V_1 + V_2 \;,\; v_5 = -V_1 - V_2 + V_3}$$

NOTE :- (a) The fundamental cut-set Matrix corresponding to links or chord is always equal to -ve of transpose of fundament - Tie-set Matrix corresponding to tree Branches and vice versa.

- This result is always valid irrespective of the type of tree selected for a given graph.

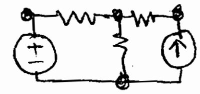
- The tree set of eq^n represent the equilibrium eq^n. Since these equation represent the given N/w. completely and uniquely.

Concluding Comment :- While designing a particular ckt and when any element value in any branch of the N/w is changed again and again Then the method of analysis using KVL & KCL technique becomes more complicated.

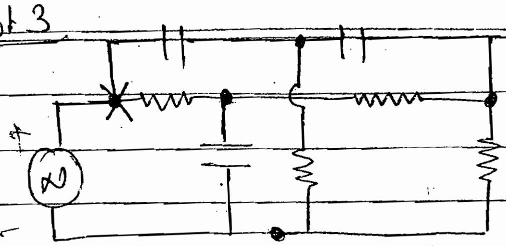
Under this condition the method of analysis using graph theory is simplify. Since

(i) Graph of the N/w
(ii) Tree of the graph.
(iii) Tie-set Matrix
(iv) Cut set Matrix

all will remain same. Only one equat^n has to be modify where the Branch voltage is represented in Terms of Branch current. All other eq^n remain unchanged.



Chapt 3

Q.6.  Twin-T N/ω.

$n = 4, b = 7$

No. of f-tie set = No. of KVL eq^n = $b-(n-1) = 4$
No. of f-cut set = No. of KCL eq^n = $n-1 = 3$

→ dependent source are not consider

Network Theorem

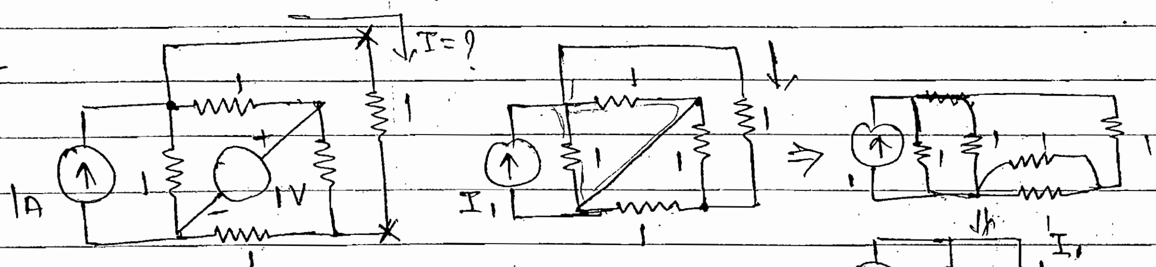
→ used to simplify elect. Networks.

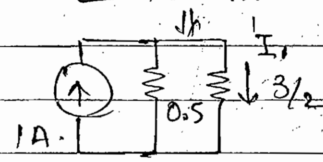
- Superposition Theorem: The response in any element of a linear bilateral RLC N/ω containing more than one independent voltages and current source is the sum of responses produce by the source each acting alone when
  (i) All other independent voltage sources are short ckt or replaced by their internal impedances.
  (ii) All other independent current sources are open ckt or replaced by their internal impedances.
  (iii) All dependent voltage and current sources remain as they are and therefore these sources are neither S.C. nor O.C.

The Theorem is not applicable for → the N/ω. containing singal independent voltage or current source
→ Non linear element
→ unilateral element ex: PN junction Diode

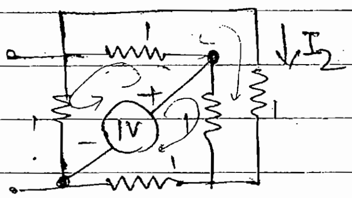
The Theorem is also not applicable to non-linear parameter such as power.

Ex:-





Solution: $I = 1 \times \dfrac{0.5}{3/2+0.5} = \dfrac{1}{4}$ A.

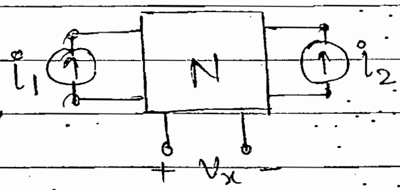


$I_2 = 0$

Showing the Balanced Bridge

$I = I_1 + I_2 = 1/4$ A.

Ex



N: linear & contains only resistors.

6/11/2014

N: linear Network & contains only resistors

If $i_1 = 8A$, $i_2 = 12A$; $V_x = 80V$.

$i_1 = -8A$, $i_2 = 4A$; $V_x = 0V$

find V_x if $i_1 = i_2 = 20A$.

Soluth: $i_2 = 0$; $V_{x1} = i_1 R_1$ $\quad$ $i_1 = 0$; $V_{x2} = i_2 R_2$

$$V_x = V_{x1} + V_{x2}$$

$$\boxed{V_x = i_1 R_1 + i_2 R_2}$$

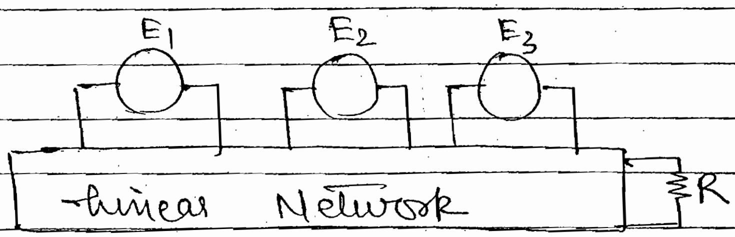
$$\left.\begin{aligned} 8R_1 + 12R_2 &= 80 \\ -8R_1 + 4R_2 &= 0 \end{aligned}\right\} \quad \begin{aligned} R_1 &= 5/2 \\ R_2 &= 5. \end{aligned}$$

$$V_x = i_1 R_1 + i_2 R_2 = 20 \times \frac{5}{2} + 20 \times 5 = 150V.$$

linear OR Resistive N/w ⇒ Superposition.

→ Once the condition is specified, there is NO change in sign (current, voltage) if the direction is reversed.

Ex :-



$P_1 = 18W$
$P_2 = 50W$
$P_3 = 98W$

When $E_1 \rightarrow$ excited, $E_2 = E_3 = 0$ power consumption By R is 18W.
Similarly for P_2, P_3.

$P_{max} \neq P_3 + P_2 + P_1$ $P_{min} \neq P_3 - P_2 - P_1$

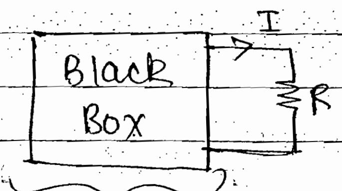
∴ Power non-linear can't use superposition Theorem.

$P = \frac{V^2}{R} = I^2R$ $V = \sqrt{P} \cdot \sqrt{R}$

$V_{max} = V_1 + V_2 + V_3$, $P_{max} = \frac{V_{max}^2}{R}$; $P_{min} = \frac{V_{min}^2}{R}$

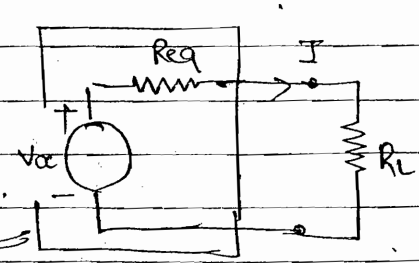
$P_{max} = (\sqrt{P_3} + \sqrt{P_2} + \sqrt{P_1})^2$, $P_{min} = (\sqrt{P_3} - \sqrt{P_2} - \sqrt{P_1})^2$
$= 450W$ $= 2W$.

Pg No. 27.
Q.22



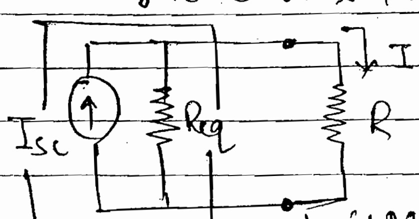
$R = 0$; $I = 3$
$R = 2$; $I = 1.5$
$R = 1$; $I = ?$

contain Resistors ~~&~~
Independent Source
Voltage Source Current Source.



$\boxed{V_{oc} = I[R_{eq} + R]}$

Thevenin's Model.



$\boxed{I = \frac{R_{eq} \cdot I_{sc}}{R_{eq} + R}}$

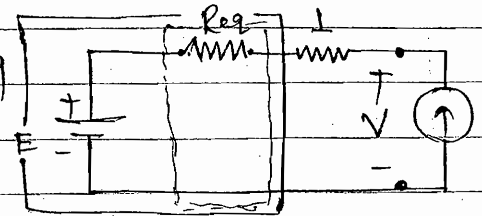
Norton's Model.

$V_{oc} = 1.5(R_{eq}+2)$ $\qquad$ $\underset{6}{V_{oc}} = I(\underset{2}{R_{eq}}+\underset{1}{R})$ $\Rightarrow I = 2A$. Ag

$\Longrightarrow V_{oc} = 6$

$R_{eq} = 2$

Q.39



Thevenin's Model.

1. $E = E_1$, $V = 5V$, I removed

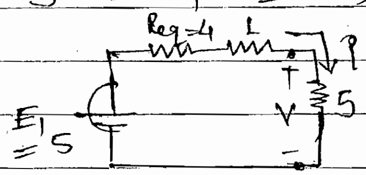
$\boxed{E_1 = V = 5V}$

2. $E = 0$; $I = 1A$; $V = 5$

$5 = I(1+R_{eq}) + E_0$

$\boxed{R_{eq} = 4}$

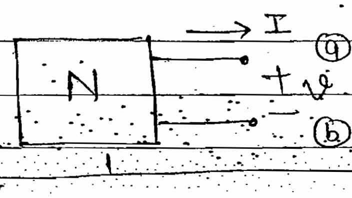
3. $E = E_1$ $\quad I \xrightarrow{\text{replaced}} 5\Omega$



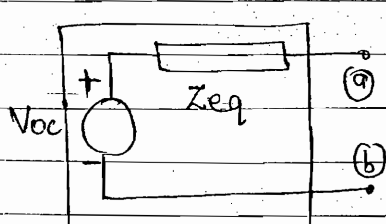
$V = \frac{5 \times 5}{5+5} = \frac{25}{10} = 2.5$

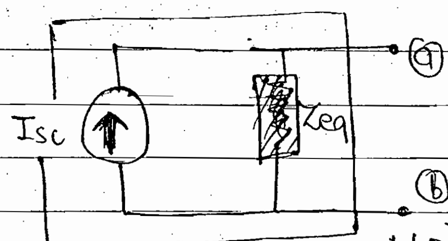
$i = \cancel{1.71/5}\ \cancel{1/0.5}\ 2.5/5 = 0.5\ A.$

- Thevenin's & Norton's Theorem:-



Thevenin's $\qquad\qquad$ Norton.





Thevenin's Model. $\qquad\qquad$ Norton's Model

V_{oc} = O.C. voltage b/w a & b when $i = 0$

I_{sc} = S.C. current b/w a & b when $V = 0$

$$\boxed{\frac{V_{oc}}{I_{sc}} = Z_{eq}}$$

- If the polarity of Terminal voltag V is $^{+}_{-}$ Then V_{oc} also V_{oc} $^{+}_{-}$

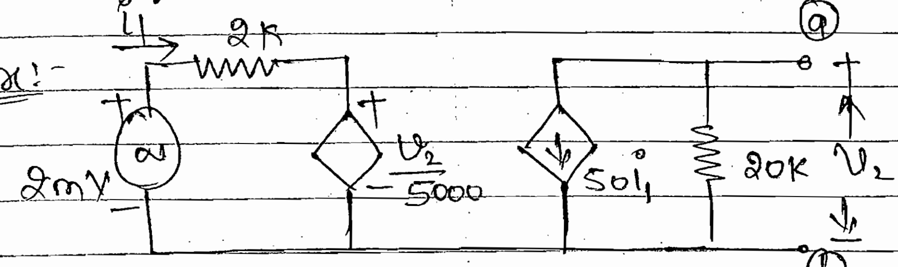
- If the polarity of terminal voltage is V $^{+}_{-}$ To satisfy it I_{sc} also (↑) I_{sc}

- Thevenin's Theorem:- A linear active RLC N/w which contains one or more independent or dependent voltage or current sources can be replaced By single voltage source Voc in series with equivalent Impedance. The Equivalent Impedance represent Z-equivalent B/w the two external terminals when
(i) All Independent voltage source are S.C. or replaced By their internal Impedances.
(ii) All Independent current source are O.C. or replaced By their internal impedances.
(iii) All dependent voltage and current sources remain as they are and therefore these sources are neither S.C. nor O.C.

The Theorem is always applicable of irrespective of
(i) Nature of the component wheather reactive or resistive
(ii) The Nature of voltage and current source wheather ac or dc
(iii) The Theorem is not applicable to the N/w. containing Non-linear elements, unilateral elements such as p-n junction diode.

- Norton's Theorem: A linear active RLC N/W which contains one or more independent or dependent voltage or current sources can be replaced By single current source I_{sc} in shunt with an impedance Z_{eq}.

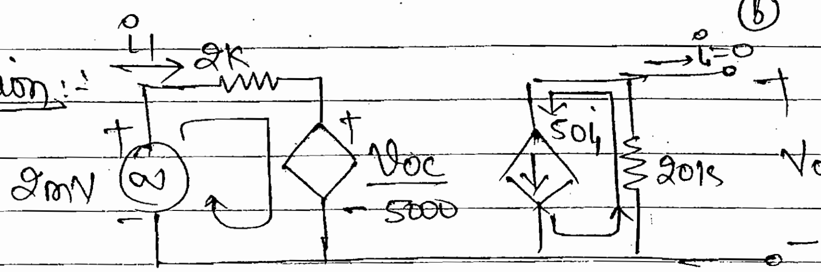
Ex:-



To find.
1. V_{oc} ... R_{eq}. Theven.
2. I_{sc} .. R_{eq}. Norton

Solution:-

V_{oc}



$$2mV = 2K i_1 + \frac{V_{oc}}{5000}$$

$$- V_{oc} = 50 i_1 \times 20 \times 10^3$$

$$2 \times 10^{-3} = i_1 2 \times 10^3 + \frac{V_{oc}}{5000}$$

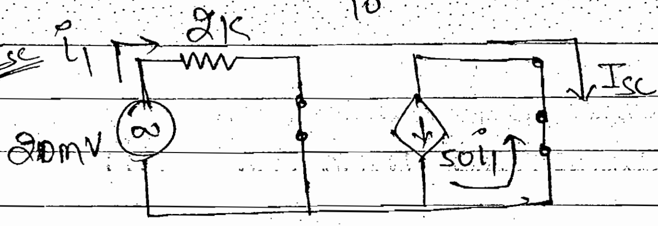
$$i_1 = \frac{-V_{oc}}{50 \times 20 \times 10^3}$$

$$\cancel{2 \times 10^{-3}} = \frac{-V_{oc}}{\cancel{50} \times \cancel{20} \times 10^3} \times \cancel{2} \times 10^3 + \frac{V_{oc}}{500}$$

10

$$\Rightarrow V_{oc} = \frac{-10}{9} V$$

I_{sc}



$$I_{sc} = -50 i_1 \quad \text{---} (1)$$

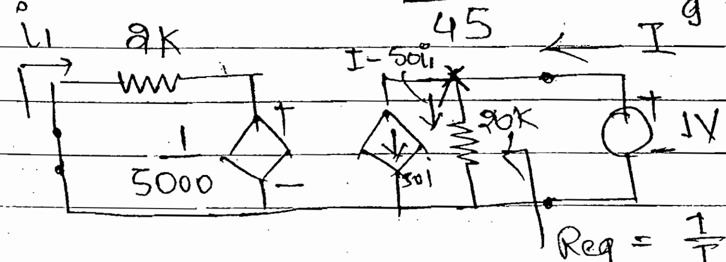
$$\cancel{2} \times 10^{-3} = \cancel{2} \times 10^3 \times i_1$$

$$i_1 = 10^{-6}$$

$$I_{sc} = -50 \times 10^{-6} = -50 \mu A$$

$$R_{eq} = \frac{V_{oc}}{I_{sc}} = \frac{-1\cancel{0}}{9 \times (-5\cancel{0} \times 10^{-6})}$$

$$= \frac{10^6}{45} = \frac{200}{9} K$$

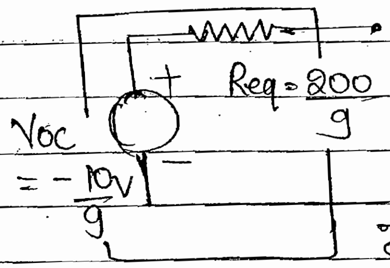


Choose any value of V, Bcoz. value of R_{eq} Remain Same.

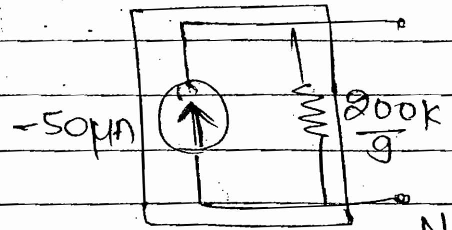
$$R_{eq} = \frac{1}{I} = ?$$

$\Rightarrow 2\times10^{3} i_1 + \frac{1}{5000} = 0$ —— (a) ; $1 = 20\times10^{3}[I - 50 i_1]$ —— (b)

from eqⁿ (a) & (b) $R_{eq} = \frac{200k}{9}$.

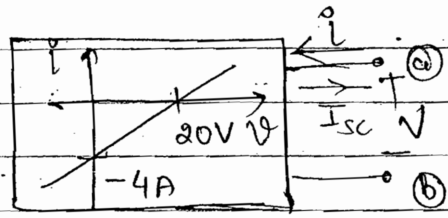


Thevenin's



Norton's.

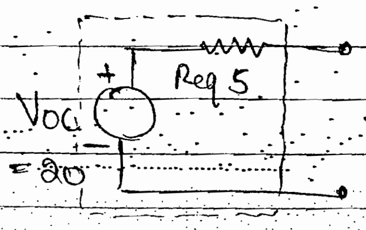
Ex



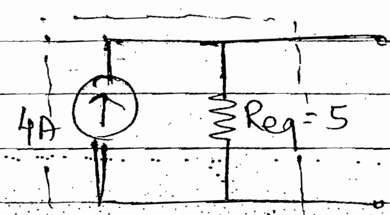
$i = 0$; $V = 20V$; $V_{oc} = 20V$

$V = 0$; $i = -4A$; $I_{sc} = 4A$

$R_{eq} = \frac{V_{oc}}{I_{sc}} = \frac{20}{4} \Rightarrow R_{eq} = 5\ \Omega$ Ans

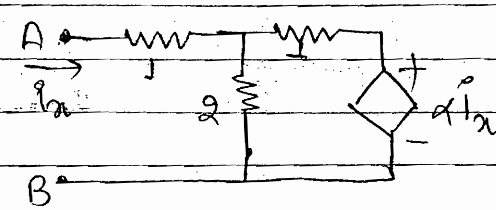


Thevenin

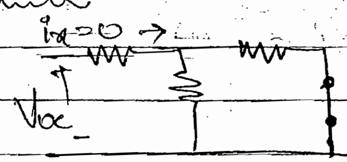


Norton.

Q.7 Conventional :-



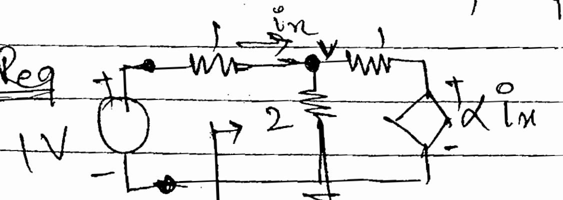
V_{oc} :- $i_x = 0$



$V_{oc} = 0$

Whenever V_{oc} is 0, Req find Independently.

Req



$R_{eq} = \frac{1}{i_x} = ?$

$\frac{V-1}{1} + \frac{V}{2} + \frac{V - \alpha i_x}{1} = 0$

$\frac{5V}{2} - \alpha i_x - 1 = 0$ —— (1)

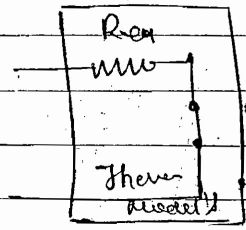
$\frac{5V}{2} - \alpha i_x - 1 = 0$
$V = \frac{(1+\alpha i_x)2}{5}$
$= \frac{2+2\alpha i_x}{5}$
$\frac{5-2+2\alpha i_x}{5}$
$i_x - \frac{2\alpha i_x}{5} = 3$
$5i_x - 2\alpha i_x = 3$
$i_x(5-2\alpha) = 15$

$i_x = \frac{1-V}{1}$ — (2) $\left\{ V = \frac{(1+i_\alpha)\times 2}{5} \right.$

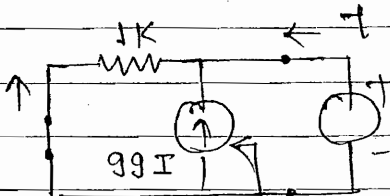
$i_x = \frac{5-(1+i_\alpha)\times 2}{5} = \frac{5-2-2i_\alpha}{5} = \frac{3-2i_\alpha}{5}$

$\frac{5i_x + 2i_x\alpha}{\cancel{5}} = \frac{3}{\cancel{5}}$ $\quad i_x = \frac{3}{(5+2\alpha)}$

$R_{eq} = \frac{1}{i_x} = \frac{5+2\alpha}{3}$



Q.29

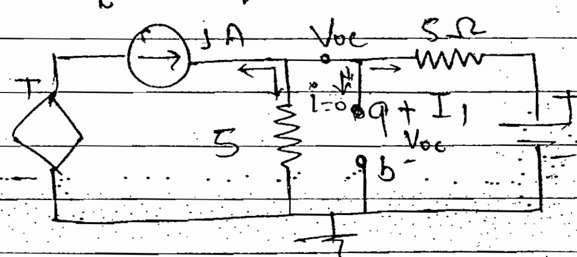


$-i = I + 99I = 100I$ — (1)
$1V = 1 \times (-I)$
$I = -1$

$R_{eq} = \frac{1}{i_1} = ?$

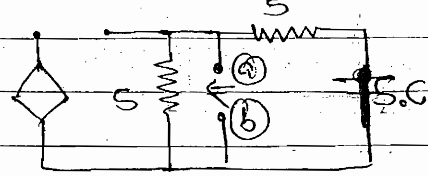
find $\frac{1}{i} = R_{eq} = 0.01\ K = 10\ \Omega$ Ans

Q.32



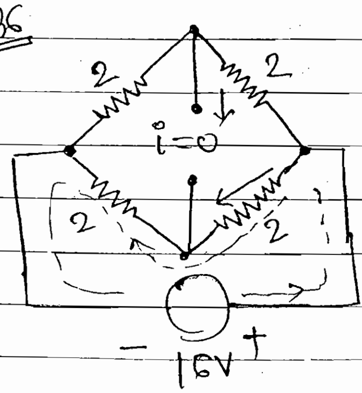
$\Rightarrow \frac{V_{oc}-10}{5} + \frac{V_{oc}}{5} = +1$
$\Rightarrow 2V_{oc} = 5 + 10$
$V_{oc} = \frac{15}{2} = 7.5$

Req



$R_{eq} = 2.5\ \Omega$ Ans

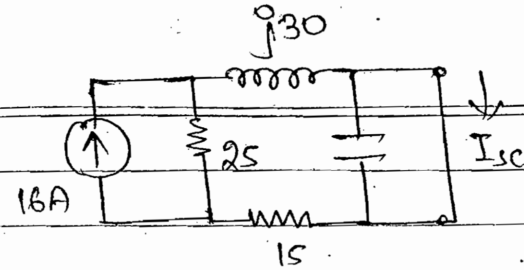
Q.36



$-16 + 2I = 0$ $\quad \boxed{I = 4}$ A.

Given Balance Condition

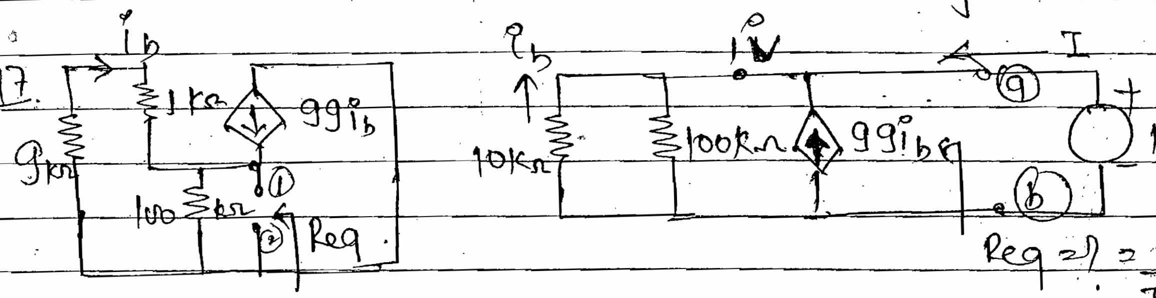
Q.45



$$I_{sc} = \frac{16 \times 25}{40 + j30}$$

$= 6.4 - j4.8$. Ans.

Q.47



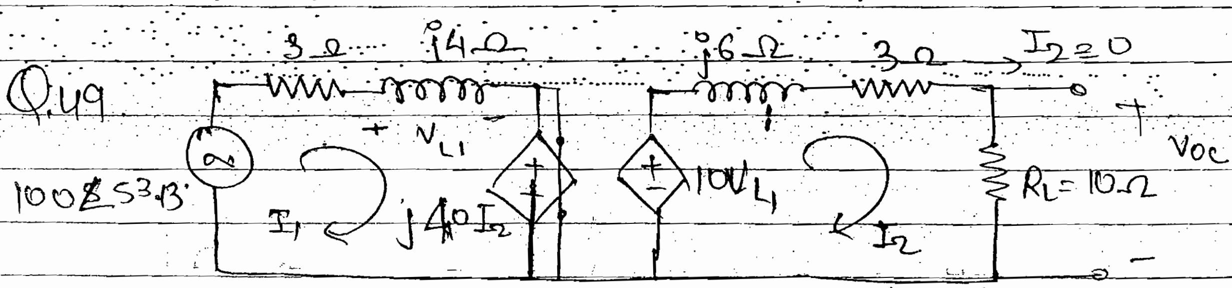
$$\Rightarrow \frac{V}{10K} + \frac{V}{100k} = -99i_b - I \quad \Rightarrow \frac{1}{10K} + \frac{1}{100K} = -(99i_b + I) \quad —(1)$$

$$\Rightarrow -i_b = \frac{V - 0}{10K} = \frac{1}{10} \quad —(2)$$

put in eq(1) $\Rightarrow$ $\frac{1}{10K} + \frac{1}{100K} = + \frac{99}{10} - I$

$\frac{1}{I} = R_{eq} = 0.1K = 100\Omega$ Ans.

Q.49



$$I_1 = \frac{100\angle 53.13}{3 + j4} = \frac{100\angle 53.13}{5\angle 53.13}$$

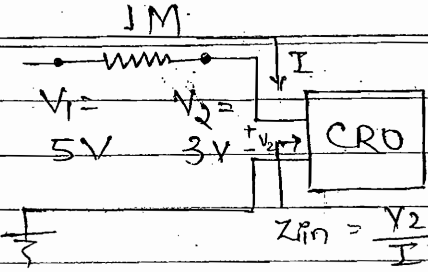
$= 20$ Amp

$V_{L1} = 20 \times j4 = j80$

$V_{oc} = 10V_{L1} = 10 \times j80 = j800$

$= 800\angle 90°$ V Ans

Q.26

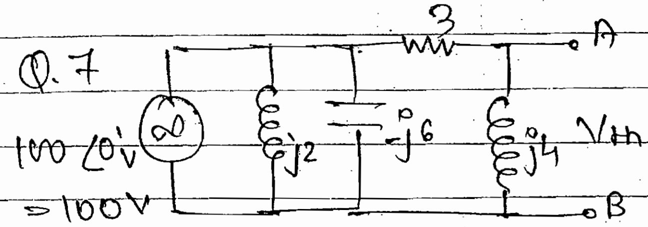


$Z_{in} = \dfrac{V_2}{I}$

$I = \dfrac{5-3}{1M} = 2\times10^{-6}$ Amp.

$Z_{in} = \dfrac{3}{2\times10^{-6}} = 1.5\ M\Omega$ Ans

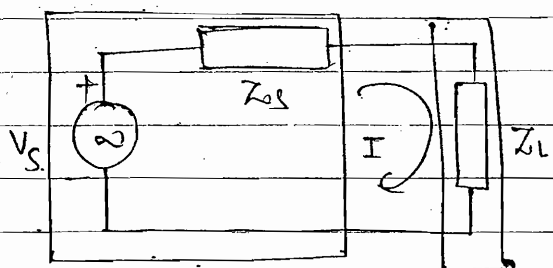
Q.7



$V_0 = 100 \times \dfrac{j4}{3+j4}$

$= j16(3-j4)$ Ans

- Max Power Transfer Theorem:



Thevenin's Model

Variable load.

To find:- Condition for transfer of Max power from the source to the load.

Consider $Z_L = R_L + jX_L$

$Z_s = R_s + jX_s$

$I = \dfrac{V_s}{|Z_s + Z_L|} = \dfrac{V_s}{(R_s+jX_s)(R_L+jX_L)} = \dfrac{V_s}{(R_s+R_L)+j(X_s+X_L)}$

$P_L = |I|^2 R_L$

$P_L = \dfrac{V_s^2 \cdot R_L}{(R_s+R_L)^2 + (X_s+X_L)^2}$ — ①

$P_L = I^2 Z_L$

$P_L = I^2 R_L$

$P_L = |I|^2 Z_L$

$\Rightarrow P_L = |I|^2 R_L$

Bcoz delivers active (Real) power.

for P_L (max).

$\dfrac{\partial}{\partial R_L}[P_L] = 0$; $\dfrac{\partial [P_L]}{\partial X_L} = 0$

$\Rightarrow \boxed{R_L = R_s}$ $\Rightarrow \boxed{X_L = -X_s}$

$Z_L = R_L + jX_L = R_s - jX_s$

$Z_s = R_s + jX_s$

$Z_s^* = R_s - jX_L$

$$\boxed{Z_L = Z_s^{*} \;;\; R_s = R_L \;;\; X_L = -X_s}$$

Condition for Maximum power Transfer:

(i) Hence, in order to transfer maximum amount of power from the source to the load, the load impedance must be complex conjugate of the source impedance. Therefore the source and the load have equal resistive parts. They must also have equal reactive parts But opposite in sign. Therefore if the source is inductive the load must be capacitive and vice-versa.

(ii) For maximum power transfer the load must be variable in nature.

(iii) To apply maximum power transfer theorem any given n/w must be first transform in the standard Thevenin's model.

(iv) The Maximum power transferred to the load is only 50% of the total power generated by the voltage source.

(a) The Theorem is also applicable irrespective of the nature of voltage source wheather it is ac or dc.

(b) Nature of source and load wheather is it resistive or reactive in nature.

case I :- General Case: $Z_L = R_L + jX_L$

$Z_s = R_s + jX_s$

$$\boxed{Z_L = Z_s^{*} \;;\; R_s = R_L \;;\; X_L = -X_s}$$

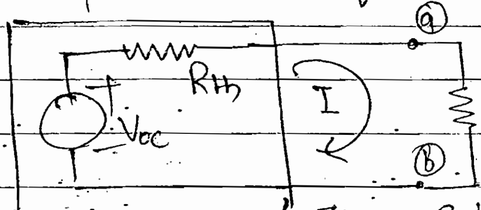
Imp case2:- $Z_L = R_L$; $Z_S = R_S + jX_S$; $P_L = \dfrac{V_S^2 R_L}{(R_S+R_L)^2+(X_S+X_L)^2}$ ⇑ 0

$$\boxed{R_L = \sqrt{R_S^2 + X_S^2} = |Z_S|}$$

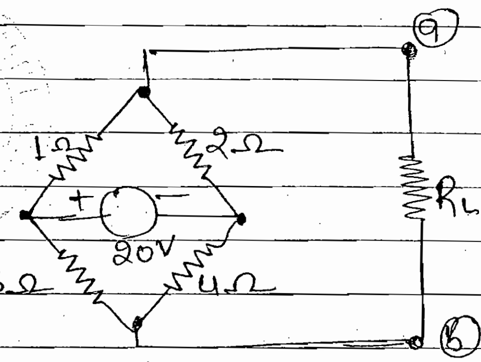
case3:- $Z_L = R_L$; $Z_S = R_S$ } ... dc case ; $P_L = \dfrac{V_S^2 R_L}{(R_S+R_L)^2+(X_S+X_L)^2}$ ⇑ 0

$$\Rightarrow \boxed{R_L = R_S}$$

Ex:- In the circuit shown. Find the value of load Resistance R_L b/w two external terminals so that maximum pow. is transferred to it. Hence calculate the maximum power transferred to load.



Thevenin's Model.



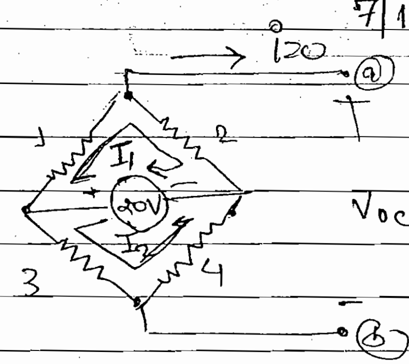
For Pmax: $R_L = R_{eq}$

$$P_{max} = I^2 R_L = \left(\frac{V_{oc}}{R_{eq}+R_L}\right)^2 R_L = \frac{V_{oc}^2}{4R_L}$$

7/11/2014

Ex:- $20 = I_1 + 2I_1$ $\quad I_1 = 20/3$

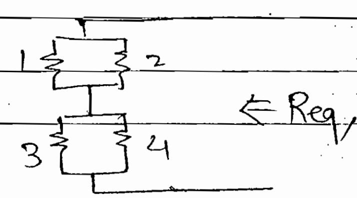
$20 = I_2(7)$ $\quad I_2 = 20/7$

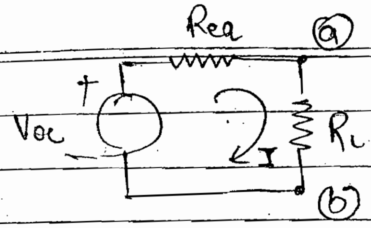


KVL

$$V_{oc} = 2I_1 - 4I_2$$
$$= 2\times\frac{20}{3} - 4\times\frac{20}{7}$$
$$= \frac{40}{21} \text{ Volt}$$

$R_{eq} = \dfrac{50}{21}\ \Omega$



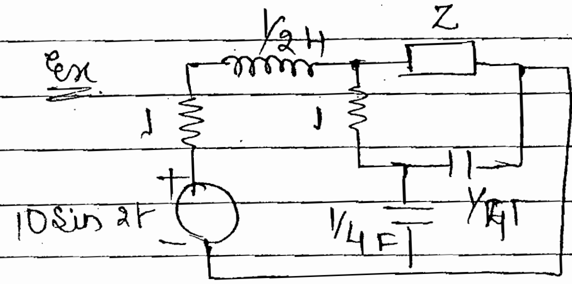


$I = \frac{V_{oc}}{R_{eq} + R_L}$; $P_{max} \Rightarrow R_L = R_{eq}$

$= \frac{50}{21}\ \Omega$

$P_{max} = \frac{V_{oc}^2}{4R_L} = \frac{8}{21}$ kW.

Ex



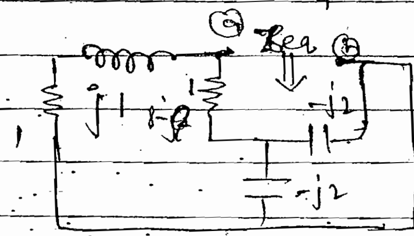
To find Z for Pmax.

$\frac{1}{2}H \Rightarrow j\omega L = j \times 2 \times \frac{1}{2} = j1\,\Omega$

$\frac{1}{4}F \Rightarrow \frac{1}{j\omega c} = \frac{1}{j \times 2 \times \frac{1}{4}} = -j2\,\Omega$

Solution:- $\omega = 2$ rad/sec.

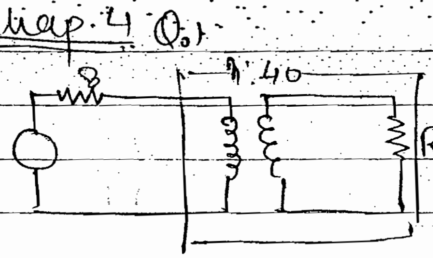
$Z_{eq} = \frac{(1+j1)(1-j1)}{(1+j1)(1-j1)} = 1 + j1$



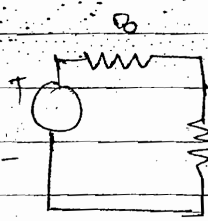
for Pmax $= Z = Z_{eq}^* = 1 \mp j0$

$= 1\,\Omega$

Chap. 4 Q.1



$\Rightarrow$



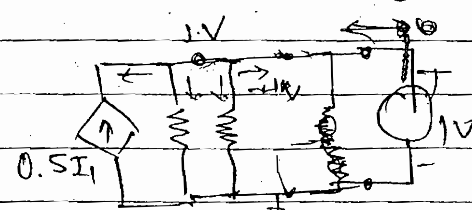
$R_L' = \left(\frac{n}{40}\right)^2 \times R_L$

for Pmax $R_L' = R_s \Rightarrow 8 = \left(\frac{n}{40}\right)^2 \times 2$ $\quad n = 80$ Ans

Q.20 ~~Req~~

K.C.L:

$0.5I_1 + \frac{1-0}{80} + \frac{1-0}{40} + (-I) = 0$ — (1)



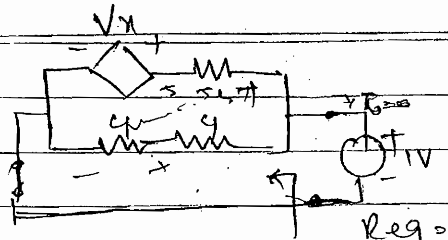
$I_1 = \frac{1-0}{40}$ — (2)

$R_{eq} = \frac{1}{I_{sc}}$

for Pmax: $R_L = R_{eq} = 16\,\Omega$

$R_{eq} = \frac{1}{I} = R_{eq} = \frac{1}{k}$ Ans

Q.40

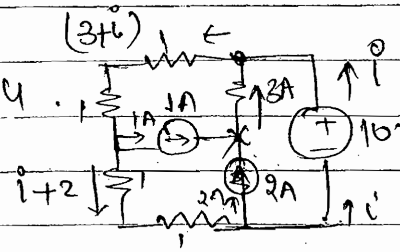


$R_{eq} = 4\Omega$

for P_{max} : $R_L = R_{eq} = 4\Omega$

$R_{eq} = \frac{V}{I}$

Q.44

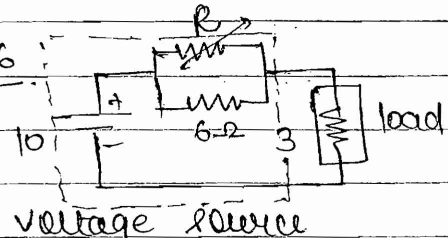


KVL ⇒ $10 = 2(3+i) + 2(i+2)$

$i = 0$ Amp.

Power delivered $= 10 \times i = 10 \times 0 = 0\,W$.

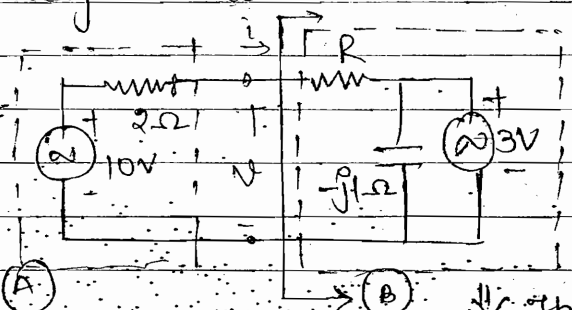
Q.46



$R_L = R_s$ is NOT applicable

$R_s = 0$ for P_{max} $\boxed{\Rightarrow R = 0}$

Q.48

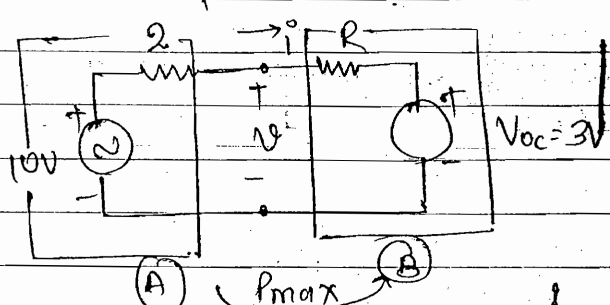


$P = vi$

$\frac{\partial [P]}{\partial R} = 0$ find R

Th. Model

$i = 0$, o.c $V_{oc} = 3\,V$. $R_{eq} = R$.



$i = \frac{10-3}{R+2}$; $v = iR + 3$

$v = \frac{7.R+3}{R+2} = \frac{7R+3R+6}{R+2}$

$= \frac{10R+6}{R+2}$

$P = vi = \frac{(10R+6) \times 7}{(R+2)^2}$

$\frac{\partial [P]}{\partial R} = 0$ $\Rightarrow R = 0.8\,\Omega$

- Tellegen's Theorem :-

In any N/w total instantaneous power absorbed By various element in different branches of N/W is always equal to zero.

In any N/w total instantaneous power delivered By various voltage and current sources is alway equal to total instantaneous power absorbed By various passive element in different Branches of N/w.

$$\boxed{\sum_{n=1}^{b} V_k \cdot i_k = 0}$$

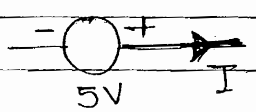
b = no. of Branch.
V_k = Branch voltages
i_k = Branch currents.

The theorem is always applicable irrespective of

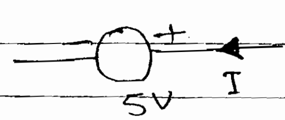
(i) Shape of N/W.
(ii) Type of elements contained.
(iii) Value of each element.

This theorem is always valid so long as the KVL & KCL eqn are applicable to the given N/w.

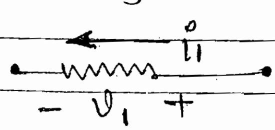
- Convention :-



Power delivered = 5 X I = 5I



Power absorbed = v X I



Power absorbed = $v_1 \times i_1$

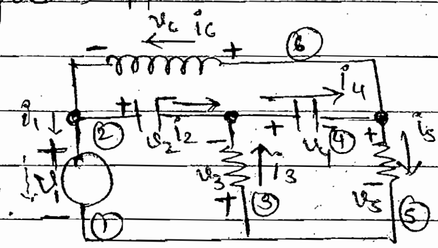
Imp.

Q In the circuit shown. Find the missing Branch voltages and Branch current. Find instantaneous power absorbed By each element in the N/W. Hence calculate total power absorbed By various element in the given N/W.

Solution. Given. $v_1 = 4V$ $\quad i_1 = 2A$

$v_2 = 2V$ $\quad i_3 = 4A$

$v_4 = 3V$ $\quad i_6 = 4A$



Take randomly the direction of current.

KVL:- ① $-v_1 + v_2 - v_3 = 0$

$-4 + 2 - v_3 = 0 \qquad v_3 = -2V$

② $-2V + v_5 + v_4 = 0$

$-2 + v_5 + 3 = 0 \qquad v_5 = -1V.$

③ $v_2 + v_6 + v_4 = 0 \qquad v_6 = -5V.$

KCL $\quad i_1 + i_2 - i_6 = 0 \qquad i_2 = 2A.$

$i_2 + i_3 - i_4 = 0 \qquad i_4 = +6A$

$i_4 - i_5 - i_6 = 0 \qquad i_5 = 2A$

	1	2	3	4	5	6
v_K	4	2	-2	3	-1	-5
i_K	2	2	4	6	2	4
$v_K i_K$	8	4	-8	18	-2	-20
	A	A	D	D	D	D

$\sum_{i=1}^{6} v_K i_K = 30 - 30$

$= 0$ Ans.

A → absorbed D → delivered

• Reciprocity Theorem:-

In a linear Bilateral single source N/W the ratio of excitation to response is constant when the positions of the excitation and response are interchanged.

(i) In verifying this Theorem the basic configuration of the N/W remains same, only the external conditions are changed.

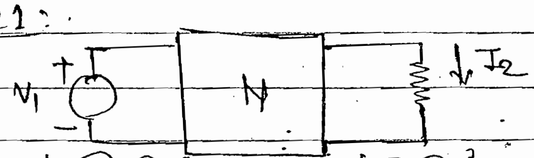
(ii) The Theorem is not applicable for the N/W containing
- (a) Non-linear elements
- (b) Unilateral elements such as p-n junction diode.
- (c) Multiple independent or dependent voltage and current source.
- (d) A single dependent voltage and current source.

- The basis of Theorem is → symmetry of the impedance and admittance Matrix.

Impedance Matrix $[Z] = \begin{bmatrix} Z_{11} & Z_{12} & Z_{13} \\ Z_{21} & Z_{22} & Z_{23} \\ Z_{31} & Z_{32} & Z_{33} \end{bmatrix}$; $\boxed{\begin{matrix} Z_{12} = Z_{21} \\ Z_{31} = Z_{13} \\ Z_{23} = Z_{32} \end{matrix}}$

Reciprocity Theorem is applicable for any Reciprocal N/W

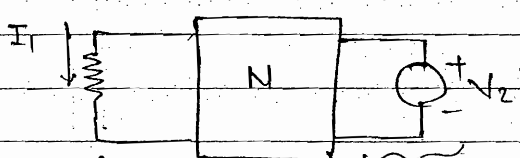
Case 1:



Excitation ~~~ Response

$\dfrac{\text{Excitation}}{\text{Response}} = \dfrac{V_1}{I_2} = Z_{12}$

Case 2:



response ~~~ excitation

$\dfrac{\text{Excitation}}{\text{Response}} = \dfrac{V_2}{I_1} = Z_{21}$

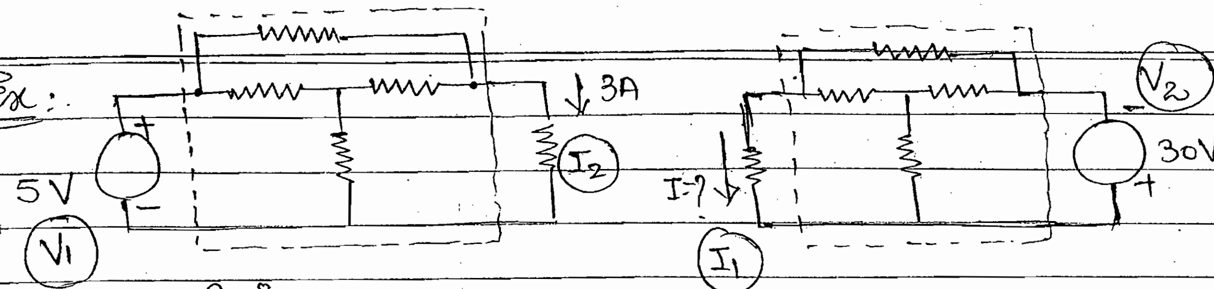
$$\boxed{\begin{matrix} Z_{12} = Z_{21} \\ \dfrac{V_1}{I_2} = \dfrac{V_2}{I_1} \end{matrix}}$$

$Z_{12} = Z_{21}$ --- for reciprocal N/W

$\dfrac{V_1}{I_2} = \dfrac{V_2}{I_1}$ --- Reciprocity theorem is applicable.

• polarity of voltage and current of Both N/w must same.

Ex:



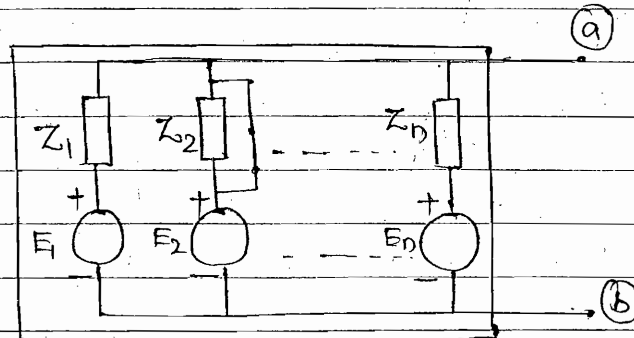
--- Bridged-T-Network.

Solution $Z_{12} = Z_{21}$; $5 \equiv V_1$, $V_2 \equiv -30$ $I = -18A$.

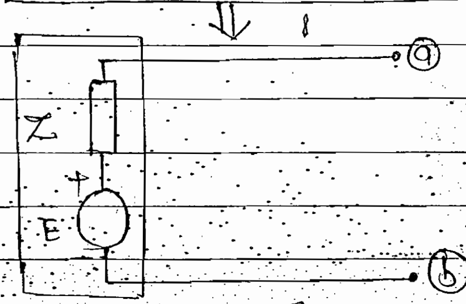
$3 \equiv I_2$ $I_1 \equiv I$

• Millman's Theorem :-

$$\boxed{E = \frac{\sum E_i Y_i}{\sum Y_i} \qquad Z = \frac{1}{\sum Y_i}}$$



Suppose $Z_2 = 0$, Then voltage across a, b terminal is E_2 and value of current will be E_2/Ampe.



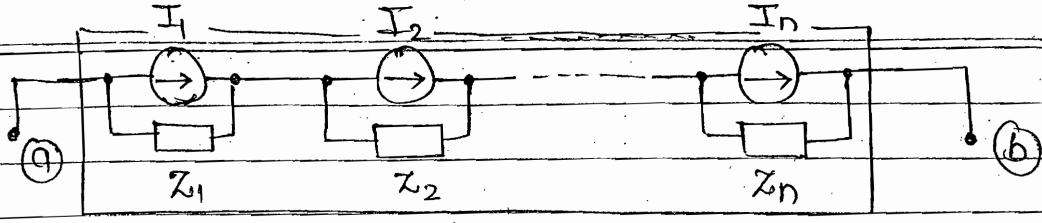
Thevenin's Model.

• The Theorem is an extension of the thevenin's theorem and is useful whenever they there are large no. of voltage sources in any given N/w.

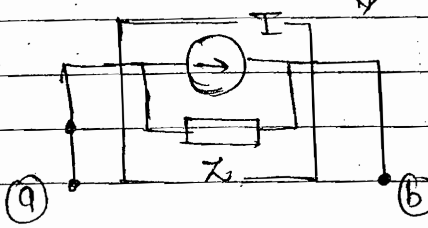
• The Theorem is applicable only when any given N/w can be rearranged in the standard format

• Dual Millman's Theorem :-

$$\boxed{I = \frac{\sum I_i Z_i}{\sum Z_i} \qquad Z = \sum Z_i}$$



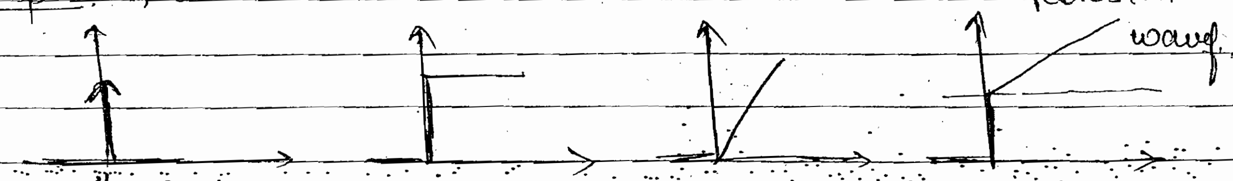
$\Downarrow$



Norton's Model

- This Theorem is an extension of the Norton's Theorem and is useful whenever there are large No. of current source in the N/w.
- The theorem is applicable only when any given N/w can be re-arranged in the standard format.

Chap. 2 Q.6

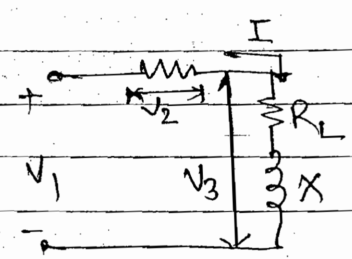


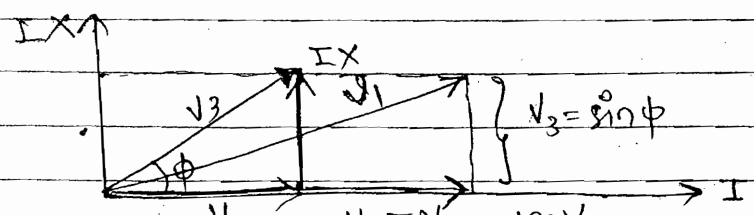
$\hookrightarrow$ $i(t)$ is max

at $t = 0^+$; $i(t)$ is max

Chap.1 Q.52 $V_1 = 220V$ $V_2 = 122V$

$V_3 = 136V$





$V_R = V_2 = 122V$

$$V_1 = \sqrt{(V_3 \cos\phi + V_2)^2 + (V_3 \sin\phi)^2}$$

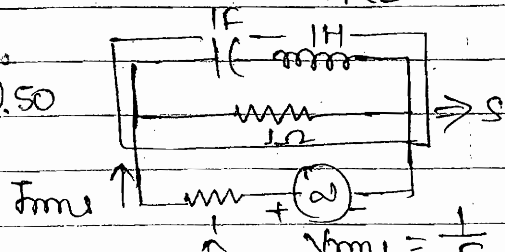
$V_R = V_3 \times \cos\phi$ $\qquad V_1^2 = V_3^2 \cos^2\phi + V_2^2 + 2V_3V_2\cos\phi + V_3^2\sin^2\phi$

$$220^2 = 136^2 + 122^2 + 2 \times 136 \times 122 \cos\phi + 136^2$$

$\cos\phi = 0.45$

(ii) $P_L = \frac{V_{RL}^2}{R_L} = \frac{(V_3 \cos\phi)^2}{R_L} = 750\,W$ Ans

Q.50



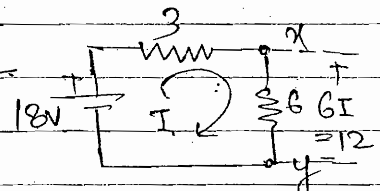
$\Rightarrow$ S.C. $1F \Rightarrow \frac{1}{j \times 1 \times 1} = -j1\,\Omega$

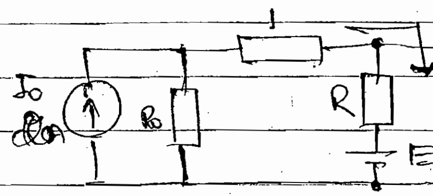
$1H \Rightarrow j\omega L = j1\,\Omega$

$\omega = 1$ rad/sec. $R_{eq.} = j1 - j1 = 0$

$I_{rms} = V_{rms}/R = 1/\sqrt{2}$ Ans.

Q.41

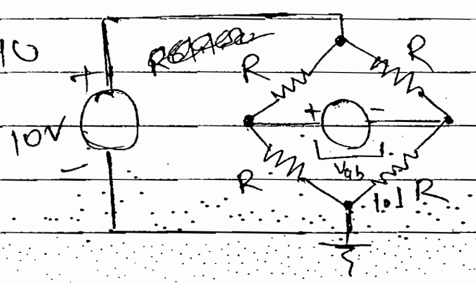




$12 = 2R + E$

$I = 18/9 = 2A$.

Q.40



$V_{ab} = V_a - V_b$

$= 5 - \frac{1.1R}{1.1R+R} \times 10 = -0.238$ V.

8/NOV/2014

Network Synthesis

Given: Any Network ~~in~~ Mathematical form $F(s)$ –
- → driving pt immittance fun^{ns} → Z_{11}, Z_{22} ; Y_{11}, Y_{22}
- → Transfer func → Y_{12}, Y_{21} ; Z_{21}, Z_{12} ; G_{12}, G_{21} ; α_{12}, α_{21}

To find:-

1. Shape of Network
   - Foster's form –
     - Type I (series type)
     - Type II (Parallel type)

     } using partial fraction expansion

   - Cauer's form –
     - Type I
     - Type II

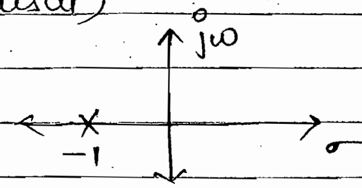
     } ladder type N/w } using continued fraction expansion C.F.E.

2. Type of Element –
   - LC
   - RC
   - RL
   - RLC

3. Value of each element.

- Conditions for a Network to be :-
1. Physically realizable (causal)
2. Stable

$\frac{1}{s+1} \longrightarrow \underbrace{e^{-t}u(t)}$



$= 0;\ t<0$

$= e^{-t};\ t>0$

- For any network to be a stable N/w following condition must be satisfied.

Condition 1. The N/w functn F(s) can't have poles in RHP.

Condition 2. The N/w functn can't have multiple poles along the jω axis including the origin

Condition 3:- The degree of Numerator functn of the N/w F(s) can't exceed the degree of denominator by more than unity

ex:- $\frac{s^2+s+1}{s^2+2s+1}$ ✓, $\frac{s+1}{s^2+2s+1}$ ✓, $\frac{s^3+2s^2+s+1}{s^2+s+1}$ ✓

$\frac{s^4+ \cdots}{s^2+ \cdots}$ ×

- For a physically realizable functn o/p can't exist Before the i/p applied.
- For a stable N/w the response must be finite for finite excitation
- Any functn F(s) with poles in LHP has inverse laplace form which is zero for $t<0$
- Stability Implies Causality. Therefore any stable N/w is always a physically realizable N/w But the reverse may not be True. Hence any physically realizable N/w may or may not be stable N/w.

• Hurwitz polynomial P(s) :-

Network funct^n $F(s) = \frac{\text{Num poly}}{\boxed{\text{Denom . poly}}}$ P(s)

...... must Be Hurwitz

...... all poles of F(s) lie in L.H.P.

...... N/w must Be stable

...... N/w must also be physically realizable

features.

$P(s) = s^6 + 2s^5 + 4s^4 + 2s^3 + 3s^2 + 5s + 9$

Even parts | odd parts

$Pe(s) = s^6 + 4s^4 + 3s^2 + 9$ | $Po(s) = 2s^5 + 2s^3 + 5s$

CFE of $\frac{Pe(s)}{Po(s)}$ ... has all +ve quotient terms Divisor) Dividend (Quotient

Remainder

• $P(s) = P_1(s) \cdot P_2(s)$ ... Both P_1 & P_2 must hurwitz.

⇒ Any poly. Even or odd → all roots lie will at imag. axis.

⇒ If odd. poly is not given. then diff. even. poly and find. CFE. $Pe(s) / P'_e(s)$

⇒ Numerical value of Quotient represent RLC element value.

(1) If s is real then P(s) is also real

(2) All the coefficients of the Hurwitz poly must Be real and +ve.

(3) NO term must Be missing in the hurwitz poly unless all the even parts or all the odd parts are missing.

CFE :- Contin. funct exp. are the ratio of even to odd part or odd to even part. has all +ve quotient terms.

- The quotient terms of the continued fra. exp. represent the numerical values of RLC components of the N/w.
- If any poly is either even or odd then it is hurwitz if the continued fraction exp. of the ratio of $P(s)/P'(s)$ has all +ve quotient terms.
- If the given poly is either even or odd then all its roots lie along the $j\omega$ axis only.
- If $P(s) = P_1(s) \cdot P_2(s)$ then poly. $P(s)$ is hurwitz if the two poly. $P_1(s)$ and $P_2(s)$ are separately hurwitz.
- The hurwitz poly. represents the denominator poly. of a stable and physically realizable N/w. This poly ensures all the poles of the N/w functions lie in LHP. This poly. all ensures no multiple poles exists along the $j\omega$ axis including the origin.

Ex: Find wheather following poly. is hurwitz or not

or Find wheather or not following poly. represents the denominator poly of a stable and physically realizable N/w.

1 Solution: $P(s) = s^4 + s^3 + 5s^2 + 3s + 4$

$P_e(s) = s^4 + 5s^2 + 4$ $\qquad\qquad$ $P_o(s) = s^3 + 3s$

$$
\begin{array}{l}
s^3+3s \,)\, s^4+5s^2+4 \,(\, s \\
\quad\; -\; s^4+3s^2 \\
\qquad\qquad 2s^2+4 \,)\, s^3+3s \,(\, s/2 \\
\qquad\qquad\qquad -\; s^3 \pm 2s \\
\qquad\qquad\qquad\qquad s \,)\, 2s^2+4 \,(\, 2s \\
\qquad\qquad\qquad\qquad\quad -\; 2s^2 \\
\qquad\qquad\qquad\qquad\qquad 4 \,)\, s \,(\, s/4 \\
\qquad\qquad\qquad\qquad\qquad\quad -\; s \\
\qquad\qquad\qquad\qquad\qquad\quad\;\; \times
\end{array}
$$

$\cancel{s^3+3s \,)\, s^4+5s^2+4 \,(\, s} \quad \cancel{s^4 + 3s^2} \quad \cancel{2s^2+4} \quad \cancel{2s^2+4 \,)\, s^3+3s \,(\, s/2}$

+ve quotient terms

$\Rightarrow P(s)$ is Hurwitz

$$CFE = \frac{P_e(s)}{P_o(s)}$$

$$= s + \cfrac{1}{\cfrac{s}{2} + \cfrac{1}{2s + \cfrac{1}{s/4}}}$$

+ve quotient terms.

ex :- $P(s) = s^4 + 4$ → s^2 (even part) is missing.

- given poly. is not hurwitz.
- poles : $s^4 + 4 = 0$ $s' = \pm j\sqrt{2}, \pm j\sqrt{2}$.
- multiple poles along $j\omega$ axis.

ex $P(s) = s^3 + 2s^2 + 3s + 6$

$P_e(s) = 2s^2 + 6$

$P_o(s) = s^3 + 3s$

CFE of $\dfrac{P_o(s)}{P_e(s)}$ $2s^2+6 \overline{) s^3 + 3s} (s/2$

$-\cancel{s^3} \pm \cancel{3s}$

$\times \quad \times$ .... CFE has been terminated abruptly.

⇒ $2s^2 + 6$ or $s^3 + 3s$ is one factor of $P(s)$

Let $s^3 + 3s$ is a factor of $P(s)$ $P_1(s) = s^3 + 3s$

To find second factor of $P(s)$

$$P(s) = \underbrace{P_1(s)}_{s^3+3s} \cdot P_2(s)$$

$P_2(s) = \dfrac{P(s)}{P_1(s)}$ $s^3+3s \overline{) s^3 + 2s^2 + 3s + 6} (1 + \underbrace{\frac{2}{s}}$

$-\cancel{s^3} \quad + \cancel{3s}$

$2s^2 \quad + 6$

$-2s^2 \quad + 6$

$\times$

$\equiv 1 + \dfrac{1}{s/2}$

$$P(s) = \underbrace{P_1(s)}_{\underbrace{s^3+3s}_{s(s^2+3)}} \cdot \underbrace{P_2(s)}_{1+\frac{1}{s/2}}$$

Hurwitz

Poles :- $s = 0, \pm j\sqrt{3}$,

also Hurwitz

↳ Both $P_1(s)$ and $P_2(s)$ hurwitz.

∴ $P(s)$ is hurwitz.

- PRF .... Positive Real funct^n

$F(s)$

Z_{11}, Z_{22}	~~$Z_{12}, Z_{21}, Y_{12}, Y_{21}$~~
$\underbrace{Y_{11}, Y_{22}}$	~~$\underbrace{G_{12}, G_{21}, \alpha_{12}, \alpha_{21}}$~~
driving pt Immi. funct^n	Transfer Function

- +ve real function represent stable and physically realisable passive driving point immittance funct^n.
- Definition of PRF .... $F(s)$:-

Any function $F(s)$ is said to be PRF if $\boxed{Re[F(s)]_{s=j\omega} \geq 0}$ for all ω.

Z_{11} → $r + j\omega$ (must +ve zero)

Features: (i) if $F(s)$ is PRF then $\frac{1}{F(s)}$ is also a PRF $F_1(s) \mid \frac{1}{F(s)}$

(ii) The sum of two +ve Real functions is also a PRF
$F_1(s) + F_2(s)$ (can be use as series)

(iii) All the poles and zeros lie in LHP or along the $j\omega$ axis. All the poles and zeros either lie along -ve real axis or occur in complex conjugate pairs

(iv) All the poles along the $j\omega$ axis are simple including a origin.

(v) The degree of Numerator cannot exceed the degree of denominator by more than unity.

(vi) The lowest power of s of the Numerator poly. can't exceed the lowest power of s of the denominator poly. by more than unity

The necessary condition and sufficient condition for any funct^n F(s) to be a +ve real funct^n are:-

(i) the poles and zeros lie in LHP Therefore all the poles and zeros Numerator and denominator poly must be hurwitz.

(ii) The poles along $j\omega$ axis have +ve and real residues

This is verify by making partial fract^n expansion and checking the residues of only those poles which lie along the $j\omega$ axis only.

(iii) $\boxed{Re\left[F(s)\right]_{s=j\omega} \geq 0}$ for all ω.

⑥ SYNTHESIS OF ELECTRICAL NETWORKS CONTAINING SPECIFIED ELEMENTS:-

case I: Driving pt LC Immittance funct^n Z_{LC}, Y_{LC}

features: (1) These Immittance funct^n represent the ratio of even to odd or odd to even poly.

(2) All the poles and zeros are simple, lie along $j\omega$ axis only and they alternate

(3) Their must be either a zero or a pole at the origin or at infinity.

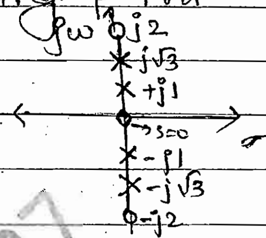
Check This condition for objec

(4) The degree of Numerator poly cannot exceed degree of denominator By more than unity.

(5) The lowest power of s of the Numerator poly and of denominator poly cannot having difference or more than unity.

✓ By for convent^n all ⑤ condition check

Q Find wheather or not following functh are physically physically realizable and stable LC driving point admittance functh.



eg:- $F_1(s) = \dfrac{s(s^2+4)}{(s^2+1)(s^2+3)}$ Zero: $s=0; \pm j2$
Poles:- $s = \pm j\sqrt{1}, \pm j\sqrt{3}$

∴ poles and zeros are not alternate.
∴ No can't realized.

$F_2(s) = \dfrac{s^5+4s^3+5s}{3s^4+6s^2} = \dfrac{s(s^4+4s^2+5)}{s^2(3s^2+6)}$ (s^0 missing)

s^0 term is missing. Double poles at origin ∴ NO

$F_3(s) = \dfrac{2(s^2+1)(s^2+9)}{(s^2+4)s}$ Zero: $s = \pm j\sqrt{1}, \pm j\sqrt{3}$
Poles:- $s = 0, \pm j2$

∴ Yes

case2 Driving point RC impedances funth RL admitt. funth

case2 Driving point RC Impedance functh RL Admittance functh Z_{RC}; Y_{RL}

features (1) all the poles and zero lie along -ve real axis and they alternate.

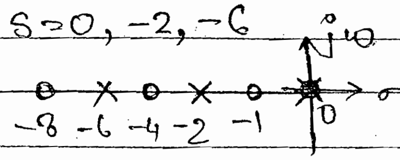
(2) The singularity nearest to or at the origin must be a pole where as the singularity nearest to or at infinity must be zero

(3) The residues of the pole must be real and +ve

Q Find wheather following functh represent physically realizable and stable driving point RL impedance functh or RLC admittance functh.

$F_1(S) = \frac{(S+1)(S+4)(S+8)}{s(s+2)(s+6)}$ Zero: $s = -1, -4, -8$

Poles: $s = 0, -2, -6$



$\frac{(S+1)(S+4)(S+8)}{s(s+2)(s+6)} = \frac{A}{s} + \frac{B}{(s+2)} + \frac{C}{(s+6)}$

$(s+1)(s+4)(s+8) = A(s+2)(s+6) + B(s)(s+6) + C(s)(s+2)$

~~at s=0~~

$F_1(s) \neq \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+6}$ Partial only possible when powers of Num is less than deno.

.... Yes.

$F_2(s) = \frac{(s+1)(s+8)}{(s+2)(s+4)}$ zeros:- $s = -1, -8$

poles :- $s = -2, -4$

— Poles and zeros do not alternate

— nearest to origin is not pole.

NO

$F_3(S) = \frac{(S+1)(s+2)}{s(s+3)}$ Zeros:- $s = -1, -2$

Poles: $s = 0, s = -3$

— Poles and zeros are not alternate

— Nearest to infinity is not a zero.

NO

Case 3: Driving point RL Impedance funct^n or RC Admittance funct^n. Z_{RL}; ~~Y~~ Y_{RC}

features: (1) All the poles and zeros lie along the -ve real axis and they alternate.

(2) The singularity nearest to or at the origin must be a zero, whereas the singularity nearest to ~~or~~ at ∞ must be pole

(3) The residues of the pole must be real and -ve

Q Find wheather following funct^n represents physically realizable and stable driving pt RLC Impedance funct^n or RC admittance funct^n.

Solut^n $F_1(s) = \dfrac{2(s+1)(s+3)}{(s+2)(s+6)}$ Zero :- $s = -1, -3$

poles :- $s = -2, -6$

Scale factor

... Yes

Due to scale factor No change in shape, only magnitude of amplitude get change.

$F_2(s) = \dfrac{4(s+1)(s+2)}{(s+3)(s+4)}$ ... poles and zeros are not alternate ... NO

Case 4 :- Driving pt RLC Immittance function Z_{RLC}; Y_{RLC}

features (1) The driving point RLC immittance function do not follow any set rule on the location of the poles and zeros and therefore these can lie any where in LHP.

Ex Following funct^n represents pure RLC driving pt Impedance funct^n. Realise it in Cauer's type I form

$Z_{RLC}(s) = Z(s) = \dfrac{s^2+2s+2}{s^2+s+1}$ ... Cauer's Type I. (ladder type)

Zeros: $s = \dfrac{-2 \pm \sqrt{4-4\times2\times1}}{2}$

$= \dfrac{-2 \pm j2}{2} = -1 \pm j1$

Poles :- $s = \dfrac{-1 \pm \sqrt{1-4\times1\times1}}{2}$

$= \dfrac{-1 \pm j\sqrt{3}}{2} = -0.5 \pm j\,0.866$

Cauer's type I → Inductor → Series } always.
Capacitor → parallel

Cauer's type 2 → Capacitor → series } always.
Inductor → parallel

CFE of $\dfrac{s^2+2s+2}{s^2+s+1}$

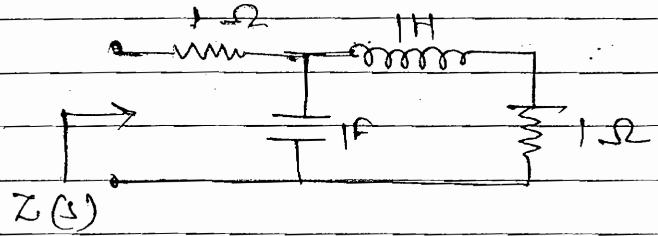
$$
\begin{array}{l}
s^2+s+1)\ s^2+2s+2\ (1 \leftarrow Z_1 \text{ (series)} \\
\quad -\cancel{s^2} \pm s \pm 1 \\
\qquad s+1)\ \cancel{s^2}+s+1\ (s \leftarrow Y_2 \text{ (parallel)} \\
\qquad\quad -\cancel{s^2} \pm s \\
\qquad\qquad 1)\ s+1\ (s \leftarrow Z_3 \text{ (series)} \\
\qquad\qquad\quad -\cancel{s} \\
\qquad\qquad\qquad 1)\ 1\ (1 \leftarrow Y_4 \text{ parallel} \\
\qquad\qquad\qquad\quad -1 \\
\qquad\qquad\qquad\quad \times
\end{array}
$$

$Z_1 = 1 = R_1$; $R_1 = 1\Omega$ (series)

$Y_2 = s = sC_2$; $C_2 = 1F$ (parallel)

$Z_3 = s = sL_3$; $L_3 = 1H$ (series)

$Y_4 = 1 = R_4$; $R_4 = 1\Omega$ (parallel)



Cauer's type - I ladder N/w.

- $\dfrac{s^2 + \ldots}{s^2 + \ldots} = 1 = R$; $R_1 = 1\Omega$ (series)

  $\dfrac{s^3 + \ldots}{s^2 + \ldots} = s = sL$; $L_1 = 1H$ (Series)

  $\dfrac{s^4 + \ldots}{s^2 + \ldots} = s^2$ ×

Cauer's - I	Cauer's - II
CFE of $\dfrac{s^2+2s+1}{s^2+s+1}$	CFE of $\dfrac{s^2+2s+1}{1+s+s^2}$ Reverse order of (I).

- If the given funct^n is an Impedance function, Then the first element in Cauer's N/w represents an Impedance element and is a series element.

By observing $\frac{s^2 ..}{s^2 ..}$ finite 2 pole 2 zero so → 4 element
only for RLC

- In general ~~two~~ total No. of elements for RLC N/W in the cauer's form is equal to the algebraic sum of No. finite zeros and poles.

- In Cauer's type 1 form capacitor is always a shunt element whereas inductor is always a series element. In cauer's type 2 N/W the inductor is always in parallel element whereas capacitor is always series element.

- In Cauer's type 1 N/W continued fraction expansion is found by arranging the Numerator and Denominator polynomials in descending power of (s). In cauer's type 2 form the continued fraction expansion is found by rearranging Numerator and denominator polyn. in ascending power s

- For any immittance funct^n cauer's type 1 and type 2 N/W are purely independent of each other.

Q Following funct^n represents physically realizable and stable LC driving point immittance form. Realise it in foster type 1 and type 2 form.

$$F(s) = \frac{2(s^2+1)(s^2+9)}{s(s^2+4)}$$

---- foster's type 1 (series form)

---- foster's type 2 (Parallel form)

$$F(s) = \frac{2s^4 + 20s^2 + 18}{s^3 + 4s}$$

As degree of Numerator > degree of denominator

$$s^3+4s \overline{)\,2s^4 + 20s^2 + 18\,}\left(2s\right.$$
$$\underline{-2s^4 \pm 8s^2}$$
$$12s^2 + 18$$

$$F(s) = \frac{2s^4 + 20s^2 + 18}{s^3+s} = 2s + \left\{\frac{12s^2+18}{s(s^2+4)}\right\} \longrightarrow \frac{A}{s} + \frac{Bx+C}{s^2+4}$$

$$\therefore A = 9/2 \; ; \; B = 15/2 \; ; \; C = 0$$

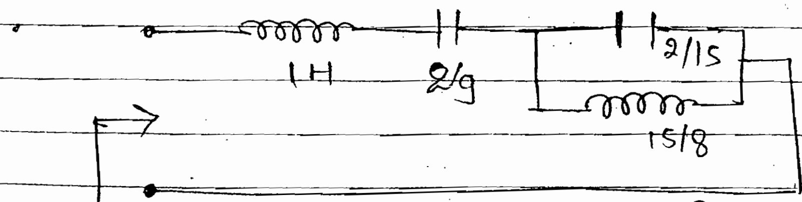
$$F(s) = 2s + \frac{9/2}{s} + \frac{(15/2)s}{s^2+4}$$

<u>case I</u> :- <u>foster's type 1</u> ... series form $\boxed{F(s) = Z(s)}$

$$Z(s) = 2s + \frac{9/2}{s} + \frac{(15/2)s}{s^2+4}$$
$$= Z_1 \quad Z_2 \quad Z_3$$

- $Z_1 = 2s = sL_1 \; ; \; L_1 = 1H \longleftarrow$
- $Z_2 = (9/2)/s = \frac{1}{(2/9)s} \; ; \; C_2 = \frac{2}{9} F \longleftarrow$
- $Z_3 = \frac{(15/2)s}{s^2+4} = \frac{1}{Y_3} \; ; \; Y_3 = \frac{s^2+4}{(15/2)s} = \frac{s}{15/2} + \frac{8}{15s}$
$$= Y_3' + Y_3''$$

$$\left.\begin{array}{l} Y_3' = \frac{2}{15}s = \; ; \; C_3 = \frac{2}{15} F \\ Y_3'' = \frac{8}{15s} = \; ; \; L_3 = \frac{15}{8} H \end{array}\right\} \longleftarrow$$

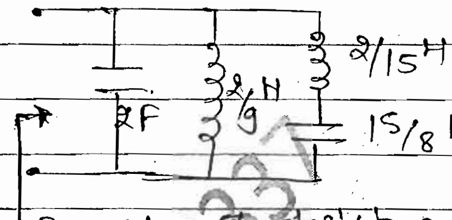


$F(s) = Z(s)$ ..... foster's type 1 N/w, series form N/w

Case 2 Foster's Type 2 Parallel form: $\boxed{F(s) \equiv Y(s)}$

$$F(s) = 2s + \frac{9/2}{s} + \frac{(15/2)s}{s^2+4}$$

$$= Y_1 + Y_2 + Y_3$$



$$Y_1 = 2s \equiv sC_1 = 2F \leftarrow$$

$$Y_2 = \frac{9/2}{s} \equiv \frac{1}{sL_2} = \frac{2}{9}H \leftarrow$$

$$Y_3 = \frac{(15/2)s}{s^2+4} \equiv \frac{1}{Z_3} \quad ; \quad Z_3 = \frac{s^2+4}{(15/2)s} = \underbrace{\frac{2}{15}s}_{Z_3'} + \underbrace{\frac{8}{15s}}_{Z_3''}$$

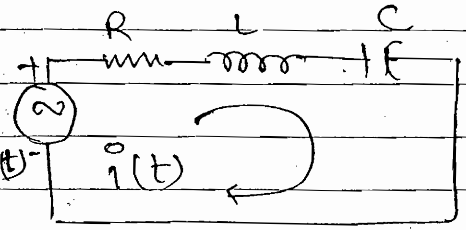
$$\left.\begin{aligned} Z' &= \frac{2}{15}s = sL_3 \ ; \ L_3 = \frac{2}{15}H \\ Z'' &= \frac{8}{15s} = \frac{1}{sC_3} \ ; \ C_3 = \frac{15}{8}F \end{aligned}\right\} \leftarrow$$

- Foster's Type I $\xleftrightarrow[N/\omega]{\text{dual}}$ Foster's Type II

Comment: The foster's type 1 and foster's type 2 N/w represent the dual N/w. These two N/w are NOT the equivalent N/w. and Therefore the Foster's type 1 N/w cannot be replaced by Type 2 N/w.

- Duality Between 2 Network:-

N1: (series RLC ckt)



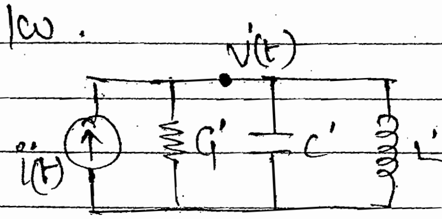
KVL

$$v(t) = R\,i(t) + L\frac{di}{dt} + \frac{1}{C}\int_0^t i\,dt \quad \text{—①}$$

N_2: (Parallel RLC ckt) dual N/w.

<u>KCL</u>:

$$i'(t) = v'(t)G' + C'\frac{dv'(t)}{dt} + \frac{1}{L}\int_0^t v(t)\,dt \quad \text{---(2)}$$

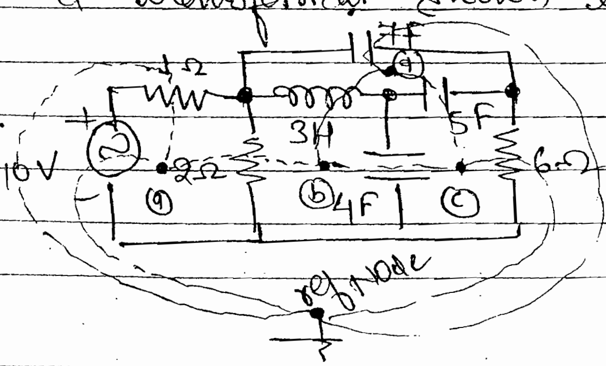


N_1	dual N/w N_2
✓ $v(t)$	$i(t)$
✓ $i(t)$	$v(t)$
✓ Series circuit	parallel circuit
✓ loop	Node
✓ KVL	KCL
✓ R	G
✓ L	C
✓ C	L
✓ Voltage source	Current source

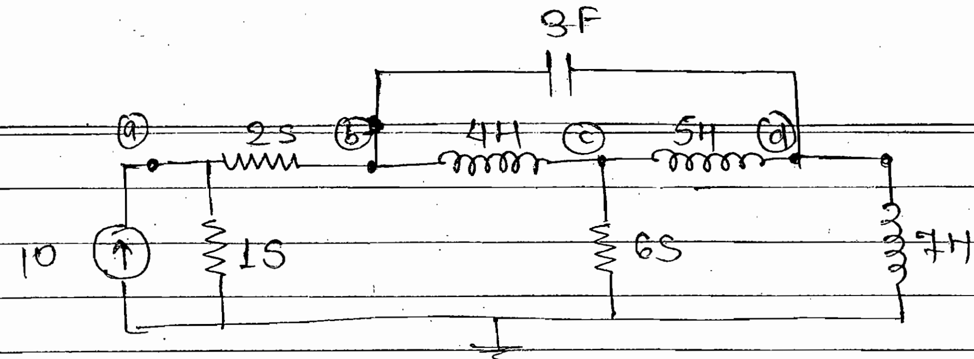
<u>Comment</u>. (1). The N/w N_1 and the N_2 are dual of each other. These two N/w are not the equivalent N/w's and therefore the N/w N_1 cannot be replaced by its dual N/w N_2 and vice-versa.

(2) If we know the response of any N/w in terms of current variable for specified excitatn then we can write the response of its dual N/w. in terms of voltage variable for similar excitatn directly by inspection <u>without actually solving it</u> using a transformatn shown in Table.

<u>Ex</u>



<u>To find</u> :- dual N/w.

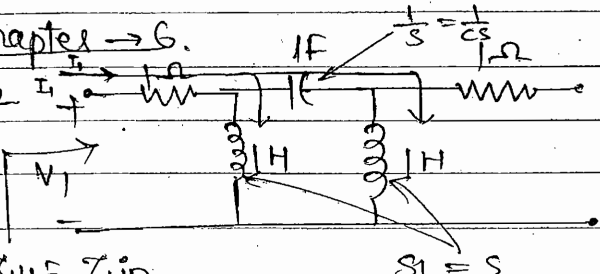


dual Network

→ Numerical values remain same.

Chapter → 6.

Q.2



$Z_{11} = Z_{in}$

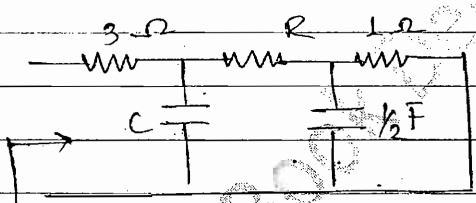
$$Z_{in} = 1 + \frac{S \times \frac{1}{CS} * SL}{S + \left(\frac{1}{CS} + SL\right)}$$

$$= 1 + \frac{S \times (1/_S \times S)}{S + \left(\frac{1}{S} + S\right)}$$

$$= \frac{S^3 + 2s^2 + S + 1}{2s^2 + 1}$$

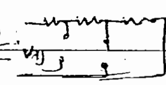
∴ Inductor → parallel
∴ Capacitor → series } Cauer's type 2.

Q.4



$$Z(s) = \frac{3(s^2 + 6s + 8)}{s^2 + 4s + 3}$$

at s = 0 Cap = O.C.



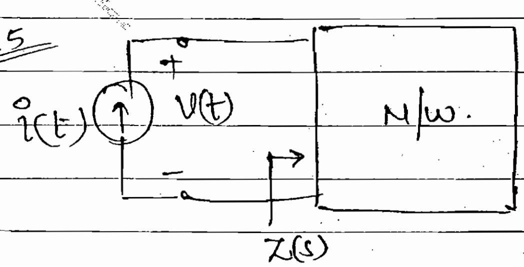
$$Z(0) \Rightarrow 3 + R + 1 = \frac{3(0 + 0 + 8)}{0 + 0 + 3}$$

$$= 21\ \Omega$$

at ∞ C = S.C. ....C cannot be found ... Cauer's – I.

CEE of = $(3s^2 + 18s + 24) / s^2 + 4s + 3$.

Q.5



$i(t) = u(t)$

$I(s) = \frac{1}{s}$

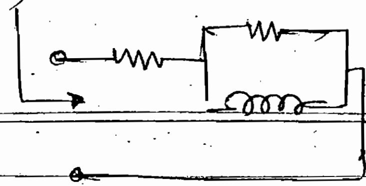
$v(t) = 1 + e^{-t/\tau}$

$v(s) = \frac{1}{s} + \frac{1}{s + \frac{1}{2}}$

$$\frac{V(s)}{I(s)} = Z(s) = 1 + \frac{S}{S + \frac{1}{2}} = Z_1 + Z_2$$

$$= Z_1 = R_1 = 1\ \Omega \quad ; \cancel{Z_2 = R_2 = 1\ \Omega}.$$

$Z(s)$

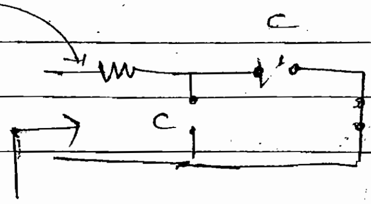


$Y_2 = \frac{1}{Z_2} = 1 + \frac{1}{s\tau} = Y_2' + Y_2''$

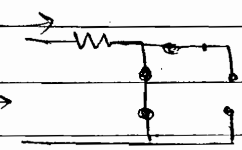
$Y_2' = 1 = \frac{1}{R_2}$; $R_2 = 1\,\Omega$ $\qquad Y_2'' = \frac{1}{s\tau} = \frac{1}{sL_2}$; $L_2 = \tau$ H

Q.8 $s = 0$ C --- O.C ; L = SC

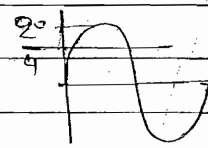
$s = \infty$ C -- S.C ; L = O.C.



$Z(0) = S$



$Z(\infty) = 0.75\,\Omega$ ⟶



$Z(0) \to \infty$ — $1/s$

$2/s$ → cannot take otherwise exceed

$V_n = \frac{V}{V_n} = \left(\frac{4}{3}\right)$

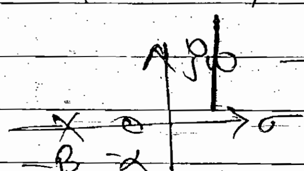
(b) → $\frac{1}{s}$; $0.75\,\Omega$ $\qquad$ By L . Not possible.

Q.10 $Z(s) = \frac{s+\alpha}{s+\beta} = \frac{V}{I} = \frac{j\omega + \alpha}{j\omega + \beta}$ as I/p is sinus.

V leads I ; $\theta \to$ +ve

$\theta = \tan^{-1}\omega/\alpha - \tan^{-1}\omega/\beta = +ve$ $\qquad \boxed{\alpha < \beta}$

OR



Inductive RL

Q.13 Zero: 60, 20 ⟹ $s = \pm j20 \pm j60$

Poles: 0, 40 ; ∞ ; $s = 0$, $\pm j40$

$Z\big|_{\omega = 10} = -j70$

$$Z(s) = K \frac{(s^2 + 20^2)(s^2 + 60^2)}{s(s^2 + 40^2)}$$

$\frac{s^4}{s^3} = s \Rightarrow sL$

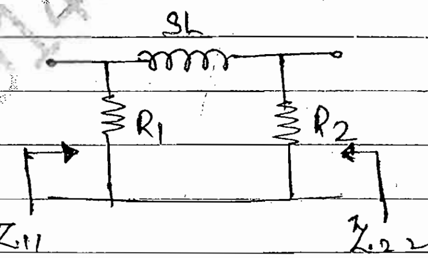
Q.15 $\frac{V(s)}{I(s)} = Z(s) = \frac{(s+1)}{(s+\sqrt{2})(s+1/\sqrt{2})}$

$i(t) = \sin t = I_m \sin \omega t$; $s = j\omega$
$\quad\quad\quad\quad\quad\quad \hookrightarrow 1 \quad \hookrightarrow 1$; $s = j1$

$Z(j\omega) = \frac{j1+1}{(j1+\sqrt{2})(j1+1/\sqrt{2})} = \frac{V}{I}$, $I = 1$

$|V| = \frac{\sqrt{2}}{(\sqrt{1+2})(\sqrt{1+1/2})} = \frac{2}{3}$ Ans.

Q.24 $Z_{11} = K_1\left(\frac{s+3}{8+s}\right)$



$Z_{11} = \frac{R_1 \times (R_2 + sL)}{R_1 + (R_2 + sL)}$

$= \frac{R_1 \times L\,(R_2/L + s)}{R_1 + L\,(R_2/L + s)} = \frac{R_1 \cancel{L}}{\cancel{L}} \frac{(R_2/L + s)}{\left(s + \frac{R_1 + R_2}{L}\right)}$

$K_1 = R_1$

$\frac{R_2}{L} = 3$; $\frac{R_1 + R_2}{L} = 8$; $\frac{R_1}{L} = 5$.

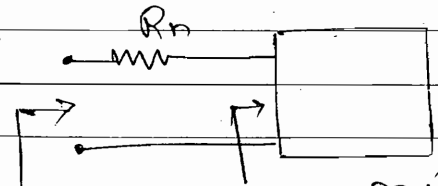
$Z_{22} = \frac{R_2 \times (sL + R_1)}{R_2 + (sL + R_1)}$

Denominator of Z_{11} and Z_{22} remain same so get direct ans.

$= \frac{R_2 \cancel{L}}{\cancel{L}} \frac{(R_1/L + s)}{\left(\frac{R_1+R_2}{L} + s\right)} = K_2\left(\frac{s+5}{s+8}\right)$

$K_2 = R_2$; $\frac{R_1+R_2}{L} = 8$; $\frac{R_1}{L} = 5$. Ans.

Q.25



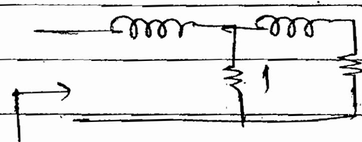
$Z_2 = R_n + Z_1$ $\quad Z_1 = r \pm jx$

PRF .... physically realizable

$Z_2 = \underbrace{R_n + Z_1}$ $\qquad |R_n| \leq r$

$= \underbrace{R_n + r}_{\geq 0} \pm jx$ $\qquad |R_n| \leq Re\,[z_1]$

Q.29 $\quad sL = 2s \qquad sL = s$



$s = -2, \infty.$ $\qquad Z(s) = 2s + \dfrac{1 \times (s+1)}{1 + (s+1)}$

Imp Question (Extra)

Q Following funcⁿ Represented physically realizable, stable of RLC driving point impedance function. Realise it in Foster's type-II N/w.

$$Z(s) = \frac{(s+1)(s+4)}{(s+2)(s+3)} = Z_{RLC}(s).$$

Solution F.T-2 is parallel N/w. — Foster's Type-2.

So, $Y(s) = \dfrac{1}{(Z)} = \dfrac{(s+2)(s+3)}{(s+1)(s+4)} = \dfrac{s^2 + 5s + 6}{s^2 + 5s + 4}$

$= 1 + \dfrac{2}{(s+1)(s+4)} = 1 + \dfrac{2/3}{s+1} + \dfrac{-2/3}{s+4}$

Due to -ve sign we cannot realised.

$\dfrac{2}{(s+1)(s+4)}$ ⇒ difference in degree is also 2

$$\frac{Y(s)}{s} = \frac{(s+2)(s+3)}{s(s+1)(s+4)} = \frac{3/2}{s} + \frac{-2/3}{s+1} + \frac{1/6}{s+4}$$

$$Y(s) = \frac{3}{2} + \frac{-2/3\, s}{s+1} + \frac{1/6\, s}{s+4}$$

• If value of Resistance coming -ve → stop there → Not Realizable

Divide only that term having -ve/ value.

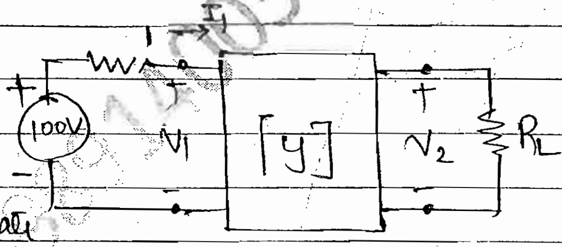
$$s+1 \overline{)\, s+4\,}(1 \quad ; \quad -s \pm 1 \quad ; \quad -1$$

$$Y(s) = \frac{3}{2} - \frac{2}{3}\left[1 - \frac{1}{s+1}\right] + \frac{1}{6}\,\frac{s}{(s+4)}$$

$$\boxed{Y(s) = \frac{5}{6} + \frac{2/3}{s+1} + \frac{(1/6)s}{s+4}}$$

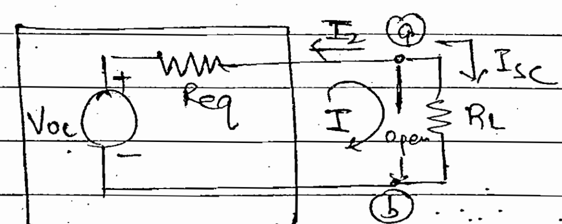
Now all terms Become +ve.

Q In the ckt shown find the value of load Resistance RL for Maximum power transferred. Hence calculate Maximum power transferred. $[Y] = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$; S



Solution: For Pmax : $R_L = R_{eq}$

$P_{max} = \frac{V_{oc}^2}{4R_L}$



Thevenin's Model.

$I_1 = 5V_1 + 4V_2$ — (1)

$I_2 = 4V_1 + 5V_2$ — (2)

$100 = 1 \times I_1 + V_1$ — (3)

Voc :- $V_2 = V_{oc}$; $I_2 = 0$

$I_1 = 5V_1 + 4V_{oc}$; $0 = 4V_1 + 5V_{oc}$; $100 = I_1 + V_1$

find Voc

Isc :- $V_2 = 0$; $I_2 = -I_{sc}$

$I_1 = 5V_1 + 0$; $-I_{sc} = 4V_1 + 0$; $100 = I_1 + V_1$

find Isc

find $R_{eq} = \frac{V_{oc}}{I_{sc}}$

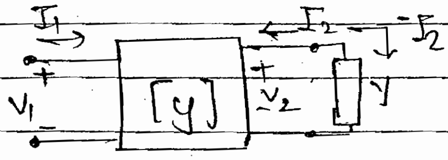
Shortcut

• $V_2 = 1\,V$; 100 V = S.C.

$R_{eq} = \frac{1}{I_2}$

$$\begin{cases} I_1 = 5V_1 + 4 \\ I_2 = 4V_1 + 5 \\ 0 = I_1 + V_1 \end{cases} \qquad \text{find } Req = \frac{1}{I_2}$$

Q. To find $G_{21} = \dfrac{V_2}{V_1}$



Solution $I_1 = y_{11}V_1 + y_{12}V_2$

$I_2 = y_{21}V_1 + y_{22}V_2$

$-I_2 = V_2 Y_L$

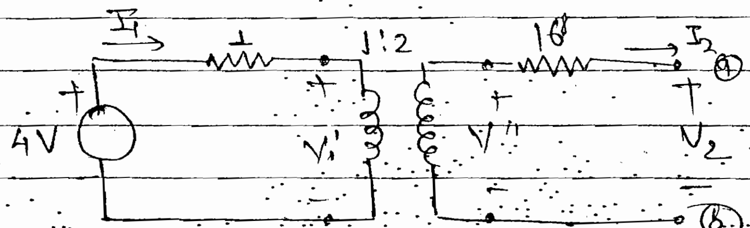
Putting value of eqn ③ in eqn ②

$-V_2 Y_L = y_{21}V_1 + y_{22}V_2$

$-y_{21}V_1 = V_2(y_{22} + Y_L)$

$\Rightarrow G_{21} = \dfrac{V_2}{V_1} = \dfrac{-y_{21}}{y_{22} + Y_L}$

Q



To Find:-

1. Thevenin's V_{oc}, Req.
2. Norton's I_{sc}, Req.

Solution: ① $4 = I_1 + V'$

② $V_2 = -16I_2 + V''$

③ $\dfrac{V'}{V''} = \dfrac{1}{2}$, $V'' = 2V'$

④ $\dfrac{I_2}{I_1} = \dfrac{1}{2}$, $I_1 = 2I_2$

V_{oc}: $I_2 = 0$; $V_2 = V_{oc}$

$V'' = 2V'$

$I_1 = 0$

$4 = V'$

$V_2 = V'' = V_{oc}$

I_{sc}: $V_2 = 0$; $I_2 = I_{sc}$

$4 = I_1 + V'$

$16 I_{sc} = V''$

$2V' = V''$

$I_1 = 2I_{sc}$

Req:- $V_2 = 1$, $4V = S.C.$

$Req = \dfrac{1}{I_{sc}} = \dfrac{1}{-I_2}$

$0 = I_1 + V'$

$1 = -16I_2 + V''$

$V'' = 2V'$

$I_1 = 2I_2$

$\boxed{V_{oc} = 8V \quad Req = 20\Omega \quad I_{sc} = 2/5 A}$